How characterisations of the trace fail: sharp violation bounds in terms of Michelson contrasts

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Abstract

A positive functional on a matrix algebra is a trace if and only if it satisfies any one of many inequalities, the first being Gardner's $|\varphi(X)| \le \varphi(|X|)$. We ask how strongly these characterisations fail for a non-tracial functional $\varphi_W = \mathrm{Tr}(W\,\cdot\,)$, measured by the Michelson contrast $\delta = \Delta(W)$ of its density. For Gardner's inequality the sharp constant depends on δ and on the contrast δ' of $|X|$ only through their product: $|\varphi_W(X)| \le (1-\delta^2\delta'^2)^{-1/2}\varphi_W(|X|)$, with equality in even dimension. As a contrast is the sine of the Krein–Gustafson angle of an operator, the constant is the secant of the angle whose sine is $\delta\delta'$. For 25 characterisations we find the largest violation exactly in sixteen cases and bound it from below in four; it vanishes at the trace to the orders δ, δ^2, δ^4 or δ^8, and on direct sums and direct integrals (finite type I algebras with separable predual) only the contrast inside the fibres counts, for all items except three that single out the canonical trace or are non-linear in φ. We also find the sharp bounds of three unnormalised trace functionals of our earlier work and sharp contrast thresholds below which subadditivity of the square root (via Gustafson's criterion for positive Jordan products) and the power-mean inequality cannot fail. The characterisations by the projection lattice extend to measures on projections; there the only new measures live in 2×2 blocks, where the envelopes are the same but attained. A numerical search, certified as a lower bound for each supremum, supports conjectured values for the remaining inequalities.

Keywords: trace characterisation; positive functional; Michelson contrast; operator angle; Kantorovich inequality; Jordan product; two projections.
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1 Introduction

Let φ be a positive linear functional on a C*-algebra. Gardner [25] proved that φ is a trace, $\varphi(XY) = \varphi(YX)$, if and only if

$$|\varphi(X)| \le \varphi(|X|) \qquad \text{for all } X. \tag{1}$$

Since then many inequalities that hold for every trace have been shown to hold *only* for traces: the triangle inequality for $\varphi(|\cdot|)$, monotonicity of $A \mapsto \varphi(A^2)$, Young's, Golden–Thompson's and Peierls–Bogoliubov's inequalities, and a family of inequalities in which only projections enter (Section 2 reviews them). Each of these theorems answers a yes/no question. For a functional that is not a trace, the inequality fails for some operators; but how strongly, and does the size of the failure measure how far the functional is from being a trace? This paper answers these questions for 25 characterisations on matrix algebras.

On M_n every positive functional is $\varphi_W = \mathrm{Tr}(W\,\cdot\,)$ with a density $W \ge 0$, and φ_W is a trace if and only if W is a multiple of 1. The natural scale-invariant distance of W from the scalars is its *Michelson contrast*

$$\Delta(W) = \frac{\lambda_{\max}(W) - \lambda_{\min}(W)}{\lambda_{\max}(W) + \lambda_{\min}(W)} \in [0,1],$$

which was studied for operators in [1]. Gibbs states show that this is a physically meaningful scale. For a Hamiltonian H with extreme eigenvalues $E_{\min} \le E_{\max}$ and the inverse temperature β, the density $W = e^{-\beta H}/\operatorname{Tr} e^{-\beta H}$ has

$$\Delta(W) = \tanh \frac{\beta(E_{\max} - E_{\min})}{2},$$

so $\Delta(W) \to 0$ at infinite temperature, where the Gibbs state becomes the normalised trace, and $\Delta(W) \to 1$ at zero temperature. Every bound in terms of $\Delta(W)$ is therefore a law in the temperature: for Gardner's inequality we shall find the largest relative violation $\cosh(\beta(E_{\max} - E_{\min})/2) - 1$. In the language of modular theory, φ_W (for invertible W) is a trace if and only if its modular group $\sigma_t(X) = W^{it} X W^{-it}$ is trivial; the derivation $X \mapsto [\log W, X]$ that generates it has the spectrum $\{\log w_i - \log w_j\}$, whose largest point is $\log(\lambda_{\max}(W)/\lambda_{\min}(W)) = 2 \operatorname{artanh} \Delta(W)$. The contrast thus measures how far the modular dynamics is from being trivial.

In [1] we plotted the violation of Gardner's inequality against $\Delta(W)$ for random real 2×2 and 3×3 matrices [1, Sect. 5.1, Fig. 1] and asked whether such violations are bounded by functions of the contrasts [1, Sect. 7]. To make the question precise, write an inequality as $\mathcal{L} \le \mathcal{R}$, where $\mathcal{L}$ and $\mathcal{R}$ depend on φ and on operator variables, and fix a normalisation $\mathcal{N} > 0$ that makes the quotient $(\mathcal{L} - \mathcal{R})/\mathcal{N}$ invariant under the scaling of W and of the variables (mostly $\mathcal{N} = \mathcal{R}$). The *violation envelope* is

$$V_n(\delta) = \sup\Big\{ \frac{\mathcal{L} - \mathcal{R}}{\mathcal{N}} : \ W \ge 0 \text{ in } M_n, \ \Delta(W) = \delta, \ \text{all operator variables free} \Big\}.$$

By the characterisation theorems $V_n(\delta) > 0$ for $\delta > 0$, and $V_n(0) = 0$. A violation needs more than a non-tracial functional: if the operators are scalars, nothing can go wrong. So we also prescribe the contrast δ' of the operator variables (of their moduli, for operators that are not positive) and write $V_n(\delta, \delta')$.

The main result. For Gardner's inequality both contrasts can be prescribed and the answer is a function of their product alone.

Theorem A. *Let $W > 0$ and X be in M_n, $n \ge 2$, with $\Delta(W) = \delta$ and $\Delta(|X|) = \delta'$, $\delta\delta' < 1$. Then*

$$|\varphi_W(X)| \le \frac{1}{\sqrt{1 - \delta^2 \delta'^2}} \, \varphi_W(|X|),$$

and for even n the constant is attained, so $V_n(\delta, \delta') = (1 - \delta^2\delta'^2)^{-1/2} - 1$. For odd n the envelope is given by an explicit one-dimensional maximum; it is smaller when $\delta' < 1$ and tends to the even value as $n \to \infty$. The same upper bound holds for the triangle inequality $\varphi(|X + Y|) \le \varphi(|X|) + \varphi(|Y|)$ and for $\varphi(A + A^*) \le 2\varphi(|A^*|)$ (Theorem 6.1).

The theorem has a simple geometric meaning (Figure 1). Krein [39] and Gustafson [28] introduced the angle $\psi(A)$ of an operator, the largest angle between a vector x and its image Ax; for $A > 0$, $\sin \psi(A) = \Delta(A)$, since A divided by the midpoint $\bar{\lambda}(A)$ of its spectrum is $1 + E$ with $\|E\| = \Delta(A)$. If $|X| = vv^*$ has rank one ($\delta' = 1$), Gardner's quotient is $|\langle Wv, Uv\rangle|/\langle Wv, v\rangle \le \|Wv\|/\langle Wv, v\rangle = \sec \angle(v, Wv) \le \sec \psi(W)$, which is the Kantorovich constant of W. For general $|X|$, the proof reduces to two-dimensional blocks on which W and $|X|$ are, up to scale, $1 + \delta S$ and $1 + \delta' S'$ with symmetries S, S'; there $\operatorname{Tr}(|X|W)$ and the trace norm $\||X|W\|_1$ are $2\operatorname{Re} z$ and $2|z|$ for $z = 1 + \delta\delta' e^{i\alpha}$, where $e^{\pm i\alpha}$ are the eigenvalues of the rotation $S'S$. Gardner's quotient is $\sec \arg z$, and the circle described by z is seen from the origin under the angle θ with $\sin\theta = \delta\delta' = \sin\psi(W) \sin\psi(|X|)$. The product of the contrasts is the radius of this circle, and the constant of Theorem A is $\sec\theta$.

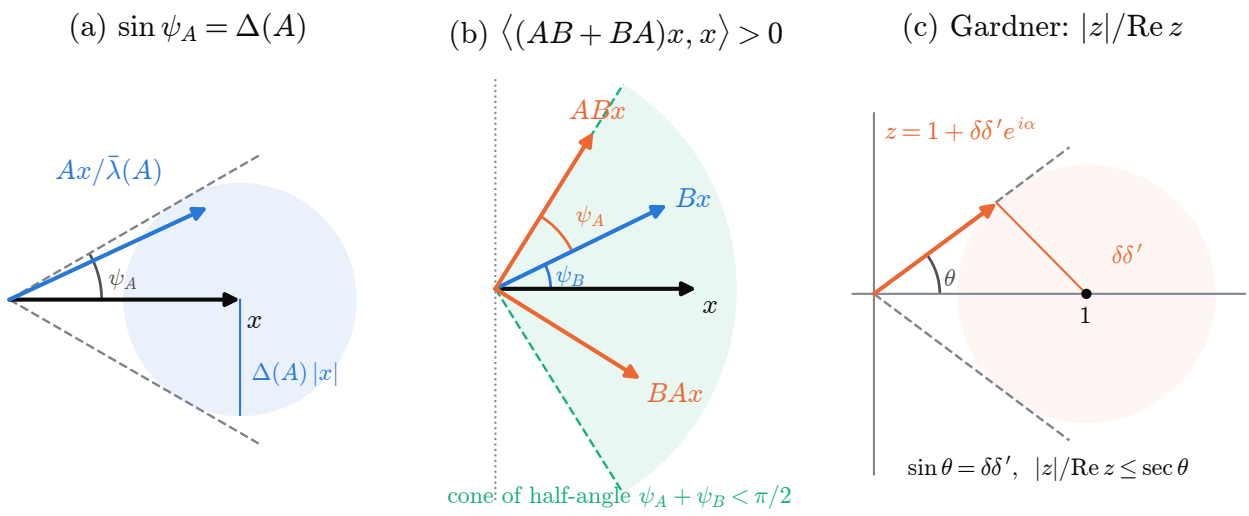


Figure 1: (a) The angle of a positive operator: $Ax/\bar{\lambda}(A)$ lies in the ball of radius $\Delta(A)|x|$ about x, so $\angle(x, Ax) \le \psi_A = \arcsin\Delta(A)$. (b) For a Jordan product the angles add; if $\psi_A + \psi_B < \pi/2$, then ABx and BAx stay in a cone around x that lies in the half-space of x, and $AB + BA > 0$ (Theorem 11.2). (c) On a two-dimensional block, Gardner's quotient is $|z|/\mathrm{Re}\,z = \sec\arg z$ for a point z on the circle of radius $\delta\delta'$ about 1 (Remark 6.2); the largest argument is $\arcsin(\delta\delta')$.

Further results. With the operators free, we determine the envelopes of 16 of the 25 inequalities exactly and bound four others from below (Theorem B in Section 4). The constants that appear are the Kantorovich constant $\kappa = \sec\psi(W)$, the condition number of W and the Specht ratio; the envelopes vanish at the trace to the orders δ, δ^2, δ^4 and δ^8 (Figure 2). So the characterisations differ greatly in sensitivity: two inequalities that are equivalent for traces may fail at different rates. Returning to the unnormalised quantities of [1], we find the sharp bound of $|\mathrm{Tr}(XZ)| - \mathrm{Tr}(X|Z|)$ with both contrasts prescribed and those of the two other trace functionals sampled there (Theorem C), which answers the question of [1]. On a direct sum $\bigoplus_c M_{k_c}$ a functional is a trace when its density is central, however different the scalars on the blocks, and accordingly only the contrast inside the blocks enters the envelopes (Theorem D). For every finite type I algebra with separable predual, i.e. for direct integrals over a σ-finite measure space, the envelope is bounded by the essential supremum of the fibre envelopes, with equality for densities that are constant on cells (Theorem 10.3).

The operators enter the inequalities through composites such as sums, congruences, Jordan products and commutators, and Section 11 records how contrasts behave under these operations: the rapidities $\operatorname{artanh}\Delta$ add under products and means (this is the triangle inequality of Hilbert's projective metric), the angles $\arcsin\Delta$ add under Jordan products (Gustafson's criterion), and the commutator test $\varphi(AB) = \varphi(BA)$ fails by exactly $2\Delta(W)\Delta(A)\Delta(B)$ in even dimension, after the natural normalisation. These rules have a consequence for the inequalities themselves. The numerical search shows that several characterisations are violated only by operators whose contrast is close to 1. For subadditivity of the square root, $\varphi((A+B)^{1/2}) \le \varphi(A^{1/2}) + \varphi(B^{1/2})$, we prove this: if $\Delta(A) = \Delta(B) \le 2\sqrt{2}/3$, then $A^{1/2}B^{1/2} + B^{1/2}A^{1/2} \ge 0$ by Gustafson's criterion, hence $(A+B)^{1/2} \le A^{1/2} + B^{1/2}$ as operators, and no positive functional can violate the inequality; above $2\sqrt{2}/3$ it is violated, for every contrast of the density close enough to 1, by suitable operators and densities. For the power-mean inequality $\varphi(A^2)^{1/2} \le \varphi(A)$ we find the envelope with both contrasts prescribed in closed form; it is never positive when $\Delta(A) \le \sqrt{(n-1)/n}$, and this threshold is sharp (Section 11.1 and Figure 6).

What is new and what is classical. The characterisation theorems are due to the cited authors. The constants κ, h and the Specht ratio, and the inequalities behind them (Kantorovich's inequality in the operator form of Greub and Rheinboldt [27], Specht's reverse arithmetic–geometric mean inequality [48]), are classical, and so are several statements of Section 11: the bounds for congruences are the submultiplicativity of the condition number, the positivity criterion for Jordan products is Gustafson's [28], and the norm bound for commutators of positive operators follows from Kittaneh's inequality [37]. What we add is that the classical constants are the exact envelopes, the dependence on the product of the contrasts in Theorem A, the exact envelopes for pairs of projections (Theorem 8.1), the constructions that vanish to the orders δ^4 and δ^8, the sharp bounds of Theorem C, the sharp trace version of the commutator bound, and the threshold results of Section 11.1. We know of no previous study of the envelopes; the closest work is [1], which samples the unnormalised Gardner violation but defines no envelope and proves no bound.

Numerical evidence. For the inequalities without a closed form we computed the envelopes by constrained optimisation over all operator variables. Every reported value is recomputed in 40-digit arithmetic at an exactly admissible point, so it is a rigorous *lower* bound for the supremum, not a certified value of it. Where the envelope is known, the search reproduces it to 1×10^{-7} (items 1–5, 8 and 11, and 9 for $n = 2$), to 3×10^{-4} (items 17, 18, 20–22 and 24) and, for the suprema of items 19 and 23, which are only approached, to 2×10^{-6}; elsewhere it supports the conjectured values of Section 13. Two observations from the search guide the theory. First, the largest violation is usually found at projections or at two-level operators $\ell + (1-\ell)P$; this is a theorem for the items to which the vertex principle (Lemma 3.5) applies and an observation for the others, with exceptions in which the extremisers have three eigenvalues. Second, random sampling, the method of [1], rarely comes close to the envelope: for monotonicity and subadditivity fewer than one random sample in a thousand violates the inequality at all, so sampling alone would suggest that these inequalities hold for every functional. The envelope searches use the open-source library described in the companion paper [41], which reproduces and extends the NumPy simulations of [1]; part of the present results appears there as a use case. The random sampling, the figure for Theorem C and the computations for composite operators are separate NumPy/SciPy programs that do not use the library (Section 12.1).

Organisation. Section 2 reviews related work. Section 3 lists the inequalities and collects the tools, and Section 4 states Theorems B–D. Section 6 proves Theorems A and C; Sections 5, 7, 8 and 9 prove Theorem B, and Section 10 Theorem D. Section 11 treats composite operators and the thresholds, Section 12 the numerical evidence, and Section 13 lists conjectures and open problems. Appendix A shows the computed envelopes and random samples item by item.

2 Related work

Characterisations of the trace. After Gardner [25], traces on M_n or among the positive normal functionals on a von Neumann algebra were characterised by the triangle inequality for $\varphi(|\cdot|)$ [25, 49], by $|\varphi(A^k)| \le \varphi(|A|^k)$ [46], by monotonicity $0 \le A \le B \Rightarrow \varphi(A^p) \le \varphi(B^p)$ [8, 47] and its weighted forms [32, 33], by Young's inequality [7, 22], by subadditivity of $\varphi \circ f$ for concave f [52], by the Peierls–Bogoliubov and Araki–Lieb–Thirring inequalities [12], by the generalised Powers–Størmer inequality [34], by inequalities for determinants and power means [15], by the spectral geometric mean [18], by inequalities for $|\varphi(A^2)|$ [17], by metrics on projections [14], by subadditivity on the projection lattice [46, Theorem 4], and by inequalities in which only projections enter [9, 10, 11, 13]. Bikchentaev and Moslehian [16] survey these results; the canonical trace on M_n is characterised

in [43, 15], and the dual problem, in which the functional is arbitrary and an operator is to be recognised as central, is treated in [45]. All these results are qualitative.

Reverse inequalities. Our upper bounds are reverse inequalities of Kantorovich type. Kantorovich's inequality [35] and its operator form due to Greub and Rheinboldt [27] bound $\omega(W^2)\omega(1)$ by $\kappa^2\omega(W)^2$; the Mond–Pečarić method [24] produces such constants systematically, and Specht's ratio [48] is the sharp constant of the reverse arithmetic–geometric mean inequality. The question here is different: not whether such a constant bounds the violation, but whether it is the smallest one.

Operator angles. Krein [39] introduced the deviation of an operator and Gustafson [28] the equivalent angle $\psi(A)$, $\cos\psi(A) = \inf_{x\neq 0} \operatorname{Re}\langle Ax, x\rangle/(\|Ax\|\|x\|)$, the first antieigenvalue; for $0 < m \leq A \leq M$ with m, M in the spectrum, $\cos\psi(A) = 2\sqrt{Mm}/(M+m)$ and $\sin\psi(A) = (M-m)/(M+m)$ [30, 29]. Thus the Michelson contrast of a positive operator is the sine of its angle, and the Kantorovich constant is $\sec\psi(W)$. Gustafson used the angle to decide when products of positive operators are accretive [28].

Jordan products. The product AB of positive operators is accretive exactly when the Jordan product $AB + BA = 2\operatorname{Re}(AB)$ is positive. Bounds for the eigenvalues of $AB + BA$ go back to Strang [50] and Nicholson [44]; Alikakos and Bates [2] bounded all eigenvalues and showed that Strang's lower bound for the smallest one is best possible [3], and Conley, Pucci and Serrin [23] gave necessary and sufficient conditions for the positivity of the symmetrised product of two positive definite matrices. Moslehian and Najafi [42, Theorem 2.1] proved that $AB + BA \geq 0$ if and only if $f(A+B) \leq f(A) + f(B)$ for every non-negative operator monotone function f on $[0,\infty)$. This links Jordan products to the subadditivity inequalities that characterise traces [52], and it is the background of Section 11.1.

Commutators and Hilbert's metric. Kittaneh [37] proved $\|XY - YX\| \leq \|X\|\|Y\|$ when $X \geq 0$, which gives $\|AB - BA\| \leq \frac{1}{2}\|A\|\|B\|$ for positive A, B; versions for unitarily invariant norms are in [36, 6]. On the cone of positive definite matrices, Thompson's metric [51] is $d_T(A,B) = \|\log(A^{-1/2}BA^{-1/2})\|$ and Hilbert's projective metric [20, 40] is $d_H(A,B) = \log\bigl(\lambda_{\max}/\lambda_{\min}\bigr)(A^{-1/2}BA^{-1/2})$. The rapidity of a positive definite A is

$$\operatorname{artanh}\Delta(A) = \tfrac{1}{2}\log\frac{\lambda_{\max}(A)}{\lambda_{\min}(A)} = \tfrac{1}{2}\,d_H(A,1) = \min_{t>0} d_T(A,t1),$$

the Thompson distance of A to the scalars; this is why rapidities behave additively under products (Proposition 11.1).

3 Setting and tools

3.1 Functionals, contrast and envelopes

Throughout, Tr is the usual trace on $M_n = M_n(\mathbb{C})$, $|X| = (X^*X)^{1/2}$, $\|X\|_1 = \operatorname{Tr}|X|$, $\operatorname{Re}X = \frac{1}{2}(X + X^*)$, and $\varphi = \varphi_W = \operatorname{Tr}(W\,\cdot\,)$ with $W \geq 0$; φ is a trace iff $W \in \mathbb{C}1$. We write $a \geq b$ for the extreme eigenvalues of W, e_1, e_2 for corresponding unit eigenvectors, and $\delta = \Delta(W)$; unless stated otherwise $W > 0$, i.e. $\delta < 1$. The quotient $(\mathcal{L} - \mathcal{R})/\mathcal{N}$ is invariant under $W \mapsto tW$, $t > 0$, and the envelope $V_n(\delta)$ is the supremum over all W with $\Delta(W) = \delta$; for a fixed density we write $V(W)$. We call a value of an envelope *attained* if the supremum is a maximum and *approached*

otherwise. Variables called P, Q, R are orthogonal projections, $A, B \geq 0$ unless said otherwise, and X, Y arbitrary; $P \wedge Q$ and $P \vee Q$ are the projections onto $\operatorname{ran} P \cap \operatorname{ran} Q$ and onto $\operatorname{ran} P + \operatorname{ran} Q$.

	inequality $\mathcal{L} \leq \mathcal{R}$	variables	$\mathcal{N}$	source
1	$\lvert\varphi(X)\rvert \leq \varphi(\lvert X\rvert)$	X	$\mathcal{R}$	[25]
2	$\varphi(\lvert X+Y\rvert) \leq \varphi(\lvert X\rvert) + \varphi(\lvert Y\rvert)$	X, Y	$\mathcal{R}$	[25, 49]
3	$\varphi(A + A^*) \leq 2\varphi(\lvert A^*\rvert)$	A arbitrary	$\mathcal{R}$	[13]
4	$\lvert\varphi(A^2)\rvert \leq \varphi(A^*A)$	A arbitrary	$\mathcal{R}$	[46, Thm. 1, $k=2$], [17]
5	$\varphi(\lvert P-R\rvert) \leq \varphi(\lvert P-Q\rvert) + \varphi(\lvert Q-R\rvert)$	P, Q, R	$\mathcal{R}$	[14]
6	$\varphi(A \natural B) \leq \sqrt{\varphi(A)\varphi(B)}$	$A, B > 0$	$\mathcal{R}$	[18]
7	$\varphi((B^{1/2}AB^{1/2})^2) \leq \varphi(BA^2B)$	A, B	$\mathcal{R}$	[12, Lemma 2, $r=2$, $p=1$]
8	$\varphi(\lvert AB^*\rvert) \leq \frac{1}{2}\varphi(\lvert A\rvert^2) + \frac{1}{2}\varphi(\lvert B\rvert^2)$	A, B arbitrary	$\mathcal{R}$	extension of [7]
9	$\varphi(PQ + QP) \leq 2\varphi(QPQ)$	P, Q	$\varphi(P) + \varphi(Q)$	[13]
10	$0 \leq A \leq B \Rightarrow \varphi(A^2) \leq \varphi(B^2)$	$A \leq B$	$\mathcal{R}$	[8, 47]
11	$(\det A)^{1/n} \leq \frac{1}{n}\varphi(A)$, $\varphi(1) = n$	A	$(\det A)^{1/n}$	[15, Thm. 3(iii)]
12	$\varphi((A+B)^{1/2}) \leq \varphi(A^{1/2}) + \varphi(B^{1/2})$	A, B	$\mathcal{R}$	[52, Cor. 1]
13	$\varphi(A + B - \lvert A-B\rvert) \leq 2\varphi(A^{1/4}B^{1/2}A^{1/4})$	A, B	$\varphi(A+B)$	[34, Lemma 3.1]
14	$\lvert\varphi(PQ)\rvert \leq \varphi(P)^{1/3}\varphi(Q)^{2/3}$	P, Q	$\mathcal{R}$	[11, Thm. 2(i)]
15	$\varphi(e^{P+Q}) \leq \varphi(e^{P/2}e^{Q}e^{P/2})$	P, Q	$\mathcal{R}$	[11, Thm. 2(iv)]
16	$\varphi(e^A)\exp\frac{\varphi(e^{A/2}Be^{A/2})}{\varphi(e^A)} \leq \varphi(e^{A+B})$	A, B self-adj.	$\mathcal{R}$	[12, Lemma 1]
17	$\varphi(PQP) \leq \varphi(Q)$	P, Q	$\mathcal{R}$	[10, Thm. 3.4(iv)]
18	$\varphi(\lvert P-Q\rvert) \leq \varphi(P+Q)$	P, Q	$\mathcal{R}$	[10, Thm. 3.4(v)]
19	$\varphi(P+Q) \leq \varphi(1 + P \wedge Q)$	P, Q	$\mathcal{R}$	[10, Thm. 3.4(iii)]
20	$\varphi(e^{QPQ}) \leq \varphi(e^P)$	P, Q	$\mathcal{R}$	[11, Thm. 2(v)]
21	$0 \leq \varphi(\operatorname{Re} PQ)$	P, Q	$\varphi(P) + \varphi(Q)$	[11, Thm. 2(vi)]
22	$\varphi(\operatorname{Re} PQ) \leq \varphi(\lvert PQ\rvert)$	P, Q	$\mathcal{R}$	[11, Thm. 2(vii)]
23	$\varphi(P \vee Q) \leq \varphi(P) + \varphi(Q)$	P, Q	$\mathcal{R}$	[46, Thm. 4]
24	$\varphi(A^2)^{1/2} \leq \varphi(A)$, $\varphi(1) = n$	A	$\mathcal{R}$	[15, Thm. 3(vii)]
25	$0 \leq A \leq B \Rightarrow \varphi(A^4) \leq \varphi(AB^2A)$	$A \leq B$	$\mathcal{R}$	[33, Thms. 4, 5, $r=1$], [32]

Table 1: The inequalities studied, with the normalisation $\mathcal{N}$ of the violation $(\mathcal{L}-\mathcal{R})/\mathcal{N}$ and the source of the characterisation. Projections are P, Q, R; positive operators A, B unless stated otherwise. Remarks on the forms are in Section 3.2.

3.2 Remarks on the forms

Several sources state families of inequalities; we use one member. In item 4, $\lvert\varphi(A^k)\rvert \leq \varphi(\lvert A\rvert^k)$ for all A and one fixed $k \geq 1$ characterises the trace on M_n [46, Theorem 1]; $k = 1$ is Gardner's inequality. Item 7 is the Araki–Lieb–Thirring inequality $\varphi((B^{1/2}AB^{1/2})^{rp}) \leq \varphi((B^{r/2}A^rB^{r/2})^p)$ with $r = 2$, $p = 1$ [12, Lemma 2]. The published form of item 8 is $\varphi(\lvert AB\rvert) \leq \frac{1}{p}\varphi(A^p) + \frac{1}{q}\varphi(B^q)$ for $A, B \geq 0$ [7]; we use $p = q = 2$ and allow arbitrary A, B, in the form of Ando's matrix Young inequality [4], and Proposition 7.1 refers to this extension. Item 9 is the member $p = 1$ of the family $\varphi(PQ + QP) \leq 2\varphi((QPQ)^p)$, $p \in (0, 1]$ [13]. Items 10 and 25 are the cases $r = 0$ and $r = 1$ of $\varphi(A^{2r+2}) \leq \varphi(A^rB^2A^r)$ for $0 \leq A \leq B$, which characterises traces for every fixed $r \geq 0$ [33, Lemma 1, Theorems 4, 5]; on M_n the case $r = 0$ is in [8, 47] and $r > 0$ in [32, Theorem 6]. Items 11 and 24 characterise the canonical trace among the positive functionals with $\varphi(1) = n$ [15, Theorem 3]; item 24 is (vii) there with $p = 2$, $q = 1$ and needs φ faithful. Item 12 is the case $f(t) = t^{1/2}$ of Tikhonov's characterisation by $\varphi(f(A+B)) \leq \varphi(f(A)) + \varphi(f(B))$ for concave f with $f(0) = 0$ [52, Corollary 1]. Item 13 is the generalised Powers–Størmer inequality $\varphi(A+B) - \varphi(\lvert A-B\rvert) \leq 2\varphi(f(A)^{1/2}g(B)f(A)^{1/2})$, $g(t) = t/f(t)$, with $f(t) = t^{1/2}$ [34, Lemma 3.1, Remark 4]; with $\mathcal{N} = \mathcal{R}$ its envelope would be infinite

(Remark 9.3), so we normalise by $\varphi(A+B)$. Item 14 is Hölder's inequality for projections with the exponent $\alpha = 3$; for $\alpha = 2$ it holds for every positive φ [11, remark after Theorem 2]. Item 16 is the Peierls–Bogoliubov inequality, stated in [12] for positive H, K; both sides scale by e^c under $A \mapsto A + c$, $B \mapsto B + c$, so the envelope over self-adjoint arguments is the same. Items 17–19 are [10, Theorem 3.4], announced in [9], and items 14, 15, 20–22 are [11, Theorem 2].

3.3 Tools

Lemma 3.1 (Kantorovich inequality [27]). *If $b \le W \le a$, $a \ge b > 0$, then $\omega(W^2)\,\omega(1) \le \kappa^2\,\omega(W)^2$ for every positive functional ω, where $\kappa = (a+b)/(2\sqrt{ab})$.*

Proof. $(a - W)(W - b) \ge 0$ gives $\omega(W^2) \le (a+b)x - ab\,y$ with $x = \omega(W)$, $y = \omega(1)$, and $((a+b)x - aby)y \le \kappa^2 x^2$ because the difference is $(\kappa x - \sqrt{ab}\,y)^2$. □

Lemma 3.2 (Two projections; Halmos [31]). *For projections P, Q on a finite-dimensional Hilbert space H, H is the orthogonal sum of subspaces of dimension 1 or 2 invariant under P and Q. On each two-dimensional one there is an orthonormal basis f, f' with $P = ff^*$ and $Q = qq^*$, $q = \cos\theta\, f + e^{i\chi}\sin\theta\, f'$, $0 < \theta < \pi/2$.*

The next lemma gives a general upper bound; it shows that no envelope of items 1–8 and 10 can exceed the order δ.

Lemma 3.3. *For items 1–8 and 10, $V(W) \le h - 1$.*

Proof. In each item $\mathcal{L} \le a\mathcal{L}_0$, where $\mathcal{L}_0$ is the expression for the trace or the trace norm (e.g. $|\varphi(X)| \le a\|X\|_1$, $\varphi(|X+Y|) \le a\,\mathrm{Tr}\,|X+Y|$, $|\varphi(A^2)| \le a\|A^2\|_1$), $\mathcal{L}_0 \le \mathcal{R}_{\mathrm{Tr}}$ by the trace inequality (for item 4 $\|A^2\|_1 \le \|A\|_2^2$; for item 7 the Araki–Lieb–Thirring inequality; for item 6, $A\natural B$ is unitarily similar to $(A^{1/2}BA^{1/2})^{1/2}$, so $\mathrm{Tr}(A\natural B) = \|A^{1/2}B^{1/2}\|_1 \le (\mathrm{Tr}\,A\,\mathrm{Tr}\,B)^{1/2}$), and $\mathcal{R} \ge b\mathcal{R}_{\mathrm{Tr}}$. Hence $\mathcal{L}/\mathcal{R} \le a/b = h$. For item 8, $\varphi(|AB^*|) \le a\|A\|_2\|B\|_2 \le \frac{a}{2}(\|A\|_2^2 + \|B\|_2^2)$. □

3.4 Contrasts of the operator variables

The envelope $V_n(\delta)$ lets the operator variables range over all operators. To see how the violation depends on the operators themselves, we also prescribe their contrast. For a positive variable A we use $\Delta(A)$, for an arbitrary or self-adjoint variable X the contrast $\Delta(|X|)$ of its modulus (for the Peierls–Bogoliubov inequality, item 16, the variables are positive, as in [12]), and we let every operator variable of an item have the same contrast $\delta' \in [0,1]$:

$$V_n(\delta,\delta') = \sup\Big\{\frac{\mathcal{L}-\mathcal{R}}{\mathcal{N}} :\ \Delta(W) = \delta,\ \text{every operator variable of contrast } \delta'\Big\}.$$

Projections other than 0 and 1 have contrast 1, so for items 5, 9, 14, 15 and 17–23 only $\delta' = 1$ occurs, and $V_n(\delta,1) \le V_n(\delta)$ with equality whenever the supremum over all operators is approached by singular ones. A positive operator of contrast δ' whose spectrum has only two points is, up to a positive factor,

$$A = \ell' + (1-\ell')P, \qquad \ell' = \frac{1-\delta'}{1+\delta'}, \qquad P \text{ a projection},\ P \ne 0,1;$$

we call such operators *two-level*. They are the affine images of projections, and as $\delta' \to 1$ they become scaled projections.

Proposition 3.4. *If every operator variable is a positive multiple of the identity (for arbitrary and self-adjoint variables: if every modulus is), then $\mathcal{L} \le \mathcal{R}$ in every item of Table 1 and for every positive φ. Hence $V_n(\delta, 0) \le 0$ for all items; and $V_n(0, \delta') \le 0$, since for $\delta = 0$ the functional is a multiple of the trace (the canonical trace in items 11 and 24, where $\varphi(1) = n$).*

Proof. Item by item, with $|X| = x$, $|Y| = y$ scalar and $X = xU$, $Y = yV$ with unitaries (symmetries for self-adjoint X, Y), and $A = \alpha$, $B = \beta$ scalars: (1) $|\varphi(U)| \le \varphi(1)$; (2) $\varphi(|xU + yV|) \le \|xU + yV\|\varphi(1) \le (x + y)\varphi(1)$; (3) and (4) as (1); (6) and (7) are equalities; (8) $|AB^*| = xy$ and $xy \le \frac{1}{2}(x^2+y^2)$; (10) and (25) follow from $\alpha \le \beta$; (11) is an equality for $\varphi(1) = n$; (12) $\sqrt{\alpha+\beta} \le \sqrt{\alpha}+\sqrt{\beta}$; (13) $\alpha + \beta - |\alpha - \beta| = 2\min(\alpha, \beta) \le 2\sqrt{\alpha\beta}$; (16) is an equality; (24) $\varphi(\alpha^2)^{1/2} = \alpha\sqrt{n} \le \alpha n$. □

So a violation needs both contrasts: the functional must be far from a trace and the operators must be far from scalars. The next lemma is the mechanism behind the two-level operators: whenever the violation depends convexly on the spectrum of a variable, the extremal spectra have only two points.

Lemma 3.5 (Vertex principle)**.** *Fix all variables except the eigenvalues $\lambda \in [\ell', 1]^n$ of one positive variable (its eigenbasis fixed), and let $F(\lambda)$ and $\mathcal{N}(\lambda) > 0$ be the resulting values of $\mathcal{L} - \mathcal{R}$ and $\mathcal{N}$. If $\lambda \mapsto F(\lambda) - c\mathcal{N}(\lambda)$ is convex for every $c \in \mathbb{R}$, then the supremum of $F/\mathcal{N}$ over the spectra with $\max_i \lambda_i = 1$ and $\min_i \lambda_i = \ell'$ equals its supremum over the two-level spectra $\lambda \in \{\ell', 1\}^n$ (both values occurring).*

Proof. The admissible set is the union of the faces $\{\lambda_i = 1,\ \lambda_j = \ell'\}$ of the cube, and the vertices of these faces are the two-level spectra. If $F/\mathcal{N} > c$ at an admissible point, then $F - c\mathcal{N} > 0$ there, and by convexity $F - c\mathcal{N} > 0$ at a vertex of the same face, where therefore $F/\mathcal{N} > c$. □

The hypothesis holds for Gardner's inequality ($F - c\mathcal{N} = |\varphi(UT)| - (1 + c)\varphi(T)$ for $X = UT$: the modulus of a linear function minus a linear one), for item 3, for item 24 ($\varphi(A^2)^{1/2} = \|W^{1/2}A\|_2$ is a norm of a linear function of λ) and for the unnormalised Gardner violation of Section 6; for these items two-level operators suffice. For the other items, whether they suffice is a question we examine numerically (Section 12.4).

4 Statement of the results

Write $a \ge b > 0$ for the extreme eigenvalues of W and

$$h = \frac{a}{b} = \frac{1+\delta}{1-\delta}, \qquad \kappa = \frac{a+b}{2\sqrt{ab}} = \frac{1}{\sqrt{1-\delta^2}} = \sec\psi, \qquad \psi = \arcsin\delta,$$

the condition number, the Kantorovich constant and the angle of W. With the operator variables free, the envelopes are as follows.

Theorem B. *(a) $V_n = \kappa - 1$ for Gardner's inequality, the triangle inequalities for $\varphi(|\cdot|)$ on operators and on projections, $\varphi(A+A^*) \le 2\varphi(|A^*|)$, $|\varphi(A^2)| \le \varphi(A^*A)$, $\varphi(|P-Q|) \le \varphi(P+Q)$ and $\varphi(\operatorname{Re} PQ) \le \varphi(|PQ|)$, in every dimension $n \ge 2$ (Theorems 5.1 and 8.1).*

(b) $V_n = \kappa^2 - 1$ for $\varphi(PQP) \le \varphi(Q)$; $V_n = \frac{1}{2}\tan^2(\psi/2)$ for $\varphi(\operatorname{Re} PQ) \ge 0$; $V_n = \delta/(1-\delta)$ for $\varphi(P \vee Q) \le \varphi(P) + \varphi(Q)$; $V_n = \delta$ for $\varphi(P + Q) \le \varphi(1 + P \wedge Q)$ (n even); and V_n is the maximum of an explicit function of one variable, $\frac{(e-1)^2}{e(e+1)}\delta^2 + O(\delta^3)$, for $\varphi(e^{QPQ}) \le \varphi(e^P)$ (n even) (Theorem 8.1).

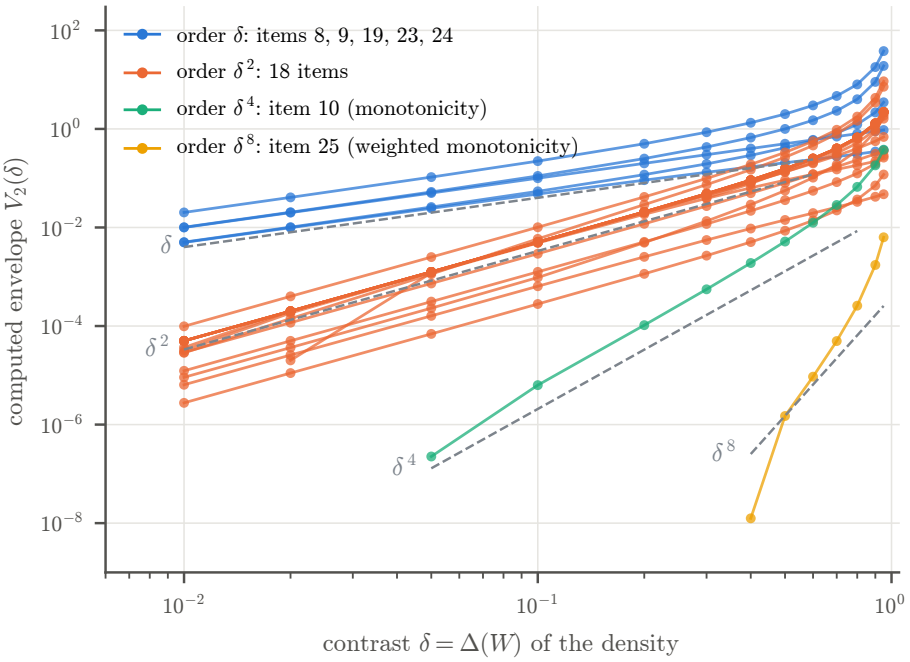


Figure 2: The ladder of orders: the computed envelopes $V_2(\delta)$ of all 25 items (certified lower bounds for the suprema, Section 12) on logarithmic axes, coloured by the order of vanishing at the trace, with guide lines of slopes 1, 2, 4 and 8. Item 25 is shown for $\delta \geq 0.4$ (the search resolves it for $\delta \geq 0.3$).

(c) $V_n = h - 1$ *for the extension of Young's inequality to arbitrary operators;* $V_2 = s/(1+s)$, $s = \sin(\psi/2)$, *for* $\varphi(PQ+QP) \leq 2\varphi(QPQ)$*; for the inequalities that characterise the* canonical *trace,* $V_2 = 1 - \sqrt{1-\delta^2}$ *for the inequality* $(\det A)^{1/n} \leq \frac{1}{n}\varphi(A)$*, increasing with* n *to* $1 - 1/S(h)$ *with the Specht ratio* S*, and* $V_n = \sqrt{(\ell+n-1)/(n\ell)} - 1$*,* $\ell = 1/h$*, for* $\varphi(A^2)^{1/2} \leq \varphi(A)$ *(Section 7).*

(d) $V_n \geq \kappa - 1$ *for the spectral geometric mean inequality,* $V_n \geq \kappa^2 - 1$ *for the Araki–Lieb–Thirring inequality,* $V_n \geq \delta^4/16 + O(\delta^5)$ *for monotonicity and* $V_n \geq \delta^8/4096\,(1 + O(\delta^2))$ *for its weighted form with* $r = 1$*; and* $V_n \leq h - 1$ *for items 1–8 and 10 (Section 9 and Lemma 3.3).*

The envelopes thus vanish at the trace to different orders, δ, δ^2, δ^4 and δ^8 (Figure 2 and Table 2), and two inequalities that are equivalent for traces may fail at different rates. For instance, $\varphi(P \vee Q) + \varphi(P \wedge Q) = \varphi(P) + \varphi(Q)$ for every trace (the two halves of this identity fail as $\delta/(1-\delta)$ and δ, Proposition 8.2), yet the lattice subadditivity fails as $\delta/(1-\delta)$, while $\varphi(|P-Q|) \leq \varphi(P+Q)$, with the same right-hand side, fails only as $\kappa - 1 \sim \delta^2/2$. The orders refer to the normalisations $\mathcal{N}$ of Table 1; a normalisation that vanishes at the extremisers can change them (Remark 9.3).

The third result returns to the unnormalised quantity of [1], in which both operators are normalised to norm one and both contrasts are prescribed. Let $\nu = \sum_c \lfloor k_c/2 \rfloor$ be the largest number of mutually orthogonal two-dimensional subspaces that fit into the blocks of $\mathcal{M}$.

Theorem C. *Let* $\mathcal{M} = \bigoplus_{c=1}^{C} M_{k_c}(\mathbb{K})$*,* $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$*, with* $\nu \geq 1$*. For* $X \geq 0$ *and* Z *in* $\mathcal{M}$ *with* $\|X\| = \|Z\| = 1$*,* $\Delta(X) = \delta < 1$ *and* $\Delta(|Z|) = \delta'$*,*

$$\sup\big(|\operatorname{Tr}(XZ)| - \operatorname{Tr}(X|Z|)\big) = \frac{\nu\,\delta^2\delta'^2}{(1+\delta)(1+\delta')},$$

and the supremum is attained (Theorem 6.4). The two other functionals of [1], $\operatorname{Tr}|XY| - \operatorname{Tr}(|X||Y|)$ *and* $\operatorname{Tr}(Y|X|Y) - \operatorname{Tr}|YXY|$*, have the sharp bounds of Proposition 6.6 and Theorem 6.9.*

On a finite-dimensional von Neumann algebra $\mathcal{M} = \bigoplus_{c=1}^{C} M_{k_c}$ a positive functional φ_W is a trace if and only if W is central, i.e. scalar on every block, however different the scalars are. The violation therefore cannot depend on the global contrast of W:

Theorem D. *Let $W = \bigoplus_c W_c > 0$ in $\mathcal{M}$. For every inequality of Table 1 except 11, 16 and 24, the largest normalised violation for φ_W on $\mathcal{M}$ is at most the largest one for φ_{W_c} on the blocks M_{k_c}, with equality for items 1–10, 12–14, 17, 18, 21–23 and 25. A density that varies only between the blocks violates none of the inequalities except 11 and 24, which characterise the canonical trace (Theorem 10.1).*

5 The Kantorovich class

Theorem 5.1. *Let $W > 0$ in M_n, $n \geq 2$, and $\varphi = \varphi_W$. For all X, Y, A and all projections P, Q, R,*

$$|\varphi(X)| \leq \kappa\,\varphi(|X|), \qquad \varphi(A + A^*) \leq 2\kappa\,\varphi(|A^*|), \qquad |\varphi(A^2)| \leq \kappa\,\varphi(A^*A),$$
$$\varphi(|X+Y|) \leq \kappa\big(\varphi(|X|) + \varphi(|Y|)\big), \qquad \varphi(|P-R|) \leq \kappa\big(\varphi(|P-Q|) + \varphi(|Q-R|)\big).$$

All five constants are attained. Hence $V_n = \kappa - 1$ for items 1–5 in every dimension $n \geq 2$.

Proof. Let $X = U|X|$ be a polar decomposition and $T = |X|$. Then $\varphi(X) = \operatorname{Tr}(WUT) = \operatorname{Tr}\big((WT^{1/2})^* U T^{1/2}\big)$, so for every contraction U

$$|\varphi(X)|^2 \leq \operatorname{Tr}(W^2 T)\,\operatorname{Tr} T \leq \kappa^2 (\operatorname{Tr} WT)^2$$

by the Cauchy–Schwarz inequality and Lemma 3.1 for $\omega = \operatorname{Tr}(T\,\cdot\,)$. The second inequality follows in the same way from $\varphi(A + A^*) = 2\operatorname{Re}\varphi(|A^*|V)$, where $A = |A^*|V$ is a polar decomposition. For the fourth, write $X + Y = S|X+Y|$; then $\varphi(|X+Y|) = \varphi(S^*X) + \varphi(S^*Y)$, and each term is bounded as above with the contraction S^*U_X or S^*U_Y in place of U. The fifth is the fourth with $X = P - Q$, $Y = Q - R$. For the third, put $X = AW^{1/2}$ in an eigenbasis of W with eigenvalues w_i. Then $\varphi(A^*A) = \sum_{ij} |X_{ij}|^2$ and

$$\varphi(A^2) = \sum_i X_{ii}^2 + \sum_{i<j} X_{ij} X_{ji} \frac{w_i + w_j}{\sqrt{w_i w_j}}, \qquad \frac{w_i + w_j}{\sqrt{w_i w_j}} \leq 2\kappa,$$

so $|\varphi(A^2)| \leq \sum_i |X_{ii}|^2 + \kappa \sum_{i<j} (|X_{ij}|^2 + |X_{ji}|^2) \leq \kappa\varphi(A^*A)$.

Equality, with e_1, e_2 eigenvectors of W for a, b: in the first two inequalities for $T = vv^*$ with $|\langle e_1, v\rangle|^2 = \frac{b}{a+b}$, $|\langle e_2, v\rangle|^2 = \frac{a}{a+b}$ and $Uv = Wv/\|Wv\|$ (equality in Cauchy–Schwarz and in Lemma 3.1); in the third for $A = \sqrt{a}\, e_1 e_2^* + \sqrt{b}\, e_2 e_1^*$; in the fourth for $X = uu^*$, $Y = -u'u'^*$ with $u, u' = \sin t\, e_1 \pm \cos t\, e_2$ and $\tan t = \sqrt{b/a}$; in the fifth for $P = uu^*$, $Q = 0$, $R = u'u'^*$. □

Remark 5.2. For a fixed density, $\sup_X |\varphi(X)|/\varphi(|X|) = \kappa(W)$ exactly. In particular the normalised violation of Gardner's inequality depends on W only through its extreme eigenvalues, not on the dimension or on the rest of the spectrum.

6 Gardner's inequality with two contrasts

For Gardner's inequality both contrasts can be prescribed and the envelope found exactly. It depends on the two contrasts only through their product.

Theorem 6.1. *Let $W > 0$ and X be in M_n, $n \geq 2$, with $\Delta(W) = \delta$ and $\Delta(|X|) = \delta'$, $\delta, \delta' \in [0,1]$, $\delta\delta' < 1$. Then*

$$|\varphi_W(X)| \leq \frac{1}{\sqrt{1-\delta^2\delta'^2}}\,\varphi_W(|X|),$$

and for even n the constant is attained: $V_n(\delta,\delta') = \kappa(\delta\delta') - 1 = (1-\delta^2\delta'^2)^{-1/2} - 1$ for Gardner's inequality. For odd $n = 2k+1$,

$$V_n(\delta,\delta') = \max_{0\leq\gamma\leq 1} \frac{k\sqrt{S^2+\gamma pq} + \ell\ell'}{k(S+\gamma p) + \ell\ell'} - 1, \qquad S = \ell + \ell',\ q = (1+\ell)(1+\ell'),\ p = (1-\ell)(1-\ell'),$$

with $\ell = \frac{1-\delta}{1+\delta}$, $\ell' = \frac{1-\delta'}{1+\delta'}$; it is smaller than $\kappa(\delta\delta') - 1$ for $\delta' < 1$ and tends to it as $n \to \infty$. For $\delta' = 1$ both give Theorem 5.1. The upper bound $\kappa(\delta\delta')$ holds also for items 2 and 3, with the contrasts of $|X|, |Y|$ and of $|A^*|$.

Proof. Write $X = UT$ with $T = |X|$. The supremum of $|\operatorname{Tr}(WUT)| = |\operatorname{Tr}(U\,TW)|$ over unitaries U is $\|TW\|_1$, so we must bound $\|TW\|_1 / \operatorname{Tr}(TW)$. For fixed eigenbases, $\|TW\|_1 - c\operatorname{Tr}(TW)$ is a convex function of the eigenvalues of T (a norm of a linear function minus a linear one), and likewise of the eigenvalues of W. By Lemma 3.5, applied to T and then to W, we may assume $W = a(\ell + (1-\ell)P)$ and $T = t(\ell' + (1-\ell')Q)$ with projections P, Q and $\ell = \frac{1-\delta}{1+\delta}$, $\ell' = \frac{1-\delta'}{1+\delta'}$. By Lemma 3.2, $\mathbb{C}^n$ splits into subspaces of dimension 1 or 2 invariant under P and Q, hence under W and T, and $\|TW\|_1$ and $\operatorname{Tr}(TW)$ are sums over them. On a one-dimensional summand the two contributions are equal. On a two-dimensional one, W and T have the eigenvalues x_1, x_2 and y_1, y_2, namely $a, a\ell$ and $t, t\ell'$; with $S = x_1y_2 + x_2y_1$, $q = (x_1+x_2)(y_1+y_2)$, $p = (x_1-x_2)(y_1-y_2)$ and γ the squared overlap of the leading eigenvectors, the proof of Proposition 6.3 gives $\|TW\|_1^2 = S^2 + \gamma pq$ and $\operatorname{Tr}(TW) = S + \gamma p$. The function $\gamma \mapsto (S^2+\gamma pq)/(S+\gamma p)^2$ is maximal at $\gamma = S/q \in [0,1]$, with the value

$$\frac{q^2}{4S(S+p)} = \frac{(x_1+x_2)^2(y_1+y_2)^2}{4(x_1y_2+x_2y_1)(x_1y_1+x_2y_2)} = \frac{1}{1-\delta^2\delta'^2},$$

the last step by inserting $x_2/x_1 = \ell$, $y_2/y_1 = \ell'$. A quotient of sums is at most the largest quotient of the summands. Equality for even n: $n/2$ copies of the 2×2 configuration $W = \operatorname{diag}(1,\ell)$, T with the eigenvalues $1, \ell'$ at $\gamma = S/q$, and U the adjoint polar factor of TW. For odd n at least one summand is one-dimensional and contributes the same amount $\mu \geq \ell\ell'$ (normalising the largest eigenvalues of W and T to 1) to $\|TW\|_1$ and to $\operatorname{Tr}(TW)$; the quotient $(\sum_j s_j + \mu)/(\sum_j t_j + \mu)$, where $s_j \geq t_j$ are the contributions of at most k two-dimensional summands, is largest for k summands and $\mu = \ell\ell'$, and, by the argument of Lemma 3.5 applied to $s - ct$, for the same γ in every summand, which gives the stated maximum; it is attained by $W = \operatorname{diag}(1,\ell)^{\oplus k} \oplus \ell$ and T correspondingly. Items 2 and 3 reduce to the first inequality as in the proof of Theorem 5.1. □

Thus the normalised violation of Gardner's inequality is $\frac{1}{2}\delta^2\delta'^2 + O(\delta^4\delta'^4)$: it vanishes to second order in each contrast and is symmetric in the density and the operator. For a Gibbs state at inverse temperature β and an operator $|X| = e^{-\beta' K}$, the constant is $(1 - \tanh^2(\beta\Delta E/2)\tanh^2(\beta'\Delta E'/2))^{-1/2}$, where ΔE and $\Delta E'$ are the spectral spreads of H and K. The two-dimensional computation in the proof has a transparent form, which explains the dependence on the product.

Remark 6.2 (Geometric form). On $\mathbb{C}^2$ let $W = w(1+\delta S)$ and $T = t(1+\delta' S')$ with $w, t > 0$ and symmetries S, S' of trace zero. The unitary $S'S$ has determinant 1, hence eigenvalues $e^{\pm i\alpha}$, and $\operatorname{Tr}(SS') = \operatorname{Tr}(S'S) = 2\cos\alpha$. Then

$$\operatorname{Tr}(TW) = 2wt\,\operatorname{Re} z, \qquad \|TW\|_1 = 2wt\,|z|, \qquad z = 1 + \delta\delta' e^{i\alpha}.$$

The first identity follows from $\operatorname{Tr}(TW) = wt\operatorname{Tr}(1+\delta S+\delta' S'+\delta\delta' S'S)$. For the second, $\|M\|_1^2 = \operatorname{Tr}(M^*M)+2|\det M|$ for 2×2 matrices, and $\operatorname{Tr}(W^2T^2) = w^2t^2\operatorname{Tr}\big((1+\delta^2+2\delta S)(1+\delta'^2+2\delta' S')\big) = 2w^2t^2\big((1+\delta^2)(1+\delta'^2)+4\delta\delta'\cos\alpha\big)$ and $|\det TW| = w^2t^2(1-\delta^2)(1-\delta'^2)$ add up to $4w^2t^2|z|^2$. So Gardner's quotient $\sup_U|\operatorname{Tr}(WUT)|/\operatorname{Tr}(WT)$ equals $|z|/\operatorname{Re} z = \sec\arg z$. As the relative position of W and T varies, z runs over the circle of radius $\delta\delta'$ about 1, whose points have $|\arg z|\le\theta$, $\sin\theta=\delta\delta'$, with equality for $\cos\alpha=-\delta\delta'$ (Figure 1(c)). This gives the 2×2 case of Theorem 6.1, with the constant $\sec\theta = (1-\delta^2\delta'^2)^{-1/2}$. Since $\delta=\sin\psi(W)$ and $\delta'=\sin\psi(T)$, the angle θ is determined by $\sin\theta=\sin\psi(W)\sin\psi(T)$.

The unnormalised violation studied in [1] behaves in the same way (Theorem 6.4 below).

In [1, Sect. 5] the violation of Gardner's inequality was studied without normalisation: for $X\ge 0$ with $\|X\|=1$ (the density) and Z with $\|Z\|=1$, the quantity

$$z(X,Z) = |\operatorname{Tr}(XZ)| - \operatorname{Tr}(X|Z|)$$

was sampled against $\Delta(X)$ and $\Delta(|Z|)$. Writing $Z=U|Z|$ with a unitary (orthogonal, in the real case) U, $z=|\operatorname{Tr}(XUY)|-\operatorname{Tr}(XY)$ with $Y=|Z|\ge 0$, and $\sup_U z = \|YX\|_1-\operatorname{Tr}(XY)$ by the duality of the trace norm, the supremum being attained at the polar unitary of YX, which is real if X,Y are. We first solve the 2×2 case.

Proposition 6.3. *Let $X,Y\ge 0$ in $M_2(\mathbb{C})$ have eigenvalues $x_1\ge x_2\ge 0$ and $y_1\ge y_2\ge 0$, $x_1,y_1>0$. For every unitary U,*

$$|\operatorname{Tr}(XUY)| - \operatorname{Tr}(XY) \;\le\; \frac{(x_1-x_2)^2(y_1-y_2)^2}{4(x_1+x_2)(y_1+y_2)}.$$

Let $S=x_1y_2+x_2y_1$, $q=(x_1+x_2)(y_1+y_2)$, and $\gamma=|\langle e_1,f_1\rangle|^2$ for unit eigenvectors e_1 of X for x_1 and f_1 of Y for y_1. If $x_1>x_2$ and $y_1>y_2$, the bound is attained iff $|\operatorname{Tr}(XUY)|=\|YX\|_1$ and $\gamma=\frac14+\frac{S}{2q}$; it is attained by real X,Y,U.

Proof. Let $p=(x_1-x_2)(y_1-y_2)$, so that $q-2S=p$. First, $|\operatorname{Tr}(XUY)|=|\operatorname{Tr}(YXU)|\le\|YX\|_1$. Expanding in the eigenbases, $\operatorname{Tr}(XY)=\sum_{ij}x_iy_j|\langle e_i,f_j\rangle|^2 = S+\gamma p$, since the matrix $(|\langle e_i,f_j\rangle|^2)$ is doubly stochastic. The singular values of YX satisfy $s_1^2+s_2^2=\operatorname{Tr}(X^2Y^2)$ and $s_1s_2=|\det YX| = x_1x_2y_1y_2$, so $\|YX\|_1^2=\operatorname{Tr}(X^2Y^2)+2x_1x_2y_1y_2 = S^2+\gamma pq$. With $\sqrt{r}\le r/q+q/4$ (equality iff $\sqrt{r}=q/2$),

$$\|YX\|_1-\operatorname{Tr}(XY)\le\frac{S^2+\gamma pq}{q}+\frac{q}{4}-S-\gamma p=\frac{(q/2-S)^2}{q}=\frac{p^2}{4q},$$

with equality iff $\gamma pq=q^2/4-S^2=\frac{p}{2}(\frac{q}{2}+S)$. Every $\gamma\in[0,1]$ is realised by a real rotation, and the polar unitary of a real YX is real. □

Theorem 6.4. *Let $\mathcal{M}=\bigoplus_{c=1}^{C}M_{k_c}(\mathbb{K})$, $\mathbb{K}\in\{\mathbb{R},\mathbb{C}\}$, and $\nu=\sum_c\lfloor k_c/2\rfloor\ge 1$. For $X\ge 0$ and $Y\ge 0$ in $\mathcal{M}$ with $\|X\|=\|Y\|=1$, $\Delta(X)=\delta\in[0,1)$ and $\Delta(Y)=\delta'\in[0,1]$, and every unitary (orthogonal) $U\in\mathcal{M}$,*

$$|\operatorname{Tr}(XUY)|-\operatorname{Tr}(XY)\le\frac{\nu\,\delta^2\delta'^2}{(1+\delta)(1+\delta')},$$

and the bound is attained. Equivalently, $\sup\{|\operatorname{Tr}(XZ)|-\operatorname{Tr}(X|Z|) : \|X\|=\|Z\|=1,\ \Delta(X)=\delta,\ \Delta(|Z|)=\delta'\}$ has this value.

Proof. Let $\ell=(1-\delta)/(1+\delta)$ and $\ell'=(1-\delta')/(1+\delta')$ be the smallest eigenvalues. It suffices to prove the bound in the complex algebra, since the maximisers below are real. By the remark above, $\sup_U z = F(X,Y):=\|YX\|_1-\operatorname{Tr}(XY)$, and the polar unitary lies in $\mathcal{M}$. For a fixed eigenbasis of X, F is a convex function of the eigenvalues of X (a norm of a linear function of them minus a linear

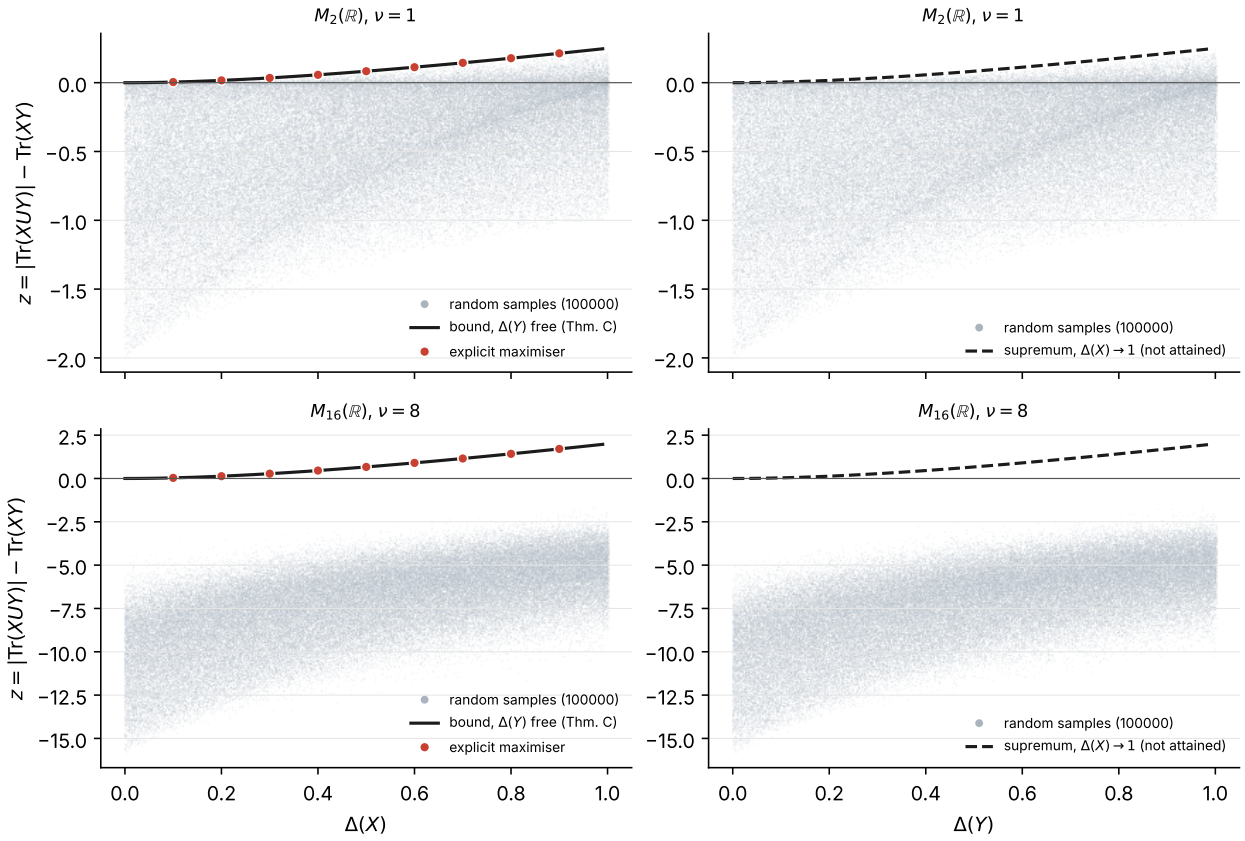


Figure 3: The unnormalised Gardner violation $z = |\operatorname{Tr}(XUY)| - \operatorname{Tr}(XY)$ for $\|X\| = \|Y\| = 1$ in $M_2(\mathbb{R})$ (top, $\nu = 1$) and $M_{16}(\mathbb{R})$ (bottom, $\nu = 8$), against $\Delta(X)$ (left) and $\Delta(Y)$ (right). Grey: 100000 random samples with Haar-orthogonal eigenbases and U and uniformly drawn contrasts, as in [1, Fig. 1]. Solid: the bound of Theorem 6.4 with $\Delta(Y)$ free, $\nu\delta^2/(2(1+\delta))$; dots: z at the explicit maximisers of its proof, which agree with it to 3×10^{-15} (printed by `trace_envelopes_figures.py`). Dashed: the supremum along $\Delta(Y)$, approached as $\Delta(X) \to 1$. In M_{16} no sample violates Gardner's inequality ($z > 0$), although violations up to almost 2 exist: the maximisers, two-level spectra on aligned pairs of eigenvectors, form a set of negligible Haar measure.

function). The admissible eigenvalues form the union of the faces $\{x_i = 1,\ x_j = \ell\}$ of the cube $[\ell, 1]^K$, $K = \sum_c k_c$, and a convex function on a face attains its maximum at a vertex; so $F(X, Y) \le F(X', Y)$ for some $X' = \ell + (1-\ell)P$ with a projection $P \in \mathcal{M}$, $P \neq 0, 1$, and likewise $F(X', Y) \le F(X', Y')$ with $Y' = \ell' + (1-\ell')Q$. By Lemma 3.2, applied in every block, X' and Y' are direct sums over subspaces of dimension 1 or 2, and so are $\|Y'X'\|_1$ and $\operatorname{Tr}(X'Y')$. One-dimensional summands contribute 0. On a two-dimensional one, $X' = \operatorname{diag}(1, \ell)$ and Y' has the eigenvalues $1, \ell'$, so by Proposition 6.3 it contributes at most $(1-\ell)^2(1-\ell')^2/(4(1+\ell)(1+\ell')) = \delta^2\delta'^2/((1+\delta)(1+\delta'))$. A block of size k_c holds at most $\lfloor k_c/2 \rfloor$ two-dimensional summands. Equality: ν blocks $\operatorname{diag}(1, \ell)$ and Y with spectrum $\{1, \ell'\}$ at the angle of Proposition 6.3, $X = Y = 1$ on the remaining one-dimensional blocks, and the polar unitary, which makes all block traces positive. □

Remark 6.5. With the contrast of $|Z|$ free, the supremum $\nu\delta^2/(2(1+\delta))$ is reached at $\delta' = 1$, i.e. for singular Z; the normalised counterpart is Theorem 5.1, $\sup_Z |\operatorname{Tr}(XZ)|/\operatorname{Tr}(X|Z|) = \kappa(X)$. The bound is additive over the blocks: it counts the mutually orthogonal two-dimensional subspaces that

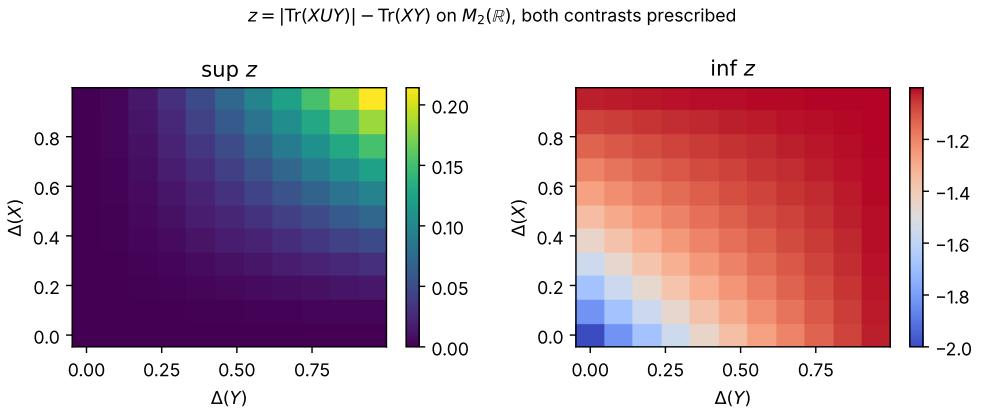


Figure 4: $M_2(\mathbb{R})$, both contrasts prescribed: the supremum (left) and infimum (right) of $z = |\operatorname{Tr}(XUY)| - \operatorname{Tr}(XY)$ found by constrained optimisation on a grid of $\Delta(Y)$ (horizontal) and $\Delta(X)$ (vertical). The left panel agrees with $\delta^2\delta'^2/((1+\delta)(1+\delta'))$ (Theorem 6.4) to 5×10^{-10} at every grid point: a violation needs both operators to be far from scalar.

fit into them and does not depend on the sizes of the blocks otherwise. Figures 3 and 4 compare it with random sampling, the method of [1], and with numerical optimisation. Sampling stays far below the bound as soon as the blocks are larger than 2×2, while the optimisation over both contrasts reproduces it.

6.1 The other two functionals of [1]

Besides Gardner's inequality, [1, Sect. 5] sampled two further unnormalised trace functionals,

$$z_2 = \operatorname{Tr}|XY| - \operatorname{Tr}(|X|\,|Y|), \qquad z_3 = \operatorname{Tr}(Y|X|Y) - \operatorname{Tr}|YXY| \quad (X = X^*,\ Y \geq 0),$$

with $|X|$ and $|Y|$ normalised to $\lambda_{\max} = 1$ and their contrasts as parameters. Both are determined here in $\mathcal{M} = \bigoplus_c M_{k_c}$, with $\nu = \sum_c \lfloor k_c/2 \rfloor$ and $\ell = \frac{1-\delta}{1+\delta}$, $\ell' = \frac{1-\delta'}{1+\delta'}$ as before; note $1 - \ell = \frac{2\delta}{1+\delta}$.

Proposition 6.6. *Let $X = UX_0$, $Y = VY_0$ with $X_0, Y_0 \geq 0$ and unitaries $U, V \in \mathcal{M}$, so that $|X| = X_0$, $|Y| = Y_0$ and $z_2 = \|X_0 V Y_0\|_1 - \operatorname{Tr}(X_0 Y_0)$. For given spectra,*

$$\sup z_2 = -\inf z_2 = \sum_c \sum_i x^{\downarrow}_{c,i}\,(y^{\downarrow}_{c,i} - y^{\uparrow}_{c,i}),$$

with $x^{\downarrow}_{c,i}$ the eigenvalues of $X_{0,c}$ in decreasing order and $y^{\downarrow}_{c,i}$, $y^{\uparrow}_{c,i}$ those of $Y_{0,c}$ in decreasing and increasing order. With $\Delta(|X|) = \delta$ and $\Delta(|Y|) = \delta'$ prescribed, $\sup z_2 = -\inf z_2 = \nu(1-\ell)(1-\ell') = \frac{4\nu\delta\delta'}{(1+\delta)(1+\delta')}$, attained by two-level spectra.

Proof. V and the eigenbases act sector by sector, and von Neumann's trace inequality $\|AB\|_1 \leq \sum_i s_i(A)s_i(B)$, its lower counterpart $\|AB\|_1 \geq \sum_i s_i(A)s_{k+1-i}(B)$ and the rearrangement inequality for $\operatorname{Tr}(X_0 Y_0)$ give the first formula; both extremes are attained by aligning or reversing the eigenbases. Pairing i with $k+1-i$, $\sum_i x^{\downarrow}_{c,i}(y^{\downarrow}_{c,i} - y^{\uparrow}_{c,i}) = \sum_{i \leq k/2}(x^{\downarrow}_{c,i} - x^{\downarrow}_{c,k+1-i})(y^{\downarrow}_{c,i} - y^{\downarrow}_{c,k+1-i}) \leq \lfloor k/2 \rfloor (1-\ell)(1-\ell')$, with equality for two-level spectra. □

So z_2 is linear in each contrast, unlike Gardner's violation, which is quadratic in each. The third functional is never negative:

Proposition 6.7. *For $X = X^*$ and $Y \geq 0$, $\operatorname{Tr}|YXY| \leq \operatorname{Tr}(Y|X|Y)$, with equality if and only if $YX_+Y^2X_-Y = 0$, in particular if Y commutes with X.*

Proof. With $X = X_+ - X_-$ and $|X| = X_+ + X_-$, the triangle inequality for the trace norm gives $\|YX_+Y - YX_-Y\|_1 \leq \operatorname{Tr}(YX_+Y) + \operatorname{Tr}(YX_-Y)$. For positive A, B, $\|A - B\|_1 = \operatorname{Tr} A + \operatorname{Tr} B$ if and only if $AB = 0$: with P the projection onto the positive part of $A - B$, $\|A - B\|_1 = \operatorname{Tr} A + \operatorname{Tr} B - 2\operatorname{Tr}((1-P)A) - 2\operatorname{Tr}(PB)$, and equality forces $(1-P)A = 0 = PB$, so $AB = AP(1-P)B = 0$; conversely, if $AB = 0$ the supports are orthogonal and $|A - B| = A + B$. Apply this to $A = YX_+Y$, $B = YX_-Y$. If $XY = YX$, then $YX_+Y^2X_-Y = Y^4X_+X_- = 0$. □

Proposition 6.8. *Let $X = X^*$ in $M_2(\mathbb{C})$, let $|X|$ have eigenvalues $x_1 \geq x_2 \geq 0$, $x_1 > 0$, and let $Y \geq 0$ have eigenvalues $y_1 \geq y_2 \geq 0$. Then*

$$\operatorname{Tr}(Y|X|Y) - \operatorname{Tr}|YXY| \leq \frac{2x_1x_2}{x_1 + x_2}\,(y_1 - y_2)^2.$$

With $S = x_1y_2 + x_2y_1$, $q = (x_1 + x_2)(y_1 + y_2)$ and $\gamma = |\langle e_1, f_1\rangle|^2$ for unit eigenvectors e_1 of X for its eigenvalue of modulus x_1 and f_1 of Y for y_1: if $x_2 > 0$ and $y_1 > y_2$, the bound is attained if and only if X is indefinite and $\gamma = S/q$; it is attained by real matrices.

Proof. z_3 is invariant under $X \mapsto -X$ and vanishes for semidefinite X. Otherwise let $Xe_1 = x_1e_1$, $Xe_2 = -x_2e_2$, $u = \langle e_1, Y^2e_1\rangle = \gamma y_1^2 + (1-\gamma)y_2^2$ and $s = y_1^2 + y_2^2$. Then $\operatorname{Tr}(Y|X|Y) = \operatorname{Tr}(|X|Y^2) = A := (x_1 - x_2)u + x_2s$, and since $\det(YXY) = -x_1x_2y_1^2y_2^2 \leq 0$, $\operatorname{Tr}|YXY| = \sqrt{\operatorname{Tr}(YXY)^2 - 4\det(YXY)} = \sqrt{L^2 + K^2}$ with $L = \operatorname{Tr}(XY^2) = (x_1 + x_2)u - x_2s$ and $K = 2\sqrt{x_1x_2}\,y_1y_2$. For the unit vector $(\alpha, \beta) = (x_1 - x_2,\, 2\sqrt{x_1x_2})/(x_1 + x_2)$ the Cauchy–Schwarz inequality gives $\sqrt{L^2 + K^2} \geq \alpha L + \beta K$, and $A - \alpha L - \beta K = \frac{2x_1x_2}{x_1+x_2}(y_1 - y_2)^2$ for every u; equality holds iff $\beta L = \alpha K$, i.e. $u - y_2^2 = (y_1 - y_2)S/(x_1 + x_2)$, i.e. $\gamma = S/q$. Real maximisers: diagonal X and Y rotated by $\arccos\sqrt{\gamma}$. □

Theorem 6.9. *Let $\mathcal{M} = \bigoplus_{c=1}^{C} M_{k_c}(\mathbb{K})$, $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, with $\nu \geq 1$, and let $|X|$ and Y be normalised to $\lambda_{\max} = 1$, with $\Delta(|X|) = \delta \in [0, 1)$ and $\Delta(Y) = \delta' \in [0, 1]$. Then*

$$\sup\bigl(\operatorname{Tr}(Y|X|Y) - \operatorname{Tr}|YXY|\bigr) = \begin{cases} (\nu - \delta)\bigl(\frac{2\delta'}{1+\delta'}\bigr)^2 & \text{if every } k_c \text{ is even,} \\ \nu\bigl(\frac{2\delta'}{1+\delta'}\bigr)^2 & \text{if some } k_c \text{ is odd,} \end{cases}$$

and the supremum is attained by real direct sums of the 2×2 maximisers of Proposition 6.8 and 1×1 blocks.

For odd blocks the eigenvalue ℓ of $|X|$ can sit on a one-dimensional summand, where it costs nothing, so the supremum does not depend on δ; for $k = 3$, $C = 1$: $(2\delta'/(1 + \delta'))^2 > (1 - \delta)(2\delta'/(1 + \delta'))^2$. The proof reduces to two-level spectra and, with Lemma 3.2, to 2×2 blocks, except in one case: the eigenvector of $|X|$ for ℓ may be spread over all two-dimensional blocks of its sector, so that X is not block diagonal. This case is settled by the following sharp bound, whose proof linearises the problem.

Proposition 6.10. *Let $X = X_+ \oplus (-1)$ on $\mathbb{C}^m \oplus \mathbb{C}^m$, where $0 < X_+ \leq 1$ has eigenvalues $x_1, \ldots, x_m$; let $c_\theta = \operatorname{diag}(\cos\theta_j)$, $s_\theta = \operatorname{diag}(\sin\theta_j)$, Q the projection onto the range of $\binom{c_\theta}{s_\theta} : \mathbb{C}^m \to \mathbb{C}^{2m}$, $b \in [0, 1)$ and $Y = b + (1 - b)Q$ (a two-level operator). Then*

$$\operatorname{Tr}(Y|X|Y) - \operatorname{Tr}|YXY| \;\leq\; (1 - b)^2 \sum_{i=1}^{m} \frac{2x_i}{1 + x_i},$$

with equality for $X_+ = \operatorname{diag}(x_1, \ldots, x_m)$ and $\cos^2\theta_i = (1 + bx_i)/((1 + x_i)(1 + b))$.

Proof. Let $\gamma = 1-b^2$, $K = 1+X_+$, $A = c_\theta^2$ (so $s_\theta^2 = 1-A$). In the decomposition $\mathbb{C}^m \oplus \mathbb{C}^m$, $Y^2 = b^2 + \gamma Q$ has the blocks $b^2+\gamma A$, $\gamma c_\theta s_\theta$, $\gamma c_\theta s_\theta$, $b^2+\gamma s_\theta^2$, all diagonal. For $\lambda > 0$ the lower right block of $\lambda - XY^2$ is the invertible diagonal $E = \lambda+1-\gamma A$, and as $(b^2+\gamma A)E - \gamma^2 A s_\theta^2 = \lambda(b^2+\gamma A)+b^2$, the Schur complement formula gives

$$\det(\lambda - XY^2) = \det\big(\lambda E - X_+(\lambda(b^2+\gamma A)+b^2)\big) = \det\big((\lambda+1)(\lambda - b^2X_+) - \lambda\gamma K A\big) = p(\lambda),$$

$p(\lambda) = \det\big((\lambda+1)(\lambda-b^2X_+) - \lambda\gamma K^{1/2} A K^{1/2}\big)$. The Schur complement of the lower right block λ gives $\det(\lambda - \Xi) = p(\lambda)$ as well, for the self-adjoint matrix

$$\Xi = \begin{pmatrix} \gamma K^{1/2} A K^{1/2} + b^2 X_+ - 1 & b X_+^{1/2} \\ b X_+^{1/2} & 0 \end{pmatrix},$$

which is affine in A. Since $\det(\lambda - YXY) = \det(\lambda - XY^2)$, the polynomials agree and YXY and Ξ have the same eigenvalues; hence $\operatorname{Tr}|YXY| = \|\Xi\|_1 \geq \operatorname{Tr}(T\Xi)$ for every self-adjoint T with $\|T\| \leq 1$. Take

$$T = \begin{pmatrix} -\Lambda & G \\ G & \Lambda \end{pmatrix}, \qquad \Lambda = \frac{1-X_+}{1+X_+}, \quad G = \frac{2X_+^{1/2}}{1+X_+},$$

so that $T^2 = 1$ (Λ and G commute and $\Lambda^2 + G^2 = 1$). With $\operatorname{Tr}(Y|X|Y) = \operatorname{Tr}(|X|Y^2) = b^2 \operatorname{Tr} X_+ + \gamma \operatorname{Tr}(X_+ A) + mb^2 + \gamma \operatorname{Tr}(1-A)$ and $K^{1/2}\Lambda K^{1/2} = 1 - X_+$, the terms containing A cancel in $\operatorname{Tr}(|X|Y^2) - \operatorname{Tr}(T\Xi)$, and

$$\operatorname{Tr}(Y|X|Y) - \operatorname{Tr}|YXY| \leq \sum_{i=1}^m \Big(b^2 x_i + 1 + \frac{1-x_i}{1+x_i}(b^2x_i - 1) - \frac{4bx_i}{1+x_i}\Big) = (1-b)^2 \sum_{i=1}^m \frac{2x_i}{1+x_i}.$$

For diagonal X_+ both operators are direct sums of 2×2 blocks, on which $X = \operatorname{diag}(x_i, -1)$, and Proposition 6.8 gives the value $(1-b)^2\, 2x_i/(1+x_i)$ at the stated angle (there $\gamma = \sin^2\theta_i$). □

Lemma 6.11. *Let $X = X^*$ be invertible, P the projection onto the positive part of X, $S = 2P-1$, and $Y \geq 0$. Then (a) if $|X| \leq |X'|$, $X' = S|X'|$ and $|X'|$ commutes with P, then $z_3(X,Y) \leq z_3(X',Y)$; (b) for fixed X, z_3 is a convex function of Y^2.*

Proof. With $A = YX_+Y$ and $B = YX_-Y$, $z_3 = \operatorname{Tr} A + \operatorname{Tr} B - \|A-B\|_1$. (a) $A = Y|X|^{1/2}P|X|^{1/2}Y$ is non-decreasing in $|X|$ among operators commuting with P, and so is B; by the triangle inequality $\|A + D - B\|_1 \leq \|A-B\|_1 + \operatorname{Tr} D$ for $D \geq 0$, so z_3 does not decrease when A or B increases. (b) As $\|H\|_1 = \max_{-1\leq T\leq 1} \operatorname{Tr}(TH)$ for self-adjoint H, with $T = 1-2E$ we get $z_3 = 2\min_{0\leq E\leq 1} \operatorname{Tr}\big(EA + (1-E)B\big)$. For invertible Y, $Z = |X|^{1/2}Y$ is invertible and $E \mapsto F = ZEZ^*$ maps $\{0 \leq E \leq 1\}$ onto $\{0 \leq F \leq G\}$, $G = ZZ^* = |X|^{1/2}Y^2|X|^{1/2}$, with $\operatorname{Tr}(EA) = \operatorname{Tr}(PF)$ and $\operatorname{Tr}((1-E)B) = \operatorname{Tr}((1-P)(G-F))$. So $z_3 = 2\min\{\operatorname{Tr}(PF + (1-P)(G-F)) : 0 \leq F \leq G\}$ is the partial minimum of a linear function over a convex set of pairs (F,G), hence convex in G, which is affine in Y^2; by continuity also at singular Y. □

Proof of Theorem 6.9. Write $a = \ell$, $b = \ell'$; it suffices to treat the complex algebra, since the maximisers are real. Let $\delta < 1$, so that X is invertible; $z_3(-X,Y) = z_3(X,Y)$. Since $|X|$ commutes with P, it has a unit eigenvector e for its eigenvalue a in one sector and in $\operatorname{ran} P$ or in $\ker P$; replacing X by $-X$ if necessary, $e \in \operatorname{ran} P$. Then $|X| \leq |X'| := 1 - (1-a)ee^*$, and $X' = S|X'| = S - (1-a)ee^*$ has contrast δ; by Lemma 6.11(a), $z_3(X,Y) \leq z_3(X',Y)$. By Lemma 6.11(b), a convex function of the eigenvalues of Y^2 on the faces $\{y_i = 1,\ y_j = b^2\}$ of $[b^2,1]^K$ is maximal at a vertex, so $z_3(X',Y) \leq z_3(X',Y')$ with $Y' = b + (1-b)Q$, $Q \neq 0, 1$ a projection in $\mathcal{M}$. Apply Lemma 3.2

to P and Q in every sector. In a sector without e, $X' = S$ is a direct sum over the subspaces, a one-dimensional summand contributes 0, and a two-dimensional one at most $(1-b)^2$ by Proposition 6.8 ($|X| = 1$ there): together at most $\lfloor k_c/2 \rfloor (1-b)^2$. In the sector of e, Lemma 6.11(a) with $|X'| \le 1$ gives the same bound for z_3 restricted to it, and, if its size k_c is odd or it contains fewer than $k_c/2$ two-dimensional subspaces, at most $\lfloor k_c/2 \rfloor (1-b)^2$, resp. $(k_c/2-1)(1-b)^2$. Otherwise its $m = k_c/2$ two-dimensional subspaces fill it, with bases f_j, f'_j, $f_j \in \operatorname{ran} P$; the unitary that multiplies f'_j by $e^{-i\chi_j}$ commutes with X' (which is -1 on $\ker P$) and removes the phases, and in the basis $(f_1, \dots, f_m, f'_1, \dots, f'_m)$ we are in the setting of Proposition 6.10 with $X_+ = 1 - (1-a)ee^*$, whose eigenvalues are $a, 1, \dots, 1$: the sector contributes at most $(1-b)^2\big(\frac{2a}{1+a} + m - 1\big) = (k_c/2 - \delta)(1-b)^2$. Summing over the sectors, $z_3 \le (\nu - \delta)(1-b)^2$ if every k_c is even and $z_3 \le \nu(1-b)^2$ otherwise. Equality: if every k_c is even, $X = \operatorname{diag}(a, -1) \oplus \operatorname{diag}(1, -1)^{\oplus(\nu-1)}$ and Y with eigenvalues $1, b$ in every block at the angle of Proposition 6.8; if some k_c is odd, $X = \operatorname{diag}(1, -1)$ on ν blocks, the eigenvalue a on a one-dimensional block of an odd sector, 1 on the other one-dimensional blocks, and Y as before ($Y = 1$ on the one-dimensional blocks). □

Before this proof was found, the cases $k = 4$ and $k = 6$ ($C = 1$) had been verified by a rigorous branch-and-bound computation over the angles of the reduced model (with gaps 10^{-7} and 10^{-5} that include floating-point rounding), and numerical optimisation reproduces all suprema of z_2 and z_3. Theorems 6.4 and 6.9 and Proposition 6.6 thus settle the three functionals of [1, Sect. 5] with both contrasts prescribed. Like Gardner's violation, they are additive over the blocks: they count the mutually orthogonal two-dimensional subspaces that fit into the sectors.

7 Young, Bikchentaev–Abed and the canonical trace

Proposition 7.1. *$\varphi_W(|AB^*|) \le h \cdot \frac{1}{2}\big(\varphi_W(|A|^2) + \varphi_W(|B|^2)\big)$ for all A, B, with equality for $A = B = e_1 e_2^*$. Hence $V_n = h - 1 = 2\delta/(1-\delta)$ for item 8 in every dimension.*

Proof. $\varphi_W(|AB^*|) \le a\|AB^*\|_1 \le a\|A\|_2\|B\|_2$ and $\varphi_W(|A|^2) = \operatorname{Tr}(WA^*A) \ge b\|A\|_2^2$. For $A = B = e_1 e_2^*$, $|AB^*| = e_1 e_1^*$ and $|A|^2 = e_2 e_2^*$, so $\mathcal{L} = a$ and $\mathcal{R} = b$. □

Remark 7.2. For the published form with $A, B \ge 0$ the violation seems to stay at the order δ^2: an independent BFGS search over $A, B \ge 0$ on M_2 (30 random starts; a test of the repository) comes within 0.1% of $\kappa - 1$ and does not exceed it (0.1546 at $\delta = 0.5$ and 1.2938 at $\delta = 0.9$, against $\kappa - 1 = 0.1547$ and 1.2942). The linear failure of item 8 is a feature of the extension to arbitrary operators.

Proposition 7.3. *Let $\Delta(W) = \delta$ and let the moduli of A and B have contrast δ'. On M_2, for $W = \operatorname{diag}(a, b)$, $T = \operatorname{diag}(\ell', 1)$ with $\ell' = \frac{1-\delta'}{1+\delta'}$, U the swap of the two basis vectors and $A = B = UT$, the extended Young inequality has the relative violation $\frac{h+\ell'^2}{1+h\ell'^2} - 1$. Hence, for $n = 2$ and, by direct sums, for every even n,*

$$V_n(\delta, \delta') \;\ge\; \frac{h + \ell'^2}{1 + h\ell'^2} - 1 = \frac{4\delta\delta'}{1 + \delta'^2 - 2\delta\delta'}.$$

For $\delta' = 1$ this is the value of Proposition 7.1.

Proof. For $A = B$, $|AB^*| = |AA^*| = AA^* = UT^2U^* = \operatorname{diag}(1, \ell'^2)$ and $|A|^2 = T^2 = \operatorname{diag}(\ell'^2, 1)$, so $\mathcal{L} = a + b\ell'^2$ and $\mathcal{R} = a\ell'^2 + b$, and $\mathcal{L}/\mathcal{R} = (h + \ell'^2)/(h\ell'^2 + 1)$. Inserting $h = \frac{1+\delta}{1-\delta}$ and ℓ' gives the second form. □

The numerical search over all A, B with this contrast finds no larger value on M_2 (agreement to four digits, Figure 16), and we conjecture equality for even n. Unlike Gardner's inequality, the extended Young inequality fails *linearly* in each contrast, $V \sim 4\delta\delta'$.

Proposition 7.4. *For $n = 2$, item 9 has the envelope $V_2 = s/(1+s)$, $s = \sin(\psi/2)$. Thus $V_2 \sim \delta/2$ as $\delta \to 0$ and $V_2 \to \sqrt{2} - 1$ as $\delta \to 1$.*

Proof. If P or Q is 0 or 1, or $P = Q$, then $\mathcal{L} = \mathcal{R}$. Otherwise $P = uu^*$, $Q = vv^*$ with unit vectors; choose phases so that $u = cv + \sigma v_\perp$ with $c = \cos\theta$, $\sigma = \sin\theta$, $0 < \theta < \pi/2$, and a unit vector $v_\perp \perp v$. Normalise $\operatorname{Tr} W = 2$, so that W has the eigenvalues $1 \pm \delta$, and put $\beta = v^*Wv$, $\gamma = v_\perp^* W v_\perp = 2 - \beta$, $t = \operatorname{Re}(v^*Wv_\perp)$ and $\rho = |v^*Wv_\perp|$; then $\beta\gamma - \rho^2 = \det W = 1 - \delta^2$. Since $PQ + QP = c(uv^* + vu^*)$ and $QPQ = c^2 Q$,

$$\mathcal{L} - \mathcal{R} = 2c\operatorname{Re}(v^*Wu) - 2c^2\beta = 2c\sigma t, \qquad \mathcal{N} = u^*Wu + \beta = \beta(1+c^2) + \sigma^2\gamma + 2c\sigma t.$$

With $D = \beta(1+c^2) + \sigma^2\gamma > 0$ the quotient $2c\sigma t/(D + 2c\sigma t)$ increases with $t \le \rho$, so it is at most $r/(1+r)$ with

$$r = \frac{2c\sigma\rho}{D} = \frac{\rho \sin 2\theta}{\beta + 1 + (\beta - 1)\cos 2\theta} \le \frac{\rho}{\sqrt{(\beta+1)^2 - (\beta-1)^2}} = \frac{\rho}{2\sqrt{\beta}},$$

using $\gamma = 2 - \beta$ and $\max_x \sin x/(p + q\cos x) = (p^2 - q^2)^{-1/2}$ for $p > |q|$. Finally $r^2 \le \rho^2/(4\beta) = \frac{1}{4}\big(2 - \beta - (1-\delta^2)/\beta\big)$, which is maximal at $\beta = \sqrt{1-\delta^2} = \cos\psi$ with the value $\frac{1}{2}(1 - \cos\psi) = \sin^2(\psi/2)$. All steps are equalities for v with $v^*Wv = \cos\psi$, a phase of $v_\perp$ with $v^*Wv_\perp = \rho > 0$, and the optimal θ, and $r \mapsto r/(1+r)$ is increasing. □

Proposition 7.5. *Let $\operatorname{Tr} W = n$. For item 11, $V(W) = 1 - \mathrm{GM}/\mathrm{AM}$ of the eigenvalues of W, attained at $A = W^{-1}$. Over the densities with contrast δ the maximum is attained at a two-level spectrum:*

$$V_n(\delta) = 1 - \min_{1 \le m < n} \frac{n\,\ell^{(n-m)/n}}{m + (n-m)\ell}, \qquad \ell = \frac{1}{h} = \frac{1-\delta}{1+\delta}.$$

In particular $V_2 = 1 - \sqrt{1-\delta^2}$, and $V_n \uparrow 1 - 1/S(h)$ as $n \to \infty$, where $S(h) = \frac{(h-1)h^{1/(h-1)}}{e \log h}$ is the Specht ratio.

Proof. By the AM–GM inequality for the eigenvalues of $W^{1/2}AW^{1/2}$, $\varphi_W(A)/n \ge (\det W \det A)^{1/n}$, with equality for $A = W^{-1}$, so $\inf_{A>0} \varphi_W(A)/(n(\det A)^{1/n}) = (\det W)^{1/n}$ and $V(W) = 1 - (\det W)^{1/n}$, which for $\operatorname{Tr} W = n$ is $1 - \mathrm{GM}/\mathrm{AM}$. In the coordinates $x_i = \log w_i$, $\mathrm{AM}/\mathrm{GM} = \frac{1}{n}\sum_i e^{x_i - \bar{x}}$ is convex, so on the box of spectra in $[b, a]$ it is maximal at a vertex, a two-level spectrum with m eigenvalues a and $n - m$ eigenvalues b; this gives the formula. Its limit is Specht's theorem [48]: $\sup \mathrm{AM}/\mathrm{GM}$ over all finite families in $[b, a]$ is $S(h)$. □

Proposition 7.6. *Let $\operatorname{Tr} W = n$. For item 24, $V(W) = w_{\min}^{-1/2} - 1$, and*

$$V_n(\delta) = \sqrt{\frac{\ell + n - 1}{n\ell}} - 1, \qquad \ell = \frac{1-\delta}{1+\delta};$$

$V_2 = (1-\delta)^{-1/2} - 1 \sim \delta/2$, $V_n \sim \frac{n-1}{n}\delta$, and $V_n \uparrow \sqrt{h} - 1$ as $n \to \infty$.

Proof. For $A \ge 0$, $\varphi(A^2) = \operatorname{Tr}(A^{1/2}WA^{1/2}A) \le \|A\|\varphi(A) \le \operatorname{Tr}(A)\varphi(A) \le \varphi(A)^2/w_{\min}$, with equality for the projection onto an eigenvector for $w_{\min}$. With $\operatorname{Tr} W = n$ and the extreme eigenvalues in the ratio h, $w_{\min} = nb/\sum_i w_i \ge nb/(b + (n-1)a) = n\ell/(\ell + n - 1)$, with equality for the spectrum $\{a, \dots, a, b\}$. For $n = 2$, $(\ell+1)/(2\ell) = 1/(1-\delta)$. □

Remark 7.7. Items 11 and 24 characterise the canonical trace among the functionals with $\varphi(1) = n$, so they are violated also by central densities on a direct sum (Section 10), and their envelopes grow with n, unlike all others with a closed form.

8 Inequalities for two projections

Theorem 8.1. *Let $W > 0$ in M_n, $n \geq 2$, with $\Delta(W) = \delta$, and let P, Q be projections. With $\mathcal{N}$ as in Table 1:*

(17) $V_n = \kappa^2 - 1$ *for* $\varphi(PQP) \leq \varphi(Q)$;

(18) $V_n = \kappa - 1$ *for* $\varphi(|P - Q|) \leq \varphi(P + Q)$;

(19) $V_n = \delta$ *for* $\varphi(P + Q) \leq \varphi(1 + P \wedge Q)$ *if n is even, and* $V_n = (n-1)\delta/(n-\delta)$ *if n is odd;*

(20) for $\varphi(e^{QPQ}) \leq \varphi(e^P)$ *and even n,* $V_n = \max_{0<g<\pi} \lambda(\delta, g)$, *where $\lambda(\delta, g)$ is the larger root of the quadratic* (2) *below, and* $V_n = \frac{(e-1)^2}{e(e+1)}\delta^2 + O(\delta^3)$;

(21) $V_n = \frac{1}{2}\tan^2(\psi/2)$ *for* $\varphi(\operatorname{Re} PQ) \geq 0$;

(22) $V_n = \kappa - 1$ *for* $\varphi(\operatorname{Re} PQ) \leq \varphi(|PQ|)$;

(23) $V_n = \delta/(1-\delta)$ *for* $\varphi(P \vee Q) \leq \varphi(P) + \varphi(Q)$.

The values are attained in 17, 18, 20–22 by rank-one P, Q, and approached in 19 and 23 as $P \neq Q$ tend to the same eigenvector of W. For odd n, V_n in item 20 lies between the M_2 value and the same maximum with $(1-\delta)/(n-1)$ added to β in (2).

Proof. Step 1: reduction to rank-one projections on M_2. By Lemma 3.2, $\mathbb{C}^n = \bigoplus_j H_j$ with $\dim H_j \in \{1, 2\}$, every H_j invariant under P and Q. Every operator in $\mathcal{L}$, $\mathcal{R}$ and $\mathcal{N}$ (namely PQP, Q, $|P-Q|$, $P+Q$, $1+P\wedge Q$, e^{QPQ}, e^P, $\operatorname{Re} PQ$, $|PQ|$, $P \vee Q$) is then block diagonal, with the block obtained from P_j, Q_j; for the lattice operations because $\operatorname{ran} P \cap \operatorname{ran} Q = \bigoplus_j (\operatorname{ran} P_j \cap \operatorname{ran} Q_j)$, the projections onto the H_j commuting with P and Q. Hence $\mathcal{L} = \sum_j L_j$, $\mathcal{R} = \sum_j R_j$ and $\mathcal{N} = \sum_j N_j$, where $\mathcal{L}_j$ is the same expression for P_j, Q_j and the functional $\operatorname{Tr}(W_j \cdot)$ on H_j, with the compression W_j of W to H_j; its spectrum lies in $[b, a]$, so $\Delta(W_j) \leq \delta$. On a one-dimensional H_j, $P_j, Q_j \in \{0, 1\}$ commute and $\mathcal{L}_j \leq \mathcal{R}_j$ in all seven items (e.g. $\max(p, q) \leq p + q$, $p + q \leq 1 + pq$, $e^{pq} \leq e^p$). On a two-dimensional H_j, $P_j \neq Q_j$ are rank-one projections. Since $\mathcal{N}_j \geq 0$, and $\mathcal{L}_j \leq \mathcal{R}_j$ wherever $\mathcal{N}_j = 0$,

$$\frac{\mathcal{L} - \mathcal{R}}{\mathcal{N}} \leq \max_j \frac{\mathcal{L}_j - \mathcal{R}_j}{\mathcal{N}_j},$$

and the quotients do not change when W_j is scaled. It therefore suffices to bound $(\mathcal{L} - \mathcal{R})/\mathcal{N}$ for $W = \frac{1}{2}(1 + \tau\sigma_3)$, $0 \leq \tau \leq \delta$, and $P = \frac{1}{2}(1 + \mu\cdot\sigma)$, $Q = \frac{1}{2}(1 + \nu\cdot\sigma)$ with unit Bloch vectors $\mu \neq \nu$ in $\mathbb{R}^3$ ($\sigma = (\sigma_1, \sigma_2, \sigma_3)$ the Pauli matrices). For fixed μ, ν the quotient is a linear-fractional function of τ (all three quantities are affine in W) with positive denominator, hence monotone on $[-\delta, \delta]$; conjugation by σ_1 maps τ to $-\tau$ and (μ, ν) to reflected vectors. So the bound for $\tau = \delta$ covers every $\tau \leq \delta$, and the M_2 envelope bounds the envelope in every dimension. In 17, 18 and 21–23, $\mathcal{N}$ vanishes on blocks where $P = Q = 0$, so the values below, attained (or approached) on M_2, are attained (approached) on M_n with P, Q supported on the span of e_1, e_2. In 19 and 20, $\mathcal{N}$ contains $\varphi(1)$, which we treat at the end.

Step 2: formulas on M_2. Write $\varphi(x_0 + x\cdot\sigma) = x_0 + \delta x_3$, $\mu\cdot\nu = \cos 2\theta$ $(0 < \theta \leq \pi/2)$ and $c = \cos\theta = |\langle u, v\rangle|$ for unit vectors u, v with $P = uu^*$, $Q = vv^*$. Then $PQP = c^2 P$, $(P - Q)^2 =$

$\sin^2\theta\cdot 1$, $P\wedge Q=0$, $P\vee Q=1$, $\operatorname{Re}PQ=\frac14(1+\mu\cdot\nu+(\mu+\nu)\cdot\sigma)$, $|PQ|=(QPQ)^{1/2}=cQ$, $e^{QPQ}=1+(e^{c^2}-1)Q$, $e^P=1+(e-1)P$, and $|\mu+\nu|=2\cos\theta$. We use the following elementary fact: *for $\beta>|z|$, $\max_{|\xi|=1}(\alpha+y\cdot\xi)/(\beta+z\cdot\xi)$ is the unique zero λ of $\alpha-\lambda\beta+|y-\lambda z|$*, since $(\alpha+y\cdot\xi)-\lambda(\beta+z\cdot\xi)\le\alpha-\lambda\beta+|y-\lambda z|$ with equality for $\xi\parallel y-\lambda z$, and the right-hand side decreases strictly in λ.

(17) $\mathcal{L}/\mathcal{R}=(1+\mu\cdot\nu)(1+\delta\mu_3)/(2(1+\delta\nu_3))$. For fixed μ, the fact with $\alpha=\beta=1$, $y=\mu$, $z=\delta e_3$ gives $\max_\nu(1+\mu\cdot\nu)/(1+\delta\nu_3)=2(1-\delta\mu_3)/(1-\delta^2)$, so $\mathcal{L}/\mathcal{R}\le(1-\delta^2\mu_3^2)/(1-\delta^2)\le\kappa^2$, with equality for $\mu_3=0$ and $\nu=(\mu-\lambda\delta e_3)/(\lambda-1)$, $\lambda=2/(1-\delta^2)$.

(18) $\mathcal{L}/\mathcal{R}=\sin\theta/(1+\frac{\delta}{2}(\mu_3+\nu_3))\le\sin\theta/(1-\delta\cos\theta)\le\kappa$, the maximum being at $\cos\theta=\delta$, $\mu+\nu\parallel -e_3$. (Alternatively, $\varphi(|P-Q|)\le\kappa\varphi(P+Q)$ is the triangle inequality of Theorem 5.1 with $X=P$, $Y=-Q$.)

(19) $(\mathcal{L}-\mathcal{R})/\mathcal{N}=\varphi(P+Q)-1=\frac{\delta}{2}(\mu_3+\nu_3)<\delta$, approached as $\mu,\nu\to e_3$.

(20) For fixed $g=2\theta$, the pair (μ_3,ν_3) ranges over the image of the unit sphere under $\xi\mapsto(\mu\cdot\xi,\nu\cdot\xi)$, a filled ellipse, and the linear-fractional quotient is maximal on its boundary, $\mu_3=\cos x$, $\nu_3=\cos(x+g)$. With $K=e-1$ and $E=e^{\cos^2(g/2)}-1$,

$$\frac{\mathcal{L}-\mathcal{R}}{\mathcal{N}}=\frac{\alpha+y\cdot\xi_x}{\beta+z\cdot\xi_x},\quad \alpha=\tfrac{E-K}{2},\ y=\tfrac{\delta}{2}(E\cos g-K,\,-E\sin g),\ \beta=1+\tfrac{K}{2},\ z=(\tfrac{\delta K}{2},0),$$

$\xi_x=(\cos x,\sin x)$. By the fact, its maximum $\lambda(\delta,g)$ is the larger root of

$$(\beta^2-|z|^2)\lambda^2-2(\alpha\beta-y\cdot z)\lambda+\alpha^2-|y|^2=0. \tag{2}$$

For small δ the maximiser has $g=2K\delta/e+O(\delta^2)$, $x=\pi/2$, and $\mathcal{L}-\mathcal{R}=-\frac{e}{8}g^2+\frac{K\delta}{2}g+O(\delta^3)$, which gives $V=K^2\delta^2/(e(e+1))+O(\delta^3)$.

(21) With $s=\mu_3+\nu_3\ge-2\cos\theta$, $(\mathcal{L}-\mathcal{R})/\mathcal{N}=-(2\cos^2\theta+\delta s)/(4(1+\frac{\delta}{2}s))$, which decreases in s; so $(\mathcal{L}-\mathcal{R})/\mathcal{N}\le(\delta C-C^2)/(2(1-\delta C))$, $C=\cos\theta$. The maximum over C is at $\delta C^2-2C+\delta=0$, $C=\tan(\psi/2)$, with the value $C^2/2=\frac12\tan^2(\psi/2)$.

(22) $\varphi(\operatorname{Re}PQ)=\operatorname{Re}\varphi(PQ)\le|\varphi(PQ)|\le\kappa\varphi(|PQ|)$ by Theorem 5.1, in every dimension. Equality: $v=\sqrt{b/(a+b)}\,e_1+\sqrt{a/(a+b)}\,e_2$, $u=Wv/\|Wv\|$; then $PQ=\langle u,v\rangle uv^*$ has the polar factor of the equality case of Theorem 5.1.

(23) $\mathcal{L}/\mathcal{R}=1/(1+\frac{\delta}{2}(\mu_3+\nu_3))<1/(1-\delta)$, approached as $\mu,\nu\to-e_3$.

Step 3: items 19 and 20 in dimension n. For even n take W with the eigenvalues a and b, each $n/2$ times, and $n/2$ copies of the M_2 extremiser, one in each block spanned by an eigenvector for a and one for b; this attains (approaches) the M_2 value, which is an upper bound by Step 1. For odd n at least one block of Lemma 3.2 is one-dimensional, so there are $k'\le k=(n-1)/2$ two-dimensional blocks. In item 19, $\mathcal{N}\ge\varphi(1)=\operatorname{Tr}W$; by Step 2 a two-dimensional block contributes $\mathcal{L}_j-\mathcal{R}_j=\operatorname{Tr}W_j\cdot\frac{\Delta(W_j)}{2}(\mu_3+\nu_3)<\lambda_1(W_j)-\lambda_2(W_j)$, and a one-dimensional one $\mathcal{L}_j-\mathcal{R}_j\le0$. By Ky Fan's maximum principle [5, Sect. III.1], $\sum_j\lambda_1(W_j)$ is at most the sum of the k' largest and $\sum_j\lambda_2(W_j)$ at least the sum of the k' smallest eigenvalues of W. The resulting bound for $(\mathcal{L}-\mathcal{R})/\mathcal{N}$ is a linear-fractional function of the spectrum of W and is maximal for $k'=k$ and the spectrum $a^{(k)},b^{(k+1)}$; hence $(\mathcal{L}-\mathcal{R})/\mathcal{N}<k(a-b)/(k(a+b)+b)=(n-1)\delta/(n-\delta)$, which is approached for this spectrum with $P=1$, $Q=0$ on the remaining eigenvector (a block with $\mathcal{L}_j=\mathcal{R}_j$). In item 20 the same configuration, with $P=Q=0$ on the remaining eigenvector, adds b/k per block to $\mathcal{N}$, i.e. $(1-\delta)/(n-1)$ to β (normalisation $\operatorname{Tr}W_j=1$); this gives the lower bound, and the M_2 value the upper bound. □

8.1 The modular law and measures on projections

For every trace the rank identity gives the modular law $\varphi(P \vee Q) + \varphi(P \wedge Q) = \varphi(P) + \varphi(Q)$. Its two halves are

$$(M_{\leq}) \quad \varphi(P \vee Q) + \varphi(P \wedge Q) \leq \varphi(P) + \varphi(Q), \qquad (M_{\geq}) \quad \varphi(P) + \varphi(Q) \leq \varphi(P \vee Q) + \varphi(P \wedge Q),$$

which we normalise, as in Table 1, by their right-hand sides. $M_{\leq}$ strengthens item 23 by the term $\varphi(P \wedge Q) \geq 0$, and $M_{\geq}$ differs from item 19 by $P \vee Q$ in place of 1.

Proposition 8.2. *Let $W > 0$ in M_n, $n \geq 2$, with $\Delta(W) = \delta$. In every dimension, $V_n = \delta/(1-\delta)$ for $M_{\leq}$ and $V_n = \delta$ for $M_{\geq}$, approached as $P \neq Q$ tend to the same eigenvector of W, for the smallest eigenvalue ($M_{\leq}$) or the largest ($M_{\geq}$), inside the span of the eigenvectors for the largest and the smallest eigenvalue. In particular each half of the modular law characterises the traces among the faithful positive functionals on M_n.*

Proof. Step 1 of the proof of Theorem 8.1 applies: on a one-dimensional block both sides agree ($\max(p,q)+\min(p,q) = p+q$) and the normalisations are non-negative, so V_n is at most the supremum over two-dimensional blocks with $\Delta(W_j) \leq \delta$, where $P_j \wedge Q_j = 0$ and $P_j \vee Q_j = 1$. There $M_{\leq}$ is item 23, whose value is less than $\delta/(1-\delta)$ (Step 2), and $M_{\geq}$ gives $(\varphi(P)+\varphi(Q)-\varphi(1))/\varphi(1) = \frac{\delta}{2}(\mu_3+\nu_3) < \delta$. Both values are approached on the block spanned by the eigenvectors of W for its largest and smallest eigenvalue, whose compression has contrast δ; the other eigenvectors form one-dimensional blocks with $P = Q = 0$, which add nothing to the violation or to the normalisation. □

Unlike item 19, $M_{\geq}$ shows no parity effect, because its normalisation does not contain $\varphi(1)$.

Measures on projections. Items 19 and 23 and the two halves of the modular law involve φ only through its values on projections. They therefore make sense for every *measure* on the projections of M_n, a function $\mu \geq 0$ with $\mu(P+Q) = \mu(P) + \mu(Q)$ whenever $PQ = 0$. By Gleason's theorem [26], for $n \geq 3$ every measure is $\mu(P) = \mathrm{Tr}(WP)$ with $W \geq 0$. On M_2 a measure is a frame function: $\mu(P) = f(\nu)$ for rank-one P with Bloch vector ν, where $f \geq 0$ and $f(\nu) + f(-\nu) = \mu(1)$, and f is otherwise arbitrary. For a measure $\mu \neq 0$ we call $\Delta(\mu) = (\sup \mu(P) - \inf \mu(P))/(\sup \mu(P) + \inf \mu(P))$, over the rank-one projections P, the contrast of μ. It equals $\Delta(W)$ for $\mu = \mathrm{Tr}(W\,\cdot)$, since $\sup_{\|e\|=1} \langle We, e\rangle = \lambda_{\max}(W)$, and on M_2, where $\mu(P) = f(\nu)$, we have $\sup f + \inf f = \mu(1)$.

Proposition 8.3. *Let μ be a measure on the projections of M_n, $n \geq 2$, with $\Delta(\mu) = \delta < 1$.*

(a) For $n \geq 3$ the envelopes of items 19 and 23 and of $M_{\leq}$ and $M_{\geq}$ over the measures of contrast δ are those over the states (Theorem 8.1, Proposition 8.2).

(b) For $n = 2$ they are, as for states, δ for item 19 and $M_{\geq}$ and $\delta/(1-\delta)$ for item 23 and $M_{\leq}$. They are attained by every measure that takes its largest value (19, $M_{\geq}$) or its smallest value (23, $M_{\leq}$) at two distinct rank-one projections. A state with $\delta > 0$ takes each extreme value at only one rank-one projection, so for states these values are only approached.

(c) Each of the four inequalities holds for all P, Q if and only if μ is a multiple of the dimension function, $\mu(P) = c \operatorname{rank} P$, i.e. if and only if $\delta = 0$.

Proof. (a) By Gleason's theorem $\mu = \mathrm{Tr}(W\,\cdot)$ with $\Delta(W) = \Delta(\mu)$. (b) Commuting P, Q satisfy all four inequalities (the modular law with equality, since $P \vee Q = (P - PQ) + (Q - PQ) + PQ$ is an orthogonal sum). For distinct rank-one P, Q we have $P \wedge Q = 0$ and $P \vee Q = 1$, and the normalised violations are $(f(P)+f(Q)-\mu(1))/\mu(1)$ for item 19 and $M_{\geq}$ and $(\mu(1)-f(P)-f(Q))/(f(P)+f(Q))$

for item 23 and $M_{\le}$. Since $\sup f = \frac{1}{2}\mu(1)(1+\delta)$ and $\inf f = \frac{1}{2}\mu(1)(1-\delta)$, they are at most δ and $\delta/(1-\delta)$, with equality if and only if $f(P) = f(Q) = \sup f$ or $= \inf f$, respectively. For a state, $f(\nu) = \frac{1}{2}\mu(1)(1+\delta\,\nu\cdot e)$ attains its maximum only at $\nu = e$ and its minimum only at $\nu = -e$. (c) If $\delta = 0$, f is constant ($n = 2$) or W is scalar ($n \ge 3$), so $\mu = c\,\mathrm{rank}$, and all four hold with equality. If $n \ge 3$ and $\delta > 0$, then $\mu = \mathrm{Tr}(W\,\cdot\,)$ with W not scalar; for rank-one $P \ne Q$ close to the eigenvector of W for its smallest (items 23, $M_{\le}$) or largest eigenvalue (items 19, $M_{\ge}$), inside the span of the eigenvectors for the largest and the smallest eigenvalue, the proofs of Theorem 8.1 and Proposition 8.2 give a positive violation for this W. If $n = 2$ and $\delta > 0$, there is a rank-one P with $f(P) < \frac{1}{2}\mu(1)$. Every antipodal pair $\pm\nu$ has a member with $f \le \frac{1}{2}\mu(1)$, so there is a rank-one $Q \notin \{P, 1-P\}$ with $f(Q) \le \frac{1}{2}\mu(1)$. Then $f(P) + f(Q) < \mu(1) = \mu(P \vee Q)$ violates item 23 and $M_{\le}$, and symmetrically with $\ge$ for item 19 and $M_{\ge}$. □

On $\mathcal{M} = \bigoplus_c M_{k_c}$ a measure is a sum $\mu = \sum_c \mu_c$ of measures on the projections of the blocks, so it is a positive linear functional (an unnormalised state) unless some k_c equals 2 (blocks of size 1 are trivial). For von Neumann algebras without a type I_2 summand positive measures extend to positive linear functionals by the theorems of Christensen and Yeadon [21, 53, 54], and Bunce and Wright [19] extended this to bounded and vector-valued measures. The non-linear measures thus live in the 2×2 blocks. The same blocks decide whether the projections of $\mathcal{M}$ carry a $\{0,1\}$-valued measure, a non-contextual assignment of values. Such a measure takes the value 1 on exactly one central projection 1_c, and on that block it exists if $k_c \le 2$ (choose one point of every antipodal pair of Bloch vectors) and not if $k_c \ge 3$, by the Kochen–Specker theorem [38], applied in a three-dimensional subspace containing a rank-one P with $\mu(P) = 1$ (alternatively by Gleason's theorem: $P \mapsto \mathrm{Tr}(WP)$ is continuous on the connected set of rank-one projections, so a $\{0,1\}$-valued one is constant, which contradicts additivity over an orthonormal basis). So $\mathcal{M}$ admits one if and only if some $k_c \le 2$. On the block where it lives such a measure has contrast 1, where the envelopes of item 23 and $M_{\le}$ are infinite. The lattice operations, measures with frame functions on the 2×2 blocks and the extremal measures of Proposition 8.3(b) are in the module `projections` of the library [41], and the test suite checks Propositions 8.2 and 8.3 (`tests/test_projections.py`).

9 Lower bounds

Proposition 9.1. *For every $n \ge 2$:*

(a) $V_n \ge \kappa - 1$ for item 6 (approached);

(b) $V_n \ge \kappa^2 - 1 = \delta^2/(1-\delta^2)$ for item 7 (approached);

(c) $V_n \ge g_{10}(\delta) := \max_{0<y<\delta} \dfrac{y^2(\delta-y)^2}{(1-2\delta y+y^2)(1-y^2)} = \dfrac{\delta^4}{16} + O(\delta^5)$ for item 10 (attained);

(d) $V_n \ge \delta^2/36 + O(\delta^3)$ for item 14, and more generally $(1-\frac{2}{p})^2\delta^2/4 + O(\delta^3)$ for Hölder's inequality $|\varphi(PQ)| \le \varphi(P)^{1/p}\varphi(Q)^{1/q}$.

By Lemma 3.3, $V_n \le h - 1$ in (a)–(c).

Proof. All constructions live on $\mathrm{span}\{e_1, e_2\}$ (extended by 0, or by a small multiple of the identity where positive definiteness is required); let $e_\pm = (e_1 \pm e_2)/\sqrt{2}$.

(a) For invertible A and a rank-one $B = vv^*$, $\|v\| = 1$, we have $A^{-1}\#vv^* = (v^*Av)^{-1/2}vv^*$, hence $(A^{-1}\#B)^{1/2} = (v^*Av)^{-1/4}vv^*$ and $A\natural B = (v^*Av)^{1/2}vv^*$. With $v = e_+$ and $A = xx^* + \varepsilon$,

$$\frac{\varphi(A\natural B)^2}{\varphi(A)\varphi(B)} = \frac{(v^*Av)(v^*Wv)}{\mathrm{Tr}(WA)} \xrightarrow[\varepsilon\to 0]{} \frac{|v^*x|^2\, v^*Wv}{x^*Wx},$$

which for $x = W^{-1}v$ equals $(v^*W^{-1}v)(v^*Wv) = \frac14(a^{-1}+b^{-1})(a+b) = \kappa^2$. By continuity the quotient $\mathcal{L}/\mathcal{R}$ approaches κ for positive definite A, $B = vv^* + \varepsilon'$.

(b) In the basis $f_1 = e_+$, $f_2 = e_-$ the density has the entries $W_{11} = W_{22} = \frac{a+b}{2}$, $W_{12} = \frac{a-b}{2}$. Take $B = \eta f_1 f_1^* + f_2 f_2^*$ and $A = uu^*$ with $u = f_1 + s\eta f_2$. Then $B^{1/2}AB^{1/2} = ww^*$ with $w = \sqrt{\eta} f_1 + s\eta f_2$, so $\mathcal{L} = \|w\|^2\, w^*Ww = \eta^2 W_{11} + O(\eta^{5/2})$, and $Bu = \eta(f_1 + sf_2)$, so $\mathcal{R} = \|u\|^2\eta^2(W_{11} + 2sW_{12} + s^2W_{22})$. As $\eta \to 0$, $\mathcal{L}/\mathcal{R} \to W_{11}/(W_{11} + 2sW_{12} + s^2W_{22})$, which at $s = -W_{12}/W_{22}$ equals $W_{11}W_{22}/(W_{11}W_{22} - W_{12}^2) = (a+b)^2/(4ab) = \kappa^2$.

(c) Normalise $a + b = 2$, so $W = \operatorname{diag}(1+\delta, 1-\delta)$, and take $A = uu^*$, $B = A + tvv^*$, $t \ge 0$, with $u = (\cos\alpha, \sin\alpha)$, $v = (-\cos\alpha, \sin\alpha)$ and $y = -\cos 2\alpha$. Then $u^*v = y$, $v^*Wu = y - \delta$ and $u^*Wu = v^*Wv = 1 - \delta y$, so

$$\mathcal{L} - \mathcal{R} = \varphi(A^2) - \varphi(B^2) = -2t\operatorname{Re}\big((u^*v)(v^*Wu)\big) - t^2 v^*Wv = 2ty(\delta - y) - t^2(1 - \delta y).$$

For $0 < y < \delta$ the best $t = y(\delta - y)/(1 - \delta y) > 0$ gives $\mathcal{L} - \mathcal{R} = G := y^2(\delta - y)^2/(1 - \delta y)$ and $(\mathcal{L} - \mathcal{R})/\mathcal{R} = G/(1 - \delta y - G) = y^2(\delta - y)^2/((1 - 2\delta y + y^2)(1 - y^2))$. Maximising over y gives $g_{10}(\delta)$; the leading order comes from $y = \delta/2$.

(d) Take $W = \frac12(1 + \delta D)$, $D = \operatorname{diag}(1, -1)$, and real unit vectors u, v at the angles $\tau \pm \varepsilon/2$ to e_1, $P = uu^*$, $Q = vv^*$. Then $\varphi(PQ) = \frac12(\cos^2\varepsilon + \delta\cos\varepsilon\cos 2\tau)$ and $\varphi(P)^{1/p}\varphi(Q)^{1/q} = \frac12\big(1 + \delta\cos 2\tau + \delta(\frac1q - \frac1p)\varepsilon\sin 2\tau\big) + O(\delta^2\varepsilon^2)$, so $\mathcal{L} - \mathcal{R} = \frac12\big(-\varepsilon^2 - \delta(1 - \frac2p)\varepsilon\sin 2\tau\big) + O(\delta^3)$ for $\varepsilon = O(\delta)$. The choice $\tau = \pi/4$, $\varepsilon = -\frac{\delta}{2}(1 - \frac2p)$ gives $\mathcal{L} - \mathcal{R} = \frac18(1 - \frac2p)^2\delta^2 + O(\delta^3)$ and $\mathcal{R} = \frac12 + O(\delta)$. □

Proposition 9.2. *For item 25 ($\varphi(A^4) \le \varphi(AB^2A)$ for $0 \le A \le B$), rank-one A gives no violation, and an explicit construction with A of rank two and $B - A$ of rank one gives*

$$V_n \ge g_{25}(\delta) = \frac{\delta^8}{4096}\big(1 + O(\delta^2)\big) \qquad \text{in every dimension } n \ge 2.$$

Proof. If $A = \alpha uu^*$ has rank one and $A \le B$, then $u^*Bu \ge \alpha$ and $AB^2A = \alpha^2\|Bu\|^2 uu^* \ge \alpha^2(u^*Bu)^2uu^* \ge A^4$, so every positive φ satisfies the inequality. For the construction take W with $\operatorname{Tr} W = 1$ and eigenvalues $(1 \pm \delta)/2$, and, in an orthonormal basis at the angle ϑ to its eigenbasis, $A = \operatorname{diag}(1, s)$, $0 < s < 1$, and $B = A + tvv^*$ with $v = (\cos\phi, \sin\phi)$; in this basis W has the entries $w_{11,22} = \frac12(1 \pm \delta\cos 2\vartheta)$ and $w_{12} = \frac{\delta}{2}\sin 2\vartheta$. Then $\mathcal{R} - \mathcal{L} = tT_1 + t^2T_2$ with $T_1 = \varphi(A^2P_vA + AP_vA^2)$, $T_2 = \varphi(AP_vA)$, $P_v = vv^*$, and whenever $T_1 < 0$ the best $t = -T_1/(2T_2)$ gives $(\mathcal{L} - \mathcal{R})/\mathcal{R} = G/(\mathcal{L} - G)$, $G = T_1^2/(4T_2)$. Let $g_{25}(\delta)$ be the maximum over (ϑ, s, ϕ); it is attained, and by embedding $\mathbb{C}^2 \subset \mathbb{C}^n$ (with A, B extended by 0) it is a lower bound in every dimension. For the leading order put $\vartheta = \pi/4$, $\sqrt{s} = \sigma\delta$ and $\tan\phi = -\varrho s^{-3/2}$. Then $T_1 = \cos^2\phi\,(1 + \varrho^2 - (1+s)\varrho/\sigma)$, $T_2 = \cos^2\phi\,(\frac12 + \frac{\varrho^2}{2s} - \frac{\varrho}{\sigma})$ and $\cos^2\phi = s^3/(s^3 + \varrho^2)$, so that

$$\frac{\mathcal{L} - \mathcal{R}}{\mathcal{R}} = \big(\sigma^2(\varsigma - \varsigma^2) - \sigma^4\big)^2\delta^8 + O(\delta^{10}), \qquad \varsigma = \sigma/\varrho,$$

which is maximal for $\varsigma = \frac12$, $\sigma^2 = \frac18$, with the value $\delta^8/4096$. □

The numerical maximisation gives $g_{25}(0.1)/(0.1^8/4096) = 1.020$. The same scaling for a general exponent $r \ge 0$ in $\varphi(A^{2r+2}) \le \varphi(A^rB^2A^r)$ gives the order $\delta^{4(r+1)}$; for $r = 0$ the supremum of the leading coefficient is $\frac{1}{16}$, that of item 10.

Remark 9.3. With $\mathcal{N} = \mathcal{R}$, item 13 would have $V = \infty$ for every $\delta > 0$: for rank-one $A = uu^*$, $B = vv^*$ with overlap $c = |u^*v| \to 0$, $\mathcal{L} \approx 2c\,|u^*Wu_\perp|$ while $\mathcal{R} = 2c^2u^*Wu$. This is why we normalise by $\varphi(A + B)$.

10 Direct sums: only the contrast inside the blocks matters

Let $\mathcal{M} = \bigoplus_{c=1}^{C} M_{k_c}$, with the trace $\mathrm{Tr} = \sum_c \mathrm{Tr}_c$ and $\varphi_W = \mathrm{Tr}(W\,\cdot\,)$, $W = \bigoplus_c W_c \geq 0$. The traces on $\mathcal{M}$ are the functionals $\sum_c t_c \mathrm{Tr}_c$, $t_c \geq 0$; so φ_W is a trace iff W is central, whatever its global contrast. The global contrast $\Delta(W)$ thus says nothing about traciality, and the following theorem shows that it says nothing about the size of the violation either.

Theorem 10.1. *Let $W = \bigoplus_c W_c > 0$ in $\mathcal{M}$, and let $V_{\mathcal{M}}(W)$ and $V_{M_{k_c}}(W_c)$ be the suprema of the normalised violation of one inequality of Table 1, other than 11, 16 and 24, for φ_W on $\mathcal{M}$ and for φ_{W_c} on M_{k_c}. Then*

$$V_{\mathcal{M}}(W) \leq \max_c V_{M_{k_c}}(W_c),$$

with equality for items 1–10, 12–14, 17, 18, 21–23 and 25. In particular, if every W_c is scalar, no inequality of Table 1 except 11 and 24 is violated, however different the scalars are; and the envelope on $\mathcal{M}$ along the intra-block contrast $\max_c \Delta(W_c) = \delta$ is at most the largest envelope $V_{k_c}(\delta)$ of the blocks.

Proof. All variables of the items are in $\mathcal{M}$, hence block diagonal, and every functional calculus acts blockwise. Let $\mathcal{L}_c$, $\mathcal{R}_c$, $\mathcal{N}_c$ be the expressions of the item for the block variables and φ_{W_c}. For all items except 11, 16 and 24, $\mathcal{L} \leq \sum_c L_c$ (with equality unless $\mathcal{L}$ contains an absolute value of a functional, as in 1, 4 and 14), $\mathcal{N} = \sum_c N_c$, and $\mathcal{R} = \sum_c R_c$, except in items 6 and 14, where $\mathcal{R} = \sqrt{\sum_c \varphi_c(A_c) \sum_c \varphi_c(B_c)} \geq \sum_c R_c$ by the Cauchy–Schwarz inequality and $\mathcal{R} \geq \sum_c R_c$ by Hölder's inequality. Since $W > 0$, $\mathcal{N}_c = 0$ forces the variables of block c to vanish where they enter $\mathcal{L}_c$, so $\mathcal{L}_c \leq \mathcal{R}_c$ there. Hence $(\mathcal{L} - \mathcal{R})/\mathcal{N} \leq \sum_c (\mathcal{L}_c - \mathcal{R}_c)/\sum_c N_c \leq \max_c (\mathcal{L}_c - \mathcal{R}_c)/\mathcal{N}_c$, the maximum over the blocks with $\mathcal{N}_c > 0$ (for $\mathcal{N} = \mathcal{R}$ in items 6 and 14 use $\mathcal{L}/\mathcal{R} \leq \sum_c L_c / \sum_c R_c$ instead). This proves the inequality. For the equality, in the listed items $\mathcal{N}$ is additive and the variables may vanish on all blocks but one without making $\mathcal{N}$ vanish; restricting a near-optimal configuration of one block and setting the variables to zero on the other blocks reproduces its quotient (in 19, 20 and 15, $\mathcal{N}$ contains $\varphi(1)$ and the other blocks dilute the violation). If every W_c is scalar, every $V_{M_{k_c}}(W_c)$ vanishes, since φ_{W_c} is then a trace on M_{k_c}; for item 16 the same conclusion holds because the Peierls–Bogoliubov inequality holds for every trace on $\mathcal{M}$. □

Remark 10.2. The theorem is a statement about the characterisations, not about the inequalities: a non-central density is detected by every characterisation of traces, but only through the variation *inside* the blocks. Items 11 and 24 characterise the *canonical* trace Tr among the functionals with $\varphi(1) = \dim$, and central W violate them; for a fixed density their violations are $1 - \mathrm{GM/AM}$ and $w_{\min}^{-1/2} - 1$ of the eigenvalues of W normalised to $\mathrm{Tr}\, W = \sum_c k_c$ (Propositions 7.5 and 7.6; the proofs do not use the block structure).

10.1 Direct integrals: continuous centres

The finite direct sum is the case of a finite centre. Every finite von Neumann algebra of type I with separable predual is a direct integral $\mathcal{M} = \int_X^{\oplus} M_{k(x)}\, d\mu(x)$ over a σ-finite measure space (X, μ), with a measurable fibre dimension $k(x) < \infty$: its elements are bounded measurable fields $x \mapsto A(x) \in M_{k(x)}$, its centre is $L^\infty(X, \mu)$, and $\mathrm{Tr}\, A = \int_X \mathrm{Tr}\, A(x)\, d\mu(x)$. For a measurable field $W(x) > 0$ with $\int \mathrm{Tr}\, W\, d\mu < \infty$ the normal positive functional is $\varphi_W(A) = \int_X \mathrm{Tr}(W(x)A(x))\, d\mu(x)$, and φ_W is a trace if and only if $W(x)$ is scalar for almost every x. If μ has no atoms, the centre has no minimal projections, and still only the fibres matter.

Theorem 10.3. *Let $\mathcal{M}$ and W be as above, fix one inequality of Table 1 other than 11, 16 and 24, and let $V_{\mathcal{M}}(W)$ and $V_{M_{k(x)}}(W(x))$ be as in Theorem 10.1.*

(a) *If $v : X \to [0,\infty]$ is measurable and $V_{M_{k(x)}}(W(x)) \le v(x)$ for almost every x, then $V_{\mathcal{M}}(W) \le \operatorname{ess\,sup}_x v(x)$.*

(b) *If W is simple, $k(x) = k_c$ and $W(x) = W_c$ on the cells X_c of a countable measurable partition with $\mu(X_c) > 0$, then $V_{\mathcal{M}}(W) \le \sup_c V_{M_{k_c}}(W_c)$, with equality for the items listed in Theorem 10.1; the measures of the cells do not enter.*

In particular, for an item whose envelope $V_n(\delta)$ is known and increasing in δ (Theorem 8.1, Proposition 8.2), $V_{\mathcal{M}}(W) \le \sup_n V_n(\delta^*)$ with the essential supremum $\delta^* = \operatorname{ess\,sup}_x \Delta(W(x))$ of the fibre contrasts, the supremum over the essential range of k, and $V_n(1) := \lim_{\delta\to 1} V_n(\delta)$ (possibly ∞). For the items listed in Theorem 10.1 this bound is the envelope over all densities with $\operatorname{ess\,sup}_x \Delta(W(x)) = \delta^*$, by (b) with one cell.

Proof. (a) The proof of Theorem 10.1 goes through with the sums over c replaced by integrals over X. All variables are measurable fields and every functional calculus acts fibrewise, as do the lattice operations ($P \wedge Q = \chi_{\{1\}}(PQP)$ and $P \vee Q = 1 - (1-P) \wedge (1-Q)$, so $(P \wedge Q)(x) = P(x) \wedge Q(x)$ almost everywhere); hence $\mathcal{L}$, $\mathcal{R}$ and $\mathcal{N}$ are integrals of the fibre quantities $\mathcal{L}_x$, $\mathcal{R}_x$, $\mathcal{N}_x$, which are integrable since the fields are bounded and $\int \operatorname{Tr} W\, d\mu < \infty$. Where $\mathcal{L}$ contains the absolute value of a functional (items 1, 4 and 14), $|\int f\, d\mu| \le \int |f|\, d\mu$ gives $\mathcal{L} \le \int \mathcal{L}_x\, d\mu$, and in items 6 and 14 the Cauchy–Schwarz and Hölder inequalities for integrals give $\mathcal{R} \ge \int \mathcal{R}_x\, d\mu$. By the definition of $V_{M_{k(x)}}(W(x))$ we have $\mathcal{L}_x - \mathcal{R}_x \le v(x)\mathcal{N}_x$ almost everywhere, and where $\mathcal{N}_x = 0$ the variables vanish where they enter $\mathcal{L}_x$, so that $\mathcal{L}_x \le \mathcal{R}_x$, since $W(x) > 0$. Hence $\mathcal{L} - \mathcal{R} \le \int v\,\mathcal{N}_x\, d\mu \le (\operatorname{ess\,sup} v)\,\mathcal{N}$; for $\mathcal{N} = \mathcal{R}$ in items 6 and 14 the same argument applies to $\mathcal{L}/\mathcal{R} \le \int \mathcal{L}_x\, d\mu / \int \mathcal{R}_x\, d\mu$. (b) The bound is (a) with $v = V_{M_{k_c}}(W_c)$ on X_c. For the equality choose a cell X_c, which has $0 < \mu(X_c) < \infty$ since $\int \operatorname{Tr} W\, d\mu < \infty$ and $W_c > 0$, and let the variables be a near-optimal configuration of the block M_{k_c} on X_c and zero elsewhere ($\varepsilon 1$ with $\varepsilon \to 0$ in item 6, where $A, B > 0$, as in Theorem 10.1). Then $\mathcal{L}$, $\mathcal{R}$ and $\mathcal{N}$ are $\mu(X_c)$ times the block quantities, and the quotient is that of the block. The corollary follows from (a) with $v(x) = V_{k(x)}(\Delta(W(x)))$, which is measurable because the eigenvalues of a measurable field are measurable. □

Theorem 10.1 is the case of a finite X with point masses. Conversely, for a simple density the envelope is that of the direct sum $\bigoplus_c M_{k_c}$, whatever the weights $\mu(X_c)$. This is how the library computes on direct integrals: its class `DirectIntegral` represents (X, μ) by nodes x_i with weights w_i (a quadrature rule, Monte Carlo samples or the cells of a partition), the algebra by $\bigoplus_i M_{k(x_i)}$, a field by its values at the nodes and Tr by $\sum_i w_i \operatorname{Tr} A(x_i)$. Fibrewise operations are exact, and for fields constant on the cells of a partition so are the integrals; the test suite checks Theorem 10.3(b) for item 23 with cells of very different measures (`tests/test_direct_integral.py`).

11 Contrasts of composite operators

The operators of Table 1 enter the inequalities through composites: sums $A+B$ (items 2, 10, 12, 13), congruences $B^{1/2}AB^{1/2}$ (items 6, 7), means (item 6), Jordan products $\operatorname{Re} PQ = \frac{1}{2}(PQ+QP)$ (items 9, 21, 22), and, behind every characterisation, the commutator $AB - BA$, on which a functional vanishes if and only if it is a trace. This section describes how the contrast of a composite depends on the contrasts of its factors, and then uses the answers to explain the thresholds observed in the numerical search. Two reparametrisations of the contrast make the answers additive. For $A > 0$ with extreme eigenvalues $M_A \ge m_A > 0$ and condition number $h_A = M_A/m_A$ let

$$\eta(A) = \operatorname{artanh} \Delta(A) = \tfrac{1}{2} \log h_A, \qquad \psi(A) = \arcsin \Delta(A) \in [0, \tfrac{\pi}{2}], \qquad \bar{\lambda}(A) = \frac{M_A + m_A}{2}.$$

The rapidity $\eta(A)$ is half the logarithm of the condition number, that is, half the Hilbert projective distance of A to 1 and the Thompson distance of A to the scalars (Section 2). The angle $\psi(A)$ is that of Krein and Gustafson [39, 28, 30], the largest angle between x and Ax: $\cos\psi(A) = \min_{x\neq 0}\langle Ax,x\rangle/(\|Ax\|\|x\|) = 2\sqrt{M_A m_A}/(M_A+m_A)$, and $\sin\psi(A) = \min_{t>0}\|tA-1\| = \Delta(A)$, attained at $t = 1/\bar{\lambda}(A)$ (Figure 1(a)). For the density, $\kappa = \cosh\eta(W) = \sec\psi(W)$, so the envelope $\kappa - 1$ of the Kantorovich class (Theorem 5.1) is $\sec\psi(W) - 1$. The angle between vectors is $\angle(x,y) = \arccos(\operatorname{Re}\langle x,y\rangle/(\|x\|\|y\|))$, the angle of the real inner product $\operatorname{Re}\langle\cdot,\cdot\rangle$ on $\mathbb{C}^n = \mathbb{R}^{2n}$; it satisfies the triangle inequality.

Proposition 11.1 (Multiplicative composites: rapidities add)**.** *Let $A, B > 0$ in M_n and $\eta_A = \eta(A)$, $\eta_B = \eta(B)$.*

(a) $\Delta(A^p) = \tanh(|p|\eta_A)$ *for real $p \neq 0$,* $\Delta(A\otimes B) = \tanh(\eta_A+\eta_B)$*, and the compression of A to a subspace has contrast at most $\Delta(A)$.*

(b) $\tanh|\eta_A - \eta_B| \le \Delta(A^{1/2}BA^{1/2}) \le \tanh(\eta_A+\eta_B)$*; the same bounds hold for the contrast of $|AB|$, and for the spectrum of AB.*

(c) For the geometric mean $A\#B = A^{1/2}(A^{-1/2}BA^{-1/2})^{1/2}A^{1/2}$, $\tanh\frac{|\eta_A-\eta_B|}{2} \le \Delta(A\#B) \le \tanh\frac{\eta_A+\eta_B}{2}$.

(d) $\Delta(sA+tB) \le \max(\Delta(A),\Delta(B))$ *for $s,t>0$, and the same holds for the harmonic mean. If $\Delta(A),\Delta(B) > 0$, then $\Delta(A+tB) = 0$ for suitable $t > 0$ and a suitable relative position.*

All bounds are attained by commuting operators, with the extreme eigenvectors aligned (upper bounds) or interchanged (lower bounds), in every dimension $n \ge 2$.

These facts are classical, and we include the short proof for the attainment statements. The upper bound in (b) is the submultiplicativity of the condition number, $h(A^{1/2}BA^{1/2}) \le h_A h_B$. In the language of Hilbert's projective metric d_H, which is invariant under congruences and under inversion, (b) reads $|d_H(B,1) - d_H(1,A^{-1})| \le d_H(B,A^{-1}) = d_H(A^{1/2}BA^{1/2},1) \le d_H(B,1) + d_H(1,A^{-1})$, i.e. it is the triangle inequality [20, 40]; this is why rapidities, the halved distances to the scalars, add.

Proof. (a) Spectral mapping, the eigenvalues of $A\otimes B$ are the products of eigenvalues, and Cauchy interlacing. (b) From $m_B A \le A^{1/2}BA^{1/2} \le M_B A$ and the monotonicity of the eigenvalues [5], $\lambda_k(A)m_B \le \lambda_k(A^{1/2}BA^{1/2}) \le \lambda_k(A)M_B$ for every k. Hence the largest eigenvalue lies in $[M_A m_B, M_A M_B]$ and the smallest in $[m_A m_B, m_A M_B]$, so the condition number lies in $[h_A/h_B, h_A h_B]$; since $A^{1/2}BA^{1/2}$ is similar to $B^{1/2}AB^{1/2}$, it is also at least h_B/h_A. With $\Delta = \tanh(\frac{1}{2}\log h)$ this is (b). For $|AB|$ use the singular values, $\sigma_k(A)\sigma_{\min}(B) \le \sigma_k(AB) \le \sigma_k(A)\sigma_{\max}(B)$; AB is similar to $A^{1/2}BA^{1/2}$. (c) The mean is monotone and homogeneous, so $\sqrt{m_A m_B} \le A\#B \le \sqrt{M_A M_B}$, which gives the upper bound. From $B = (A\#B)A^{-1}(A\#B)$, whose spectrum is that of $A^{-1/2}(A\#B)^2A^{-1/2}$, (a) and (b) give $\eta_B \le \eta_A + 2\eta(A\#B)$, and symmetrically. (d) The mediant inequality $\frac{sM_A+tM_B}{sm_A+tm_B} \le \max(h_A,h_B)$ bounds the condition number of $sA+tB$; for the harmonic mean apply it to $A^{-1}+B^{-1}$. For the last claim take $B = c - A$ with $c > M_A$ in the eigenbasis of A and with the intermediate eigenvalues reflected; $\Delta(c-A) = (M_A - m_A)/(2c - M_A - m_A)$ takes every value in $(0,1)$. Attainment: for commuting A, B with aligned extreme eigenvectors all extreme eigenvalues multiply (add, for sums), and with interchanged ones the quotients do. □

The Jordan product. For $AB + BA$ the picture changes: it need not be positive, and angles, not rapidities, add. The reason is geometric. A positive operator A turns every vector by at most its angle: $\angle(x,Ax) \le \psi_A$, where $\sin\psi_A = \Delta(A)$. Hence ABx and BAx differ from x by at most

$\psi_A + \psi_B$, and so does their sum $(AB + BA)x$, as long as $\psi_A + \psi_B < \frac{\pi}{2}$. Then $(AB + BA)x$ stays in the half-space of x, i.e. $\langle (AB + BA)x, x\rangle > 0$. If $\psi_A + \psi_B > \frac{\pi}{2}$, a suitable rotation of B against A turns some x by more than a right angle, and $AB + BA$ has a negative eigenvalue. In terms of contrasts: *$AB + BA$ is positive for every relative position if and only if the point $(\Delta(A), \Delta(B))$ lies in the unit disc* (Figure 5). The sufficiency of this criterion is Gustafson's [28], whose angle argument we reproduce. Its sharpness is in the spirit of the work on eigenvalues of Jordan products by Strang [50], Alikakos and Bates [2, 3] and Conley, Pucci and Serrin [23]; we include the short proof of both directions, since we need the extremal configuration, and add that the same angles bound the contrast of $AB + BA$.

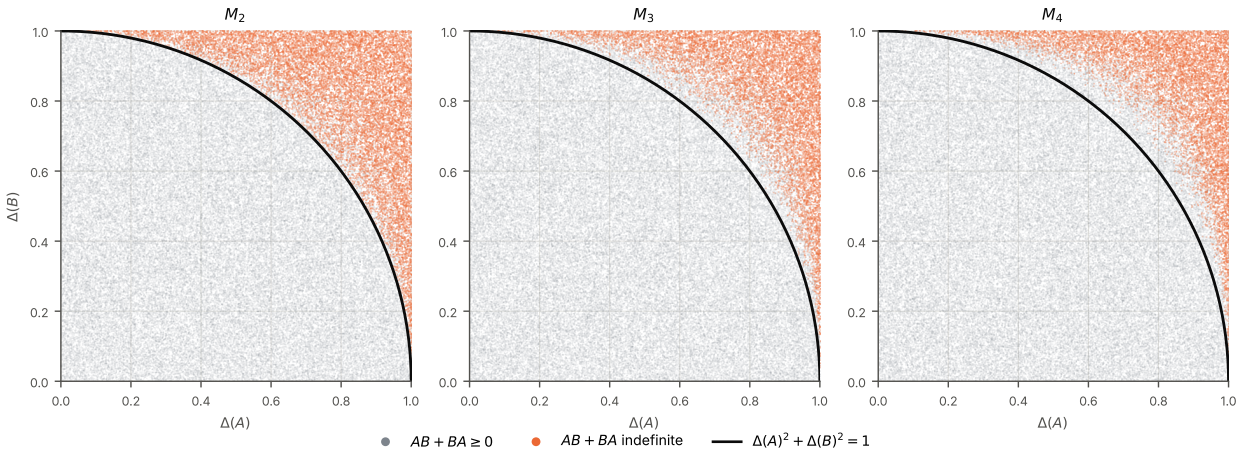


Figure 5: Is $AB + BA$ positive? 100000 random pairs $A, B > 0$ per dimension, with contrasts uniform in $[0, 1]^2$ and Haar random relative position: positive (grey) or indefinite (orange). No indefinite product occurs inside the quarter circle $\Delta(A)^2 + \Delta(B)^2 = 1$ (Theorem 11.2(a)); outside it occurs for some positions, more often the further out.

Theorem 11.2 (Jordan product)**.** *Let $A, B > 0$ in M_n, $n \geq 2$, with contrasts δ_A, δ_B, angles $\psi_A = \arcsin \delta_A$, $\psi_B = \arcsin \delta_B$ and rapidities $\eta_A = \operatorname{artanh} \delta_A$, $\eta_B = \operatorname{artanh} \delta_B$.*

(a) $AB + BA \geq 0$ for every relative position of A and B if and only if $\delta_A^2 + \delta_B^2 \leq 1$. If $\delta_A^2 + \delta_B^2 > 1$, $AB + BA$ is indefinite for some relative position, in every dimension.

(b) If $\delta_A^2 + \delta_B^2 < 1$, then $\tanh|\eta_A - \eta_B| \leq \Delta(AB + BA) \leq \sin(\psi_A + \psi_B)$, and the lower bound is attained by commuting A, B whose extreme eigenvectors are interchanged.

Proof. (a) and the upper bound in (b). We use the real angle $\angle(x, y)$ defined above. By the formula for $\cos\psi$, $\angle(x, Bx) \leq \psi_B$ and $\angle(Bx, ABx) \leq \psi_A$, so $\angle(x, ABx) \leq \psi_A + \psi_B$ by the triangle inequality; likewise $\angle(x, BAx) \leq \psi_A + \psi_B$. If $\psi_A + \psi_B < \frac{\pi}{2}$, the cone $\{y : \angle(x, y) \leq \psi_A + \psi_B\}$ is convex and contains $(AB + BA)x$, so $\langle (AB + BA)x, x\rangle > 0$; moreover the largest angle of $AB + BA$ is at most $\psi_A + \psi_B$, and its sine is $\Delta(AB + BA)$. The case $\psi_A + \psi_B = \frac{\pi}{2}$ follows by continuity.

Converse in (a). On M_2 take $A = \operatorname{diag}(1, \ell_A)$ and B with eigenvalues $1, \ell_B$ and eigenvector $(\cos\theta, \sin\theta)$ for 1, where $\ell_A = \frac{1-\delta_A}{1+\delta_A}$, $\ell_B = \frac{1-\delta_B}{1+\delta_B}$. With $c = \cos^2\theta$ and $p = (1 - \ell_A)(1 - \ell_B)$ a direct computation gives

$$\tfrac{1}{2}\operatorname{Tr}(AB + BA) = \ell_A + \ell_B + pc, \qquad \det(AB + BA) = 4\ell_A\ell_B - p^2c(1 - c).$$

At $\theta = \pi/4$ the determinant is negative if and only if $16\ell_A\ell_B < p^2$; with $\sqrt{\ell_A} = \sqrt{1-\delta_A^2}/(1+\delta_A)$ and $1-\ell_A = 2\delta_A/(1+\delta_A)$ this reads $(1-\delta_A^2)(1-\delta_B^2) < \delta_A^2\delta_B^2$, i.e. $\delta_A^2+\delta_B^2 > 1$. On M_n add the direct summands $\ell_A 1_{n-2}$ and $\ell_B 1_{n-2}$.

Lower bound in (b). If x, y are unit eigenvectors of A for its extreme eigenvalues M_A, m_A, then $\lambda_{\max}(AB+BA) \geq 2\mathrm{Re}\,\langle Bx, Ax\rangle = 2M_A\langle Bx, x\rangle \geq 2M_A m_B$ and likewise $\lambda_{\min}(AB+BA) \leq 2m_A M_B$. So the condition number of $AB+BA$ is at least h_A/h_B, and by symmetry at least h_B/h_A. □

How large can $\Delta(AB+BA)$ be? The upper bound $\sin(\psi_A+\psi_B)$ is reached only at the boundary of the disc. The true maximum depends on the dimension, because in M_2 both extreme eigenvalues of $AB+BA$ come from the same pair of 2×2 matrices, while in M_3 they can come from different blocks.

Proposition 11.3 (Largest contrast of the Jordan product)**.** *In the notation of the proof above, let* $\delta_A^2+\delta_B^2 < 1$.

(a) On M_2, $\max\Delta(AB+BA) = \max\limits_{0\leq c\leq 1}\Big(1 - \dfrac{4\ell_A\ell_B - p^2c(1-c)}{(\ell_A+\ell_B+pc)^2}\Big)^{1/2}$.

(b) On M_n, $n\geq 3$, $\max\Delta(AB+BA) \geq (2-\mu)/(2+\mu)$, *where* $\mu = \min_{0\leq c\leq 1}\big(\ell_A+\ell_B+pc - \sqrt{(\ell_A+\ell_B+pc)^2 - 4\ell_A\ell_B + p^2c(1-c)}\,\big)$ *is the smallest eigenvalue of* $AB+BA$ *over the* 2×2 *pairs of (a).*

Numerically, both values equal the commuting value $\tanh(\eta_A+\eta_B)$ *for small contrasts and exceed it for large ones.*

Proof. (a) On M_2 every pair is unitarily equivalent to the pair of the proof of Theorem 11.2 for some θ, and $\Delta^2 = 1 - 4\det/\mathrm{Tr}^2$. (b) Take $A = 1\oplus A_2$ and $B = 1\oplus B_2$ (padded with ones) with the 2×2 pair A_2, B_2 of smallest eigenvalue μ. Then $AB+BA = 2\oplus(A_2B_2+B_2A_2)$ has the largest eigenvalue 2, since $\|A_2B_2+B_2A_2\| \leq 2$, and the smallest eigenvalue μ. □

For $\delta_A = \delta_B = 0.6$: the commuting value is 0.8824, the maximum on M_2 is 0.8839, the value of (b) for $n\geq 3$ is 0.8963, and $\sin(\psi_A+\psi_B) = 0.9600$. A numerical optimisation over all relative positions and intermediate eigenvalues (24 starts per point) finds on M_3 and on M_4 exactly the value of (b) (0.8963 and 0.8963 at $\delta_A = \delta_B = 0.6$, and largest difference 3×10^{-7} on a grid of δ_A with $\delta_B = 0.6$; Figure 7). Equality for every $n\geq 3$ would follow if the smallest eigenvalue of a Jordan product with spectra in $[\ell_A, 1]$ and $[\ell_B, 1]$ were always attained on a two-dimensional subspace. Strang [50] bounded this smallest eigenvalue from below, and Alikakos and Bates showed that his bound is best possible [3]; we have not checked whether it coincides with μ.

Proposition 11.4 (Commutators: three contrasts multiply)**.** *Let* $A, B > 0$ *in* M_n.

(a) $\|AB-BA\| \leq 2\Delta(A)\Delta(B)\,\bar\lambda(A)\bar\lambda(B)$, *with equality for suitable* A, B *of any prescribed contrasts in every* M_n, $n\geq 2$. *In particular* $\|AB-BA\| \leq \frac{2\Delta(A)\Delta(B)}{(1+\Delta(A))(1+\Delta(B))}\|A\|\|B\| \leq \frac{1}{2}\|A\|\|B\|$.

(b) For a state φ_W *(*$\mathrm{Tr}\,W = 1$*) with* $\Delta(W) = \delta$, $h = \frac{1+\delta}{1-\delta}$,

$$|\varphi_W(AB-BA)| \leq 2\,c_n(\delta)\,\Delta(A)\Delta(B)\,\bar\lambda(A)\bar\lambda(B), \qquad c_n(\delta) = \begin{cases} \delta, & n \text{ even},\\ \dfrac{k(h-1)}{k(h+1)+1}, & n = 2k+1, \end{cases}$$

and the bound is attained for every δ *and all prescribed contrasts of* A *and* B.

Part (a) is a consequence of Kittaneh's inequality $\|XY - YX\| \le \|X\|\|Y\|$ for $X \ge 0$ [37], applied to $X = A - m_A \ge 0$ and $Y = B - \bar\lambda(B)$, whose norms are $2\Delta(A)\bar\lambda(A)$ and $\Delta(B)\bar\lambda(B)$; equivalently, of the bound $\|AB - BA\| \le \frac12\|A\|\|B\|$ for positive operators applied to $A - m_A$ and $B - m_B$. Our contribution is the sharp trace version (b), the commutator test of traciality with all three contrasts prescribed.

Proof. (a) Write $A = \bar\lambda(A)(1+E)$, $B = \bar\lambda(B)(1+F)$ with self-adjoint E, F, $\|E\| = \Delta(A)$, $\|F\| = \Delta(B)$. Then $AB - BA = \bar\lambda(A)\bar\lambda(B)(EF - FE)$ and $\|EF - FE\| \le 2\|E\|\|F\|$. Equality holds on M_2 for $E = \Delta(A)\sigma_z$, $F = \Delta(B)\sigma_x$, since $\sigma_z\sigma_x - \sigma_x\sigma_z = 2i\sigma_y$; in M_n add a direct summand $\bar\lambda(A) \oplus \bar\lambda(B)$. The second form uses $\|A\| = \bar\lambda(A)(1 + \Delta(A))$. (b) $C = i(AB - BA)$ is self-adjoint with $\operatorname{Tr} C = 0$ and $\|C\| \le K$, the bound of (a). By von Neumann's trace inequality [5], $|\operatorname{Tr} WC|$ is at most the maximum of $\sum_i w_i c_i$ over decreasingly ordered w_i (the eigenvalues of W) and $c_i \in [-K, K]$ with $\sum_i c_i = 0$, which is $K\sum_{i\le k}(w_i - w_{n+1-i})$, $k = \lfloor n/2 \rfloor$. Let $a = w_1$, $b = w_n = a/h$ and $S = \sum_{i\le k} w_i \le ka$. Then $\sum_{i\le k}(w_i - w_{n+1-i}) \le S - kb$ and $1 = \operatorname{Tr} W \ge S + (n-k)b$. Maximising $S - kb$ under $S \le khb$ and $S \le 1 - (n-k)b$ gives the value at $khb = 1 - (n-k)b$, namely $k(h-1)/(kh + n - k)$, which is $c_n(\delta)$ for even and odd n. For equality take k copies of the 2×2 pair of (a), W diagonal in the eigenbasis of the k blocks of C with the eigenvalues a on the positive and b on the negative eigenvectors of C, and, for odd n, the eigenvalue b of W and scalars A, B on the remaining one-dimensional block. □

So the most basic characterisation of the trace, $\varphi(AB) = \varphi(BA)$, fails exactly at the order of the product of the three contrasts, $2\Delta(W)\Delta(A)\Delta(B)$ in even dimension; this is the simplest instance of the principle that a violation needs a functional far from a trace and operators far from scalars. The odd dimensions are slightly weaker, as for Gardner's inequality (Theorem 6.1), and $c_{2k+1}(\delta) \to \delta$ as $k \to \infty$.

Random samples (100000 per bound and dimension, Haar random relative positions, the intermediate eigenvalues uniform between the extreme ones) never exceed the bounds of this section on M_2, M_3 and M_4 (the largest value of sample minus bound is -5×10^{-8}), and among 100000 random pairs per dimension with contrasts uniform in $[0,1]^2$ the number of indefinite Jordan products with $\delta_A^2 + \delta_B^2 \le 1$ is 0. As in Section 12.5, random positions rarely come close to the extremes in higher dimension: for the commutator test on M_4 the samples stay far below the bound, which needs two aligned 2×2 blocks (Figure 8).

11.1 Thresholds in the contrast of the operators

The numerical search of Section 12.3 shows that several characterisations are not violated at all unless the operators have contrast close to 1. The composition rules above explain this for two of them. The simplest example is the analogue of item 21 for positive operators, $\varphi(\operatorname{Re} AB) \ge 0$: it holds for every positive functional as long as $\Delta(A)^2 + \Delta(B)^2 \le 1$, because then $\operatorname{Re} AB = \frac12(AB + BA) \ge 0$ by Theorem 11.2, and for operators of equal contrast δ' it can fail only for $\delta' > 1/\sqrt{2}$. Projections have the angle $\frac{\pi}{2}$, $\operatorname{Re} PQ$ is indefinite in general, and accordingly item 21 is violated by every non-tracial functional, with the envelope $\frac12\tan^2(\psi/2)$ of Theorem B.

For subadditivity of the square root (item 12) the same mechanism works one level deeper, through the link between Jordan products and subadditivity found by Moslehian and Najafi [42].

Proposition 11.5 (Sharp contrast threshold for item 12). *Let $A, B > 0$ with $\Delta(A^{1/2})^2 + \Delta(B^{1/2})^2 \le 1$, i.e.*

$$\Big(\frac{\Delta(A)}{1 + \sqrt{1 - \Delta(A)^2}}\Big)^2 + \Big(\frac{\Delta(B)}{1 + \sqrt{1 - \Delta(B)^2}}\Big)^2 \le 1.$$

Then $(A+B)^{1/2} \le A^{1/2} + B^{1/2}$, *and hence* $\varphi((A+B)^{1/2}) \le \varphi(A^{1/2}) + \varphi(B^{1/2})$ *for every positive functional* φ. *In particular, if* $\Delta(A) = \Delta(B) = \delta' \le 2\sqrt{2}/3 \approx 0.943$, *item 12 is not violated for any density, so* $V_n(\delta, \delta') \le 0$ *for all* δ *and* n. *The threshold is sharp: if* $\delta' > 2\sqrt{2}/3$, *then for every* $n \ge 2$ *and every* δ *close enough to* 1 *there are a density of contrast* δ *and operators of contrast* δ' *that violate item 12, i.e.* $V_n(\delta, \delta') > 0$.

Proof. By Proposition 11.1(a), $\Delta(A^{1/2}) = \tanh(\eta_A/2) = \Delta(A)/(1 + \sqrt{1 - \Delta(A)^2})$, which gives the two forms of the hypothesis. By Theorem 11.2(a) applied to $A^{1/2}$ and $B^{1/2}$, $A^{1/2}B^{1/2} + B^{1/2}A^{1/2} \ge 0$, hence

$$(A^{1/2} + B^{1/2})^2 = A + B + A^{1/2}B^{1/2} + B^{1/2}A^{1/2} \ge A + B,$$

and, the square root being operator monotone, $A^{1/2} + B^{1/2} \ge (A+B)^{1/2}$. A positive functional preserves this inequality. For $\Delta(A) = \Delta(B) = \delta'$ the hypothesis reads $\tanh(\eta/2) \le 1/\sqrt{2}$ with $\tanh \eta = \delta'$, i.e. $\delta' = 2t/(1+t^2)$ at $t = 1/\sqrt{2}$, which is $2\sqrt{2}/3$.

Sharpness. Let $\delta' > 2\sqrt{2}/3$, so that $t = \Delta(A^{1/2}) > 1/\sqrt{2}$, and on M_2 let $A = \operatorname{diag}(1, \ell')$ and $B = RAR^*$ with R the rotation by $\pi/4$. The reflection F in the line at angle $\pi/8$ interchanges A and B, hence $A^{1/2}$ and $B^{1/2}$, so it commutes with $A + B$, with $S = A^{1/2} + B^{1/2}$ and with $J = A^{1/2}B^{1/2} + B^{1/2}A^{1/2}$. Since F has the simple eigenvalues ± 1, these three matrices are diagonal in its eigenbasis u_1, u_2, and $S^2 = A + B + J$ gives $\langle u_i, Su_i\rangle^2 = \langle u_i, (A+B)u_i\rangle + \langle u_i, Ju_i\rangle$. The pair $A^{1/2}$, $B^{1/2}$ is the configuration of the converse in Theorem 11.2(a) with $\delta_A = \delta_B = t$, so J is indefinite since $2t^2 > 1$, and for the u_i with $\langle u_i, Ju_i\rangle < 0$ we get $\langle u_i, (A^{1/2} + B^{1/2})u_i\rangle < \langle u_i, (A+B)^{1/2}u_i\rangle$. On M_n replace A, B by $A \oplus 1_{n-2}$ and $B \oplus 1_{n-2}$, which have the same contrast, and let W, normalised by $\operatorname{Tr} W = n$, have u_i (padded by zeros) as eigenvector for its largest eigenvalue and all other eigenvalues equal. As $\delta \to 1$, $\varphi_W \to n\langle u_i, \cdot\, u_i\rangle$, so $\varphi_W((A+B)^{1/2}) > \varphi_W(A^{1/2}) + \varphi_W(B^{1/2})$ for all δ close enough to 1. □

The proved threshold is consistent with the computation: on M_2 and M_3 the search finds no violation of item 12 for $\delta' \le 0.95$ and $\delta \in \{0.3, 0.6, 0.9\}$, and the first violation above 10^{-4} at $\delta' = 0.999$ for $\delta = 0.3$, 0.99 for $\delta = 0.6$, 0.99 for $\delta = 0.9$ (M_2 and M_3 alike); all these grid values lie above $2\sqrt{2}/3$ (Figure 6(a)). Unlike item 24 below, whose whole threshold curve we find for $n = 2$, here sharpness is proved only in the limit $\delta \to 1$; at $\delta = 0.3$, 0.6 and 0.9 the search finds the onset of violations higher. The criterion of Moslehian and Najafi [42, Theorem 2.1] is that $AB + BA \ge 0$ if and only if $f(A+B) \le f(A) + f(B)$ for *all* non-negative operator monotone functions f. For the square root alone the hypothesis $A^{1/2}B^{1/2} + B^{1/2}A^{1/2} \ge 0$ suffices, which is weaker in terms of contrasts since $\Delta(A^{1/2}) < \Delta(A)$, and the proof of sharpness shows that in the extremal configuration it is also necessary.

For the power-mean inequality $\varphi(A^2)^{1/2} \le \varphi(A)$ with $\varphi(1) = n$ (item 24) a Kantorovich-type argument gives the envelope with both contrasts prescribed in closed form, and with it a sharp threshold.

Proposition 11.6 (Item 24 with both contrasts). *Let* $n \ge 2$, $\Delta(W) = \delta < 1$, $\operatorname{Tr} W = n$, *and let* $A \ge 0$ *have* $\Delta(A) = \delta'$. *Put* $\ell = \frac{1-\delta}{1+\delta}$, $\ell' = \frac{1-\delta'}{1+\delta'}$,

$$x_0 = n\ell' + (1-\ell')\frac{n\ell}{n-1+\ell}, \qquad x_* = \frac{2n\ell'}{1+\ell'}, \qquad f(x) = \frac{\sqrt{(1+\ell')x - n\ell'}}{x}.$$

Then $\varphi_W(A^2)^{1/2}/\varphi_W(A) \le f(\max(x_0, x_*))$, *so* $V_n(\delta, \delta') \le f(\max(x_0, x_*)) - 1$, *with equality if* $x_0 \ge x_*$ *or* $(n-1)\ell\ell' \le 1$, *in particular always for* $n = 2$. *Consequently:*

(a) item 24 is violated for no density if and only if $\delta' \le \sqrt{(n-1)/n}$; *for* $\delta' > \sqrt{(n-1)/n}$ *it is violated for every* δ *close enough to* 1;

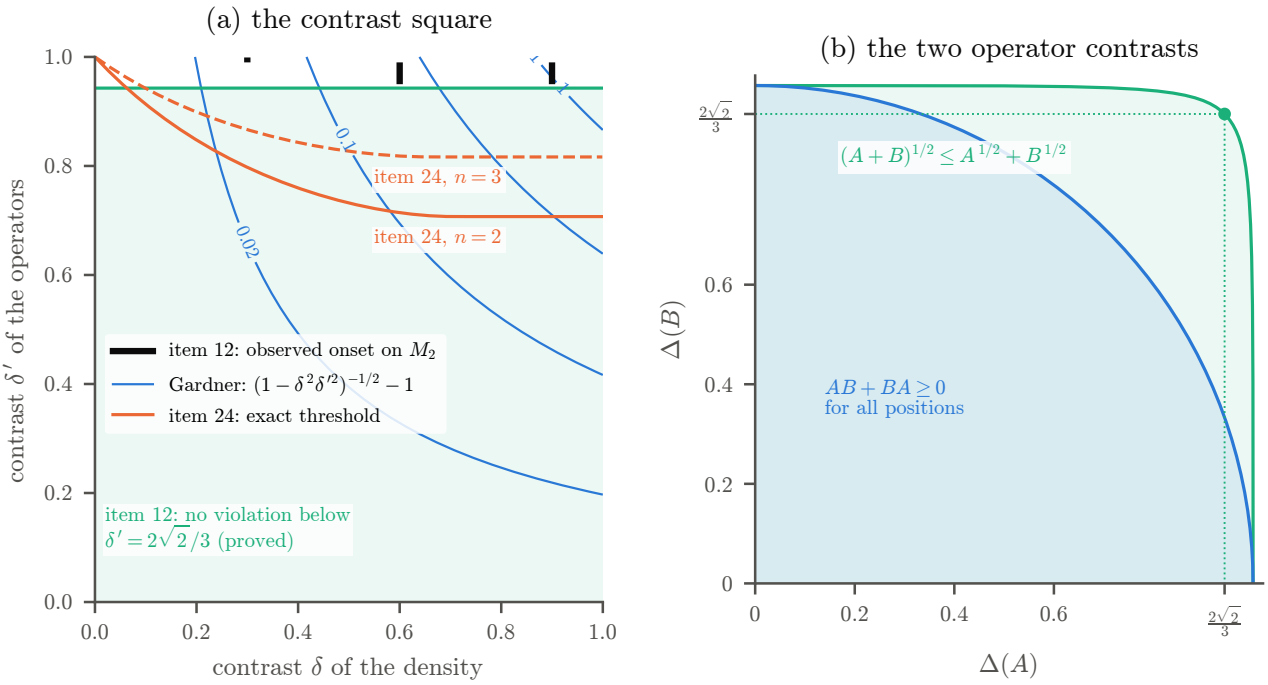


Figure 6: (a) The contrast square: level curves of the Gardner envelope $(1-\delta^2\delta'^2)^{-1/2}-1$ (blue; the values 0.02, 0.1, 0.3, 1), the band in which item 12 cannot be violated (green, Proposition 11.5), the exact violation thresholds of item 24 for $n=2$ and $n=3$ (orange, Proposition 11.6), and, in black, the interval of grid values of δ' in which the numerical search on M_2 first finds a violation of item 12, at $\delta=0.3$, 0.6 and 0.9. (b) The plane of the two operator contrasts: $AB+BA\ge 0$ for all relative positions inside the quarter disc (blue, Theorem 11.2), and $(A+B)^{1/2}\le A^{1/2}+B^{1/2}$ in the larger green region (Proposition 11.5).

(b) for $n=2$ the violation threshold is the curve $f(\max(x_0,x_*))=1$ in the (δ,δ')-square, and it equals $1/\sqrt{2}$ for all $\delta\ge 1/\sqrt{2}$.

Proof. Both sides are homogeneous of degree one in A, so let $\|A\|=1$; then the spectrum of A lies in $[\ell',1]$ and contains both end points. From $(1-A)(A-\ell')\ge 0$ we get $A^2\le(1+\ell')A-\ell'$, hence, with $x=\varphi(A)$,

$$\frac{\varphi(A^2)^{1/2}}{\varphi(A)}\le\frac{\sqrt{(1+\ell')x-n\ell'}}{x}=f(x),$$

with equality for two-level A. If e is a unit eigenvector of A for the eigenvalue 1, then $A\ge\ell'+(1-\ell')ee^*$, so $x\ge n\ell'+(1-\ell')\langle We,e\rangle\ge n\ell'+(1-\ell')\lambda_{\min}(W)$, and $\lambda_{\min}(W)\ge n\ell/(n-1+\ell)$ for $\operatorname{Tr}W=n$ (proof of Proposition 7.6); so $x\ge x_0$. The function f increases on $[n\ell'/(1+\ell'),x_*]$ and decreases on $[x_*,\infty)$, since $\frac{d}{dx}f(x)^2=x^{-3}\big(2n\ell'-(1+\ell')x\big)$; hence $f(x)\le f(\max(x_0,x_*))$. For equality take $A=\ell'+(1-\ell')ee^*$ with a unit vector e, so that $x=n\ell'+(1-\ell')\langle We,e\rangle$. If $x_0\ge x_*$, let W have the spectrum $\{a,\dots,a,b\}$ and e an eigenvector for b; then $x=x_0$. Otherwise we need $\langle We,e\rangle=n\ell'/(1+\ell')$; the values $\langle We,e\rangle$ for the spectra $\{a,\dots,a,b\}$ and $\{a,b,\dots,b\}$ (normalised to $\operatorname{Tr}W=n$) fill the interval $[n\ell/(n-1+\ell),\,n/(1+(n-1)\ell)]$, whose left end is at most $n\ell'/(1+\ell')$ because $x_0\le x_*$, and whose right end is at least $n\ell'/(1+\ell')$ if $(n-1)\ell\ell'\le 1$.

(a) Since $f(\max(x_0,x_*))\le f(x_*)=(1+\ell')/(2\sqrt{n\ell'})$, there is no violation if $(1+\ell')^2\le 4n\ell'$, i.e. $\ell'\ge(\sqrt{n}-\sqrt{n-1})^2$, i.e. $\delta'\le\sqrt{(n-1)/n}$. Conversely, as $\delta\to 1$ we have $\ell\to 0$, so $x_0\to n\ell'<x_*$ and $(n-1)\ell\ell'\le 1$, and the envelope tends to $f(x_*)-1>0$ when $(1+\ell')^2>4n\ell'$. (b) For $n=2$

equality always holds; $x_0 \le x_*$ means $1-\delta \le 2\ell'/(1+\ell')$, and at $\ell' = (\sqrt{2}-1)^2$ this is $\delta \ge 1/\sqrt{2}$. $\square$

The computed envelopes of item 24 on M_2 and M_3 with both contrasts prescribed agree at every grid point with Proposition 11.6 to 2×10^{-11} (absolute difference). The threshold rises above $\sqrt{(n-1)/n}$ for smaller δ and tends to 1 as $\delta \to 0$ (Figure 6(a)): the less tracial the functional, the more moderate the operators that detect it. For monotonicity (item 10) an argument of this kind is not available: for $0 \le A \le B = A + tvv^*$ with a unit vector v that is not an eigenvector of A, $B^2 - A^2 = t\big((A+\frac{t}{2})vv^* + vv^*(A+\frac{t}{2})\big)$ is indefinite for every $t > 0$, however small the contrasts of A and B, so the operator inequality fails and only the functional can prevent a violation. Correspondingly, the numerically observed onset of violations of item 10 depends on δ: on M_2 the first grid value of δ' with a violation above 10^{-4} is 0.99 for $\delta = 0.3$, 0.95 for $\delta = 0.6$, 0.8 for $\delta = 0.9$.

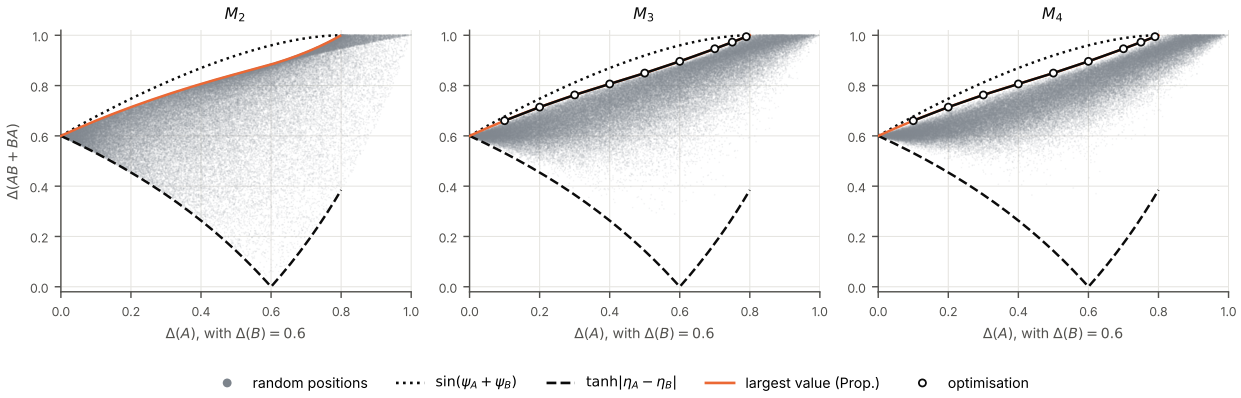


Figure 7: Contrast of the Jordan product $AB + BA$ against $\Delta(A) < 0.8$, with $\Delta(B) = 0.6$ (for $\Delta(A) \le 0.8$ it is positive, Theorem 11.2(a)): the bounds of Theorem 11.2(b), the largest value of Proposition 11.3 (orange; on M_3 and M_4 the value $(2-\mu)/(2+\mu)$), and the largest value found by optimisation (markers).

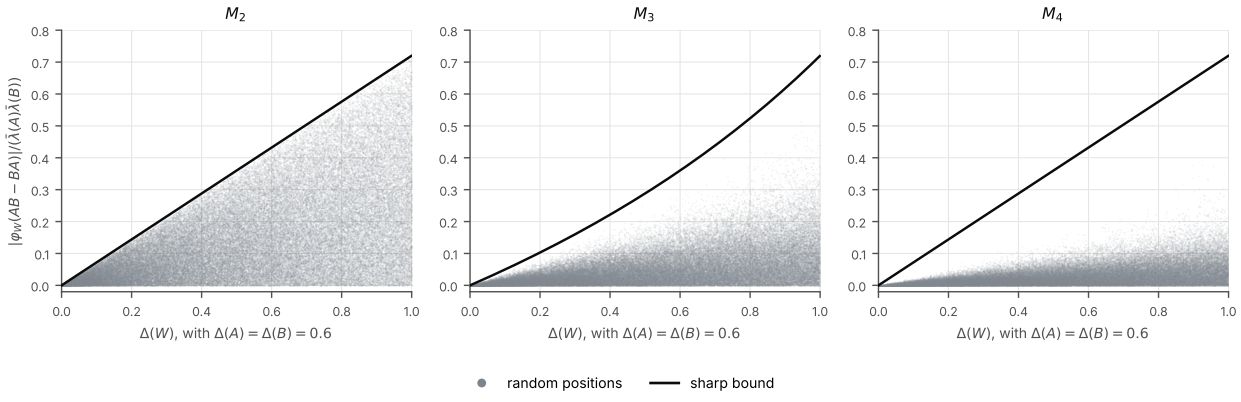


Figure 8: The commutator test $|\varphi_W(AB-BA)|/(\bar{\lambda}(A)\bar{\lambda}(B))$ for random states φ_W and random A, B against $\Delta(W)$, with $\Delta(A) = \Delta(B) = 0.6$, and the sharp bound $2c_n(\delta)\cdot 0.36$ of Proposition 11.4(b).

12 Numerical evidence

12.1 Method

The envelopes were computed by constrained gradient ascent with the library of [41] (the programs `trace_characterizations.py` and `trace_contrast.py` of its repository). The random sampling (`trace_montecarlo.py`), the computations for composite operators (`contrast_operations.py`) and Figure 3 (`trace_envelopes_figures.py`) are plain NumPy/SciPy programs in the same repository that do not use the library. The density W is diagonal with a spectrum parametrised so that $\Delta(W) = \delta$ holds exactly, the other eigenvalues being free. Positive variables have a free contrast in $[0, 1]$, so rank-deficient operators are included; there are self-adjoint and unitary variables (polar factors, and projections UDU^* with every rank pattern spread over the starts) and positive scalars. In item 2 the search runs over self-adjoint X, Y only, which contain the extremiser of Theorem 5.1. Half of the starts begin at structured points (two-level spectra, rotations in pairs of eigenvectors of W). The lattice operations are the spectral projections of $P + Q$ for the eigenvalue 2 ($P \wedge Q$) and for its range ($P \vee Q$), computed by functional calculus with an indicator function. In items 19 and 23 the suprema are approached as P and Q merge, where $P \wedge Q$ and $P \vee Q$ jump; there the search maximises $(\mathcal{L} - \mathcal{R})/\mathcal{N} - 10^{-12}/\|P - Q\|_{\mathrm{F}}^2$, which keeps P and Q about 10^{-3} apart, while the reported values are those of $(\mathcal{L} - \mathcal{R})/\mathcal{N}$. A grid of 14 contrasts times 64 starts forms one batch per item and dimension (800 gradient steps, $n = 2, 3, 4$).

The best final point for every contrast is recomputed with 40-digit arithmetic (mpmath), after the unitaries are re-orthonormalised and the contrast is recomputed from the stored eigenvalues. Every reported value is therefore the exact violation at an exactly admissible point, and so a rigorous *lower* bound for the supremum; we call it the *computed envelope*. It is not an upper bound: where no theorem is available, the true envelope may be larger. The double-precision and high-precision values agree to 5×10^{-6} (relative), and at $\delta = 0$ no value exceeds 1×10^{-16}. For items 12–16, independent NumPy/SciPy searches (BFGS from 40 random starts on M_2 at $\delta = 0.3, 0.5, 0.9$) give values such that the ratio of the batched search to them lies in $[0.994, 1.097]$; the ratios above 1 come from item 13, where BFGS stops earlier. The whole computation takes 475 s on 4 CPU processes.

12.2 Envelopes with the operators free

Tables 2 and 3, Figure 2 and the top rows of Figures 9–33 summarise the computation. Where an envelope is known exactly and attained (items 1–5, 8, 11, and 9 for $n = 2$), the search reproduces it to 1×10^{-7} for $\delta \geq 0.05$ and $n = 2, 3, 4$; for items 17, 18, 20–22 and 24 it does so to 3×10^{-4}, and for the approached suprema of 19 and 23 to 2×10^{-6}. The values of 17, 18 and 21–23 on M_3 and M_4 agree with those on M_2 to 4×10^{-7}, as Theorem 8.1 predicts, and on M_3 the search finds the odd-dimensional value 0.0694 of item 20 at $\delta = 0.5$ (M_2: 0.0802). These agreements are a check of the method rather than new information.

The new information concerns the items without a proved envelope. All five items 12–16 fail quadratically, with $V \approx 0.5\,\delta^2$ (items 12 and 13), $\delta^2/36$ (item 14, equal to the lower bound of Proposition 9.1(d) to leading order), $0.064\,\delta^2$ (item 15) and $0.09\,\delta^2$ (item 16). Item 15 decreases with n, because the identity part of $e^{P/2} e^{Q} e^{P/2}$ enters $\mathcal{N}$, whereas item 13 increases with n at large δ (on M_3 at $\delta = 0.9$ the value 0.3064 against 0.2197 on M_2); items 12, 14 and 16 give the same values for $n = 2, 3, 4$ (to 4×10^{-3}). In item 16 the best points have large $\|A\|$ and $\|B\|$, and the supremum is approached as the scale grows. Among the items with a proved lower bound, no computed value exceeds the bound (the largest relative difference, search minus bound, is -1×10^{-13}) except for item 6 in dimension $n \geq 3$: on M_2 the search never exceeds $\kappa - 1$, but for $n \geq 3$ it finds larger violations, $V \geq 0.847$ at $n = 4$, $\delta = 0.8$, where $\kappa - 1 = 0.667$. At $n = 3$, $\delta = 0.9$ the present run

items	$V^\star(\delta)$	as $\delta \to 0$	$V^\star(0.5)$	search $V(0.5)$	status
8	$h-1 = 2\delta/(1-\delta)$	2δ	2.0000	2.0000	sharp
9	$s/(1+s)$, $s = \sin\frac{\psi}{2}$	$\delta/2$	0.2056	0.2056	sharp for $n = 2$
19	δ	δ	0.5000	0.5000	sharp, approached
23	$\delta/(1-\delta)$	δ	1.0000	1.0000	sharp, approached
24	$(1-\delta)^{-1/2} - 1$	$\delta/2$	0.4142	0.4142	sharp for $n = 2$; grows with n
1–5, 18, 22	$\kappa - 1$	$\delta^2/2$	0.1547	0.1547	sharp
17	$\kappa^2 - 1$	δ^2	0.3333	0.3333	sharp
11	$1 - \sqrt{1-\delta^2}$	$\delta^2/2$	0.1340	0.1340	sharp for $n = 2$; grows with n
20	max. of one variable	$\frac{(e-1)^2}{e(e+1)}\delta^2$	0.0802	0.0802	sharp
21	$\frac{1}{2}\tan^2(\psi/2)$	$\delta^2/8$	0.0359	0.0359	sharp
6	$\geq \kappa - 1$	$\geq \delta^2/2$	0.1547	0.1430	lower bound
7	$\geq \kappa^2 - 1$	$\geq \delta^2$	0.3333	0.2653	lower bound
12–16	–	$c\,\delta^2$, $c \in [\frac{1}{36}, \frac{1}{2}]$	–	0.0086–0.1338	search
10	$\geq g_{10}(\delta)$	$\geq \delta^4/16$	0.0052	0.0052	lower bound
25	$\geq g_{25}(\delta)$	$\geq \delta^8/4096$	1.7×10^{-6}	1.5×10^{-6}	lower bound

Table 2: The items with a proved envelope or lower bound, ordered by the order of vanishing at the trace. $V^\star$: the sharp value or, where marked $\geq$, a proved lower bound; "search": the computed envelope on M_2 (a range over the items of a row). Items 11 and 24 are given for $n = 2$.

reaches only $1.2372 < \kappa - 1 = 1.2942$, but a point found by an earlier run and stored in the test suite of the repository gives $V > 2.1$ there (recomputed with 60 digits). For item 6 we therefore only know $\kappa - 1 \leq V_n \leq h - 1$. Items 9 and 10 give the same values for $n = 3, 4$ as for $n = 2$, to 3×10^{-10} and 3×10^{-8}.

Except for items 6 ($n \geq 3$), 11 ($A = W^{-1}$) and 25 (A of rank two with eigenvalue ratio $\approx \delta^2/8$), the best points are rank-one operators or matrix units in the span of eigenvectors of W for its extreme eigenvalues, and they do not commute with W. The search has two limitations. The suprema of items 6 and 7 are approached only as an operator becomes singular; gradient ascent reaches 0.924 and 0.796 of the lower bound at $\delta = 0.5$ and finds no violation of item 6 for $\delta \leq 0.2$. Item 25 is resolved for $\delta \geq 0.3$; the search reaches g_{25} to 10^{-3} for $\delta \geq 0.7$ and does not exceed it. Finally, on $M_2 \oplus M_3$ with $W = 1 \oplus s1$ and $W = \mathrm{diag}(1, \ell) \oplus s1$, $s \in \{\frac{1}{4}, 4\}$, and intra-block contrast $\delta \in \{0.3, 0.5, 0.7\}$, the computation confirms Theorem D: for central W the largest value found for the items that hold for every trace is 1×10^{-16}; for items 11 and 24 the search reproduces the exact values for the fixed density to 2×10^{-11}; with intra-block contrast no value exceeds the M_2 envelope (largest relative difference -7×10^{-14}), and where Theorem D gives equality the search reaches it to 10^{-3} in 63 of 72 cases.

12.3 Both contrasts

For the 14 items with operator variables that are not projections, a second computation prescribes the contrast δ' of all operator variables as well (in items 10 and 25 the variables of prescribed contrast are A and $B - A$; 64 starts per grid point, 800 gradient steps, best point recomputed with 40 digits; 760 s on four CPU processes). The bottom rows of Figures 9–33 (Appendix A) show the computed $V(0.9, \delta')$. At $\delta' = 0$ no item is violated, as Proposition 3.4 requires, but beyond that the dependence on δ' takes several forms. For item 1 the values follow Theorem 6.1 to 1×10^{-9} (relative, $n = 2, 3$, all grid points). For items 2 and 3 the theorem gives only the upper bound $\kappa(\delta\delta') - 1$; on M_2 the search reaches it (to 9×10^{-10}), and on M_3 it reaches the odd-dimensional value of item 1 up to the relative differences -2×10^{-4} (item 2) and -2×10^{-9} (item 3). Item 8 follows Proposition 7.3 on M_2; here the violation grows smoothly with both contrasts. Items 6, 10, 12, 13, 24 and 25, in contrast, show no violation above 10^{-4} for $\delta' < 0.8$: at $\delta = 0.9$ the first grid value of δ' with a violation above 10^{-4}

	inequality $\mathcal{L} \le \mathcal{R}$	ref.	$V(0.1)$	$V(0.5)$	$V(0.9)$	$V^\star(0.5)$	order	slope	n-dep.
1	$\lvert\varphi(X)\rvert \le \varphi(\lvert X\rvert)$	[25]	0.0050	0.1547	1.2942	0.1547	δ^2	2.03	4×10^{-10}
2	$\varphi(\lvert X+Y\rvert) \le \varphi(\lvert X\rvert)+\varphi(\lvert Y\rvert)$	[25, 49]	0.0050	0.1547	1.2942	0.1547	δ^2	2.03	1×10^{-9}
3	$\varphi(A+A^*) \le 2\varphi(\lvert A^*\rvert)$	[13]	0.0050	0.1547	1.2942	0.1547	δ^2	2.03	5×10^{-10}
4	$\lvert\varphi(A^2)\rvert \le \varphi(A^*A)$	[46, 17]	0.0050	0.1547	1.2942	0.1547	δ^2	2.03	1×10^{-8}
5	$\varphi(\lvert P-R\rvert) \le \varphi(\lvert P-Q\rvert)+\varphi(\lvert Q-R\rvert)$	[14]	0.0050	0.1547	1.2942	0.1547	δ^2	2.03	1×10^{-8}
6	$\varphi(A\natural B) \le \sqrt{\varphi(A)\varphi(B)}$	[18]	n.r.	0.1430	1.1979	0.1547^b	δ^2	–	$n\ge3$
7	$\varphi((B^{1/2}AB^{1/2})^2) \le \varphi(BA^2B)$	[12]	0.0059	0.2653	3.4322	0.3333^b	δ^2	2.31	9×10^{-2}
8	$\varphi(\lvert AB^*\rvert) \le \frac12\varphi(\lvert A\rvert^2)+\frac12\varphi(\lvert B\rvert^2)$	[7]	0.2222	2.0000	18.00	2.0000	δ^1	1.17	9×10^{-12}
9	$\varphi(PQ+QP) \le 2\varphi(QPQ)$	[13]	0.0477	0.2056	0.3469	0.2056^a	δ^1	0.94	3×10^{-10}
10	$0\le A\le B \Rightarrow \varphi(A^2)\le\varphi(B^2)$	[8, 47, 33]	6.3×10^{-6}	0.0052	0.1825	0.0052^b	δ^4	4.04	3×10^{-8}
11	$(\det A)^{1/n} \le \frac1n\varphi(A)$, $\varphi(1)=n$	[15]	0.0050	0.1340	0.5641	0.1340	δ^2	2.01	yes
12	$\varphi((A+B)^{1/2}) \le \varphi(A^{1/2})+\varphi(B^{1/2})$	[52]	0.0050	0.1338	0.9678	–	–	2.00	5×10^{-4}
13	$\varphi(A+B-\lvert A-B\rvert) \le 2\varphi(A^{1/4}B^{1/2}A^{1/4})$	[34]	0.0048	0.0911	0.2197	–	–	1.93	4×10^{-1}
14	$\lvert\varphi(PQ)\rvert \le \varphi(P)^{1/3}\varphi(Q)^{2/3}$	[11, 13]	2.8×10^{-4}	0.0086	0.0711	–	–	2.03	4×10^{-3}
15	$\varphi(e^{P+Q}) \le \varphi(e^{P/2}e^Qe^{P/2})$	[11]	6.4×10^{-4}	0.0142	0.0420	–	–	1.98	2×10^{-1}
16	$\varphi(e^A)e^{\varphi(e^{A/2}Be^{A/2})/\varphi(e^A)} \le \varphi(e^{A+B})$	[12]	9.7×10^{-4}	0.0561	0.8890	–	–	2.35	2×10^{-3}
17	$\varphi(PQP) \le \varphi(Q)$	[10]	0.0101	0.3333	4.2632	0.3333	δ^2	2.04	7×10^{-9}
18	$\varphi(\lvert P-Q\rvert) \le \varphi(P+Q)$	[10]	0.0050	0.1547	1.2942	0.1547	δ^2	2.03	1×10^{-8}
19	$\varphi(P+Q) \le \varphi(1+P\wedge Q)$	[10]	0.1000	0.5000	0.9000	0.5000^c	δ^1	1.00	odd n
20	$\varphi(e^{QPQ}) \le \varphi(e^P)$	[11]	0.0029	0.0802	0.3115	0.0802	δ^2	2.02	odd n
21	$\varphi(\mathrm{Re}\,PQ) \ge 0$	[11]	0.0013	0.0359	0.1964	0.0359	δ^2	2.02	8×10^{-9}
22	$\varphi(\mathrm{Re}\,PQ) \le \varphi(\lvert PQ\rvert)$	[11]	0.0050	0.1547	1.2942	0.1547	δ^2	2.03	5×10^{-9}
23	$\varphi(P\vee Q) \le \varphi(P)+\varphi(Q)$	[46]	0.1111	1.0000	9.0000	1.0000^c	δ^1	1.17	4×10^{-7}
24	$\varphi(A^2)^{1/2} \le \varphi(A)$, $\varphi(1)=n$	[15]	0.0541	0.4142	2.1623	0.4142	δ^1	1.13	yes
25	$0\le A\le B \Rightarrow \varphi(A^4)\le\varphi(AB^2A)$	[33]	n.r.	1.5×10^{-6}	0.0017	1.7×10^{-6b}	δ^8	–	3×10^{-6}

Table 3: Computed envelopes (certified lower bounds) on M_2 at $\delta = 0.1, 0.5, 0.9$ and the proved values $V^\star$; ^{a}: sharp for $n=2$ only, ^{b}: a lower bound, ^{c}: approached but not attained. "order": leading order of $V^\star$ as $\delta\to0$; "slope": $\log_2(V(0.2)/V(0.1))$ of the search; "n-dep.": the largest $\lvert V_n/V_2-1\rvert$ for $n=3,4$ and $\delta\ge0.1$ (for item 7 it reflects search shortfalls, for item 13 a genuine increase with n; items 19 and 20 differ for odd n, Theorem 8.1). "n.r.": no violation resolved by the search (full run on 4 CPU processes with one thread each).

is 6 ($n=2$): 0.9; 6 ($n=3$): 0.8; 10 ($n=2$): 0.8; 10 ($n=3$): 0.8; 12 ($n=2$): 0.99; 12 ($n=3$): 0.99; 13 ($n=2$): 0.9; 13 ($n=3$): 0.9; 24 ($n=2$): 0.8; 24 ($n=3$): 0.9; 25 ($n=2$): –; 25 ($n=3$): 0.99 ("–": none on the grid). For items 12 and 24 Section 11.1 proves such thresholds, and for item 24 the computed values agree with the exact envelope of Proposition 11.6; for items 6, 10, 13 and 25 the thresholds remain observations. The determinant inequality (item 11) behaves differently again: it is violated most strongly when the contrast of A is close to that of W, since its extremiser is W^{-1}, and in M_2 items 4 and 7 also have an interior maximum in δ' (for item 4 at the contrast of $\lvert A\rvert$ for the extremiser $A=\sqrt{a}\,e_1e_2^*+\sqrt{b}\,e_2e_1^*$), which the additional dimension of M_3 removes. Finally, item 16 does not see the contrast of the operators at all: it is invariant under $A\mapsto A+c$, $B\mapsto B+c$, which makes both contrasts arbitrarily small without changing the quotient, so $V(\delta,\delta')$ equals the envelope with free operators for every $\delta'>0$ and jumps to 0 at $\delta'=0$.

The same computation restricted to two-level operators of contrast δ' reproduces the violations found with general spectra on M_3 (to 1%, wherever they exceed 10^{-4}) except in the items and at the grid values 2 ($\delta'\in\{0.4,0.6\}$); 4 ($\delta'\ge0.2$); 6 ($\delta'\ge0.8$); 7 ($\delta'\ge0.6$); 11 ($\delta'\in\{0.4,0.8,0.9,0.99\}$); 13 ($\delta'\ge0.9$); 25 ($\delta'\ge0.99$). In item 4 the extremisers on M_3 apparently need a third eigenvalue (with the contrast free, the extremiser has the eigenvalues $\sqrt{a},\sqrt{b},0$; Section 12.4), and in items 6 and 7 the two-level values fall further behind as $\delta'\to1$. In item 25 two-level operators give no violation at $\delta=0.9$, $\delta'=0.99$ and 0.999, where general spectra give 1.9×10^{-4} and 1.5×10^{-3}; in item 11 they fall short most at $\delta=0.6$, $\delta'=0.9$ (0.017 against 0.045), and in item 2 by about 10% at $\delta=0.3$. In item 13 at $\delta=0.9$ two-level operators give no violation for $\delta'=0.9$–0.99, where general spectra give

0.019–0.11, but this is a failure of the restricted search rather than evidence: on M_3 it returns the same values at every $\delta' \le 0.99$, i.e. it does not move away from its starting points. On M_2, where every operator of contrast δ' is two-level, the two searches should agree; they differ in item 7 ($\delta = 0.3$, $\delta' = 0.9$) and in item 13 ($\delta = 0.9$, $\delta' = 0.9$ and 0.95, where the restricted search finds the larger value), which indicates the reliability of the search near $\delta' = 1$. In item 10 the two classes differ only where neither violates the inequality.

12.4 Projections and two-level operators

For the items to which the vertex principle (Lemma 3.5) applies, two-level operators attain the supremum, and in eleven items the variables are projections by definition. To see how far this extends, we repeated the search for the other 14 items with the positive variables (the moduli, for the others) restricted to two-level operators $c(\ell + (1-\ell)P)$ with free $\ell \in [0,1]$, $c > 0$ and a projection P of any rank (for self-adjoint variables $\alpha + \beta P$ with free real α, β), and to scaled projections cP (Table 4).

	M_2						M_3					
	$\delta = 0.5$			$\delta = 0.9$			$\delta = 0.5$			$\delta = 0.9$		
item	all	two-level	proj.	all	two-level	proj.	all	two-level	proj.	all	two-level	proj.
1	0.1547	0.1547	0.1547	1.2942	1.2942	1.2942	0.1547	0.1547	0.1547	1.2942	1.2942	1.2942
2	0.1547	0.1547	0.1547	1.2942	1.2941	1.2942	0.1547	0.1547	0.1547	1.2942	1.2941	1.2942
3	0.1547	0.1547	0.1547	1.2942	1.2942	1.2942	0.1547	0.1547	0.1547	1.2942	1.2942	1.2942
4	0.1547	0.1547	0.0774	1.2942	1.2942	0.6471	0.1547	0.1328	0.0774	1.2942	1.2612	0.7393
6	0.1427	0	0.1501	1.2139	1.1707	1.2924	0.1359	0	0.1491	1.2125	1.1405	1.2922
7	0.2656	0.2730	0	3.4387	3.5241	0	0.2733	0.2703	0	3.4965	3.4358	0
8	2.0000	2.0000	2.0000	18.00	18.00	18.00	2.0000	2.0000	2.0000	18.00	18.00	18.00
10	0.0052	0.0052	0.0052	0.1825	0.1825	0.1825	0.0052	0.0052	0.0052	0.1825	0.1825	0.1825
11	0.1340	0.1340	–	0.5641	0.5641	–	0.1347	0.1347	–	0.6188	0.6188	–
12	0.1338	0.1338	0.1338	0.9678	0.9679	0.9680	0.1338	0.1338	0.1338	0.9678	0.9675	0.9680
13	0.0911	0.0329	0.0911	0.2197	0.2171	0.2197	0.0911	0.0911	0.0911	0.3048	0.2057	0.2197
16	0.0560	0.0561	0.0560	0.8885	0.8885	0.8873	0.0561	0.0561	0.0560	0.8894	0.8887	0.8864
24	0.4142	0.4142	0.4142	2.1623	2.1623	2.1623	0.5275	0.5275	0.5275	2.6056	2.6056	2.6056
25	$1.6 \cdot 10^{-6}$	$1.4 \cdot 10^{-6}$	0	0.0017	0.0017	0	$5.3 \cdot 10^{-7}$	$2.2 \cdot 10^{-8}$	0	0.0017	0.0017	0

Table 4: Computed envelopes (certified lower bounds) over all operators, over two-level operators and over scaled projections, for the items whose variables are not projections. In M_2 every operator is two-level, so there the middle column only checks the restricted parametrisation. "–": item 11 needs an invertible A.

The result is an observation with clear exceptions, not a general principle. Scaled projections attain the computed envelope for items 1, 2, 3, 8, 10, 12, 16 (to within 0.73%) and 24 in M_2 and M_3, and for item 13 in M_2. The restricted searches are less reliable than the unrestricted one: two-level operators include the scaled projections, so every entry of the two-level column below the projection column is a failure of the restricted search, as are, on M_2, where every operator is two-level, the values 0 for item 6 and 0.0329 for item 13 at $\delta = 0.5$. In items 6 and 7 the supremum is approached only in a degenerate limit in which an operator becomes a projection and both sides of the inequality vanish; restricting the search to projections (item 6) brings it closer to the proved lower bound $\kappa - 1$ than the unrestricted search, on M_2 and on M_3, and restricting it to two-level operators (item 7) brings it closer to $\kappa^2 - 1$ on M_2 (but not on M_3); for item 7 the violation at projections themselves is zero. Two-level operators that are not projections are needed for item 4 (the extremiser has $|A|$ with the eigenvalues $\sqrt{a}, \sqrt{b}$), item 11 ($A = W^{-1}$) and item 25 (rank-one A gives no violation, Proposition 9.2). In M_3 the extremiser of item 4 lives on a two-dimensional subspace and vanishes on its complement, so it has the three eigenvalues $\sqrt{a}, \sqrt{b}, 0$ and the restricted search misses it. Finally, general operators give strictly larger violations than the restricted searches find in two cases: in M_3 at $\delta = 0.9$ item 13

is violated by 0.3048, against 0.2197 for projections and for M_2, and for item 6 with $n \geq 3$ the unrestricted search of the previous sections exceeds $\kappa - 1$, while the restricted searches stay below it. In short, projections and their affine images are a good first test of a characterisation, but they do not always realise the largest violation.

12.5 Random sampling

Random sampling is the method of [1, Sect. 5]. We extended it to all items: for $n = 2, 3, 4$ and each item we draw 200000 random functionals and operators (W with a uniformly distributed contrast and the other eigenvalues uniform in between, positive operators $U\mathrm{diag}(\lambda)U^*$ with Haar-random U, uniformly distributed contrast and random scale, arbitrary operators $V|X|$ with Haar-random V, self-adjoint ones with random signs, and Haar-random projections), and a second sample in which every positive operator (every modulus) is replaced by a scaled projection of random rank. Appendix A shows all samples together with the computed envelopes, and Table 5 the fraction of violating samples.

		M_2		M_3		M_4	
item	inequality	random	proj.	random	proj.	random	proj.
1	Gardner	4.4	19.0	< 0.1	2.9	< 0.1	0.6
2	triangle	1.1	6.3	< 0.1	6.2	< 0.1	4.5
3	$\varphi(A{+}A^*)$	0.6	3.1	< 0.1	0.4	0	< 0.1
4	$\lvert\varphi(A^2)\rvert$	6.6	6.8	0.2	0.4	< 0.1	< 0.1
5	metric	0	–	< 0.1	–	< 0.1	–
6	geom. mean	< 0.1	–	0	–	0	–
7	ALT	19.5	0	14.3	0	11.0	0
8	Young (ext.)	7.6	8.4	3.0	2.1	1.4	0.6
9	Bikchentaev-Abed	50.2	–	50.1	–	49.8	–
10	monotonicity	< 0.1	0.7	0	< 0.1	0	< 0.1
11	AM-GM	20.5	–	13.2	–	10.0	–
12	subadditivity	< 0.1	21.3	0	9.1	0	5.1
13	Powers-Stormer	< 0.1	9.0	0	3.0	0	1.4
14	Hoelder	1.3	–	< 0.1	–	0	–
15	Golden-Thompson	25.6	–	16.8	–	11.5	–
16	Peierls-Bogoliubov	2.4	1.8	0.8	0.4	0.4	0.1
17	$\varphi(PQP)$	12.0	–	4.0	–	1.9	–
18	$\varphi(\lvert P - Q\rvert)$	13.0	–	4.3	–	2.1	–
19	$P \wedge Q$	49.9	–	37.8	–	33.3	–
20	e^{QPQ}	8.0	–	2.2	–	1.0	–
21	$\mathrm{Re}PQ \geq 0$	5.6	–	1.4	–	0.6	–
22	$\mathrm{Re}PQ \leq \lvert PQ\rvert$	14.6	–	7.7	–	4.3	–
23	$P \vee Q$	49.8	–	37.7	–	33.5	–
24	power mean	7.2	50.0	0.2	25.1	< 0.1	16.8
25	weighted mon.	0	0	0	0	0	0

Table 5: Random sampling, 200000 samples per item, class and dimension: the percentage of samples that violate the inequality, for random operators and for scaled projections. "–": not applicable (the variables are projections, or an invertible operator is needed).

No sample exceeds the computed envelope by more than the interpolation between grid values allows: the largest ratio of a sampled violation to the envelope, interpolated linearly in $\Delta(W)$ at the sample's own contrast, is 1.003 over all items and dimensions. Ratios slightly above 1 are possible because the linear interpolation underestimates the envelope where it is concave between grid values, and for the items without a closed form the computed envelope is only a lower bound. This is

an independent consistency check of the computed values. The samples also show why sampling alone is a poor test of a characterisation. Random operators rarely come close to the envelope, and for monotonicity, subadditivity, the Powers–Størmer inequality and the weighted monotonicity fewer than 0.1% of the samples (or none) violate the inequality at all, so a Monte Carlo study alone would suggest that these inequalities hold for every φ_W. Sampling also deteriorates with the dimension: for Gardner's inequality the fraction of violating random samples drops from 4.4% on M_2 to below 0.1% on M_3 and M_4, and the best sample stays far below the envelope (Figure 9), because the extremal configurations, rank-one operators on a two-dimensional subspace aligned with the extreme eigenvectors of W, have a Haar measure that decreases rapidly with n. Sampling scaled projections helps: the fraction of violating samples rises, for Gardner's inequality from 4.4% to 19% on M_2 and from below 0.1% to 0.6% on M_4, for subadditivity from below 0.1% to 21%, 9% and 5% on M_2, M_3, M_4, and for the power-mean inequality from 7% to 50% on M_2, and the best sample comes close to the envelope (Figures 9, 20 and 32). The exceptions are those of Section 12.4: items 4 and 7, whose extremisers are two-level operators that are not projections, and items 6 and 11, which need invertible operators.

13 Conclusions and open problems

A characterisation of the trace by an inequality becomes quantitative once the distance of a functional from the traces is measured by the Michelson contrast of its density. For Gardner's inequality the answer is as simple as it could be: the sharp constant is the secant of the angle whose sine is the product of the contrasts of the density and of the operator. For the other characterisations the envelopes vanish at the trace to the orders δ, δ^2, δ^4 or δ^8, with the classical reverse-inequality constants as exact values where we could determine them; on direct sums and direct integrals only the contrast inside the fibres counts. The rules by which contrasts compose (rapidities add for products and means, angles add for Jordan products, three contrasts multiply in the commutator test) explain some of the thresholds in the contrast of the operators that the computation reveals, and prove them for subadditivity of the square root and for the power-mean inequality. The characterisations by the projection lattice (items 19, 23 and the modular law) hold verbatim for measures on the projections of M_n, which by Gleason's theorem are states unless $n = 2$; for $n = 2$ the envelopes do not change but are attained. The computation supports the following values.

Conjecture 13.1. $V_n = \kappa^2 - 1$ for item 7, $V_n = g_{10}(\delta)$ for item 10 and $V_n = g_{25}(\delta)$ for item 25 in every dimension; $V_2 = \kappa - 1$ for item 6; the value of Proposition 7.4 holds for item 9 in every dimension; and the lower bound of Proposition 7.3 is the two-contrast envelope of item 8 for even n.

Several questions remain open. What is the threshold of item 12 for $\delta < 1$, where the search places it above the value $2\sqrt{2}/3$ of Proposition 11.5, and what are the thresholds of items 6, 10, 13 and 25, which the search shows but which no operator inequality of the kind used in Section 11.1 can explain for item 10? What is the envelope of item 6 for $n \geq 3$, where the search exceeds $\kappa - 1$, and does it grow with n? Do items 12–16 have closed forms? The search gives the leading coefficients $\frac{1}{2}$, $\frac{1}{2}$, $\frac{1}{36}$, 0.064 and 0.09; for item 14 the coefficient $(1 - \frac{2}{p})^2/4$ is a proved lower bound. Beyond matrix algebras, an unbounded density on an infinite-dimensional von Neumann algebra may have contrast 1 while the functional is close to a trace in other senses, and a different measure of non-traciality may be needed. Finally, the same questions can be asked for other characterisations, such as those of [43, 18] with the arithmetic mean, and for the dual problem of characterising central elements [45].

Code availability. The library, the programs that compute all numbers, tables and figures, and the tests are available at `https://github.com/andre-a-no/von_neumann_type_i` (MIT licence).

Reproducing the computations. The programs are in the directory `scripts` of the repository and run in this order, since later ones read the results of earlier ones: `trace_characterizations.py -mode full` (the envelopes with free operators, Tables 2 and 3; 475 s), `trace_contrast.py -mode full` (both contrasts and the restricted searches, Table 4; 760 s), then `trace_montecarlo.py` (Table 5 and Appendix A), `contrast_operations.py` (Section 11), `trace_envelopes_figures.py` (Figure 3) and `trace_paper_schematics.py` (Figures 1, 2 and 6). Figure 4 comes from `bounds_search.py -mode full` of the pipeline of the companion paper [41]. Finally, `make_arxiv_trace.sh` compiles this paper from the generated numbers and figures.

A Random samples and computed envelopes, item by item

Each figure shows one inequality, with one panel per dimension. Top row: the normalised violation $(\mathcal{L}-\mathcal{R})/\mathcal{N}$ of the random samples (grey) and, where the variables are not projections, of the scaled-projection samples (orange) against $\Delta(W)$ on M_2, M_3, M_4; black: the computed envelope $V(\delta)$, a certified lower bound for the supremum, joined by straight lines. Bottom row (items with operator variables): samples with $\Delta(W) = 0.9$ and all operator variables of contrast δ', against δ', on M_2 and M_3, with the computed $V(0.9, \delta')$. Samples far below zero are cut off. Where an envelope is concave between grid points (item 11 below its maximum at $\delta' = \delta$ and after its steep drop near $\delta' = 1$), a few samples lie above the straight segments; on M_2 they lie below the closed form $1-(w_1\ell'+w_2)/(2\sqrt{\ell'})$, $\ell' = \frac{1-\delta'}{1+\delta'}$, $w = \frac{2}{1+\ell}(1,\ell)$.

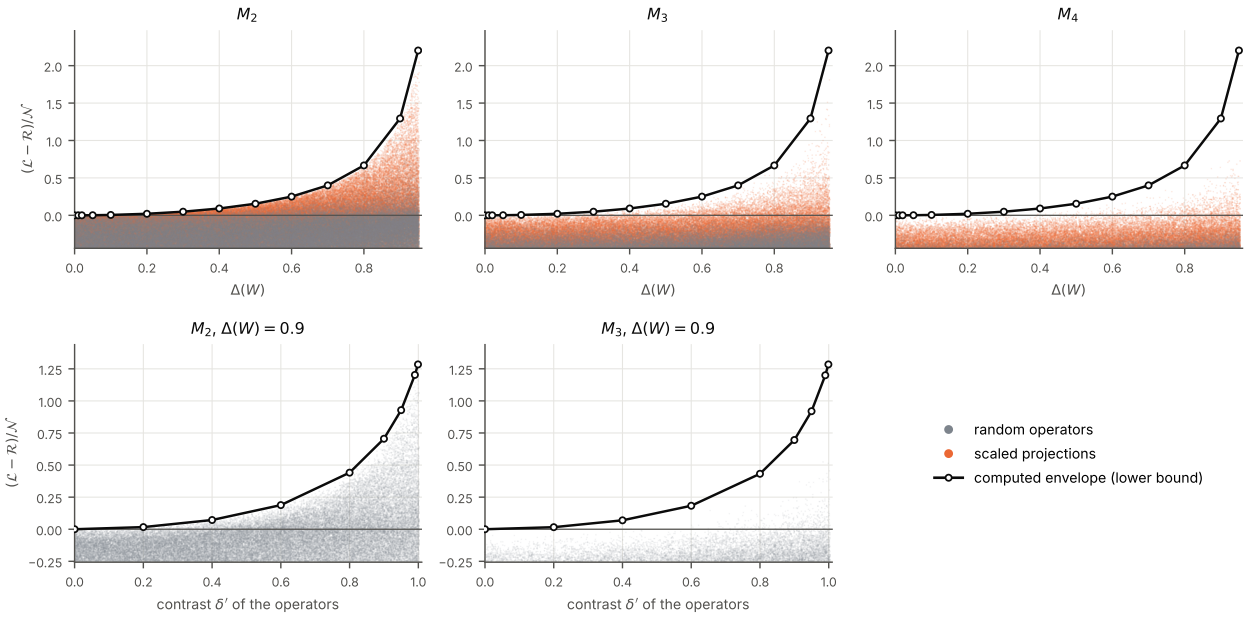


Figure 9: Item 1, $|\varphi(X)| \leq \varphi(|X|)$: random samples and the computed envelope (a certified lower bound); top: against $\Delta(W)$, bottom: against the contrast δ' of the operators at $\Delta(W) = 0.9$.

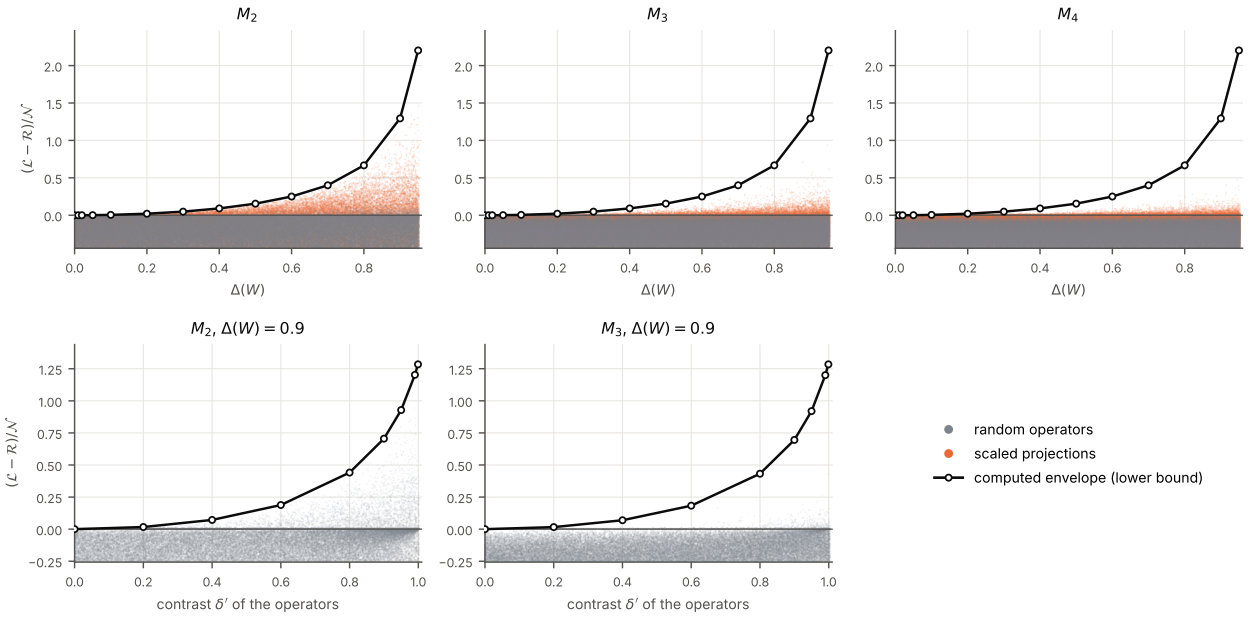


Figure 10: Item 2, $\varphi(|X+Y|) \leq \varphi(|X|) + \varphi(|Y|)$: random samples and the computed envelope (a certified lower bound); top: against $\Delta(W)$, bottom: against the contrast δ' of the operators at $\Delta(W) = 0.9$.

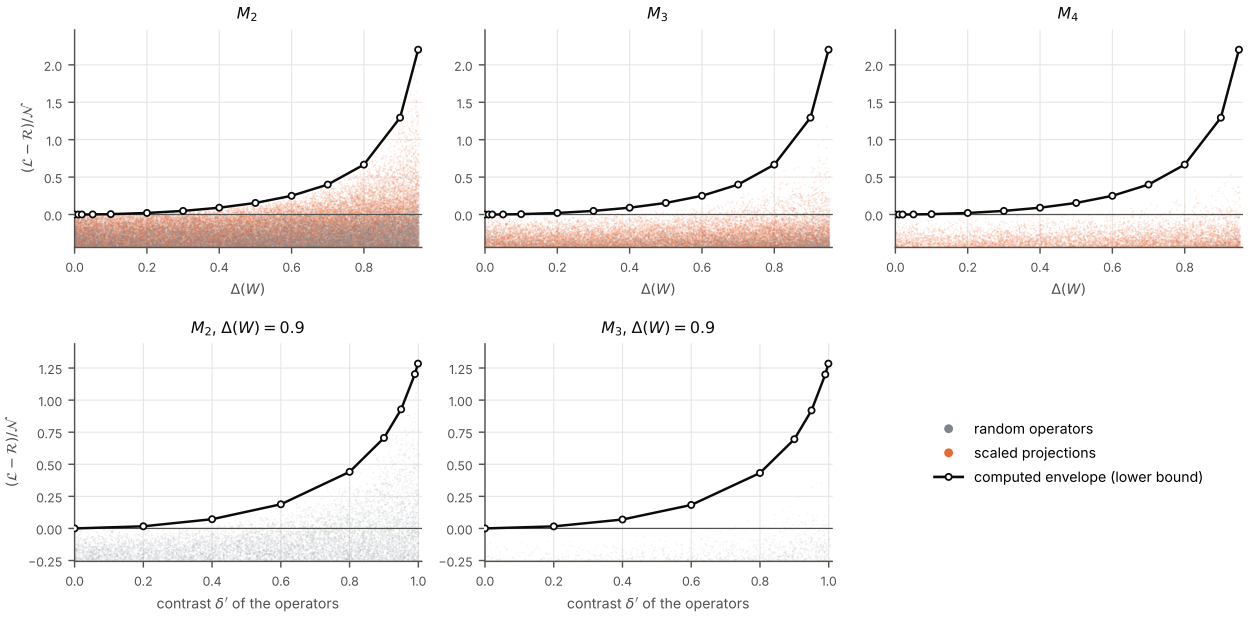


Figure 11: Item 3, $\varphi(A + A^*) \leq 2\varphi(|A^*|)$: random samples and the computed envelope (a certified lower bound); top: against $\Delta(W)$, bottom: against the contrast δ' of the operators at $\Delta(W) = 0.9$.

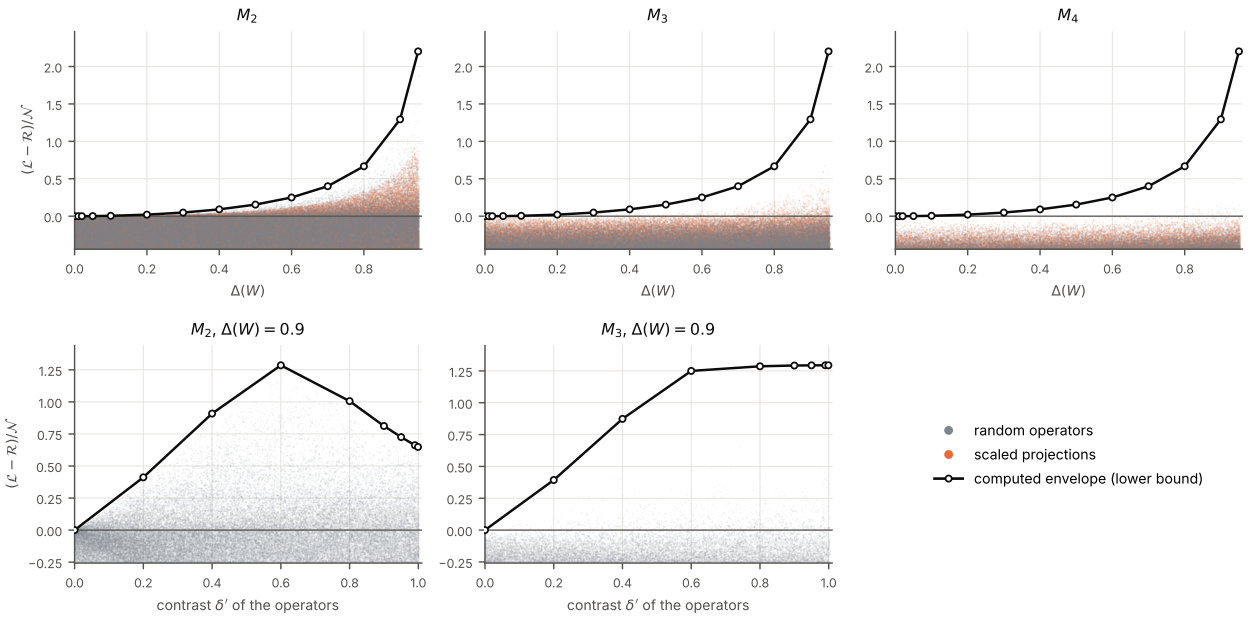


Figure 12: Item 4, $|\varphi(A^2)| \leq \varphi(A^*A)$: random samples and the computed envelope (a certified lower bound); top: against $\Delta(W)$, bottom: against the contrast δ' of the operators at $\Delta(W) = 0.9$.

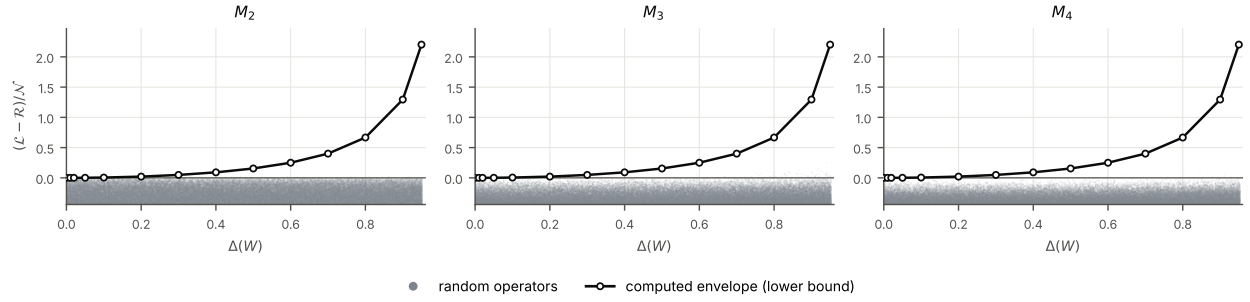


Figure 13: Item 5, $\varphi(|P-R|) \leq \varphi(|P-Q|)+\varphi(|Q-R|)$: random samples and the computed envelope (a certified lower bound) against $\Delta(W)$.

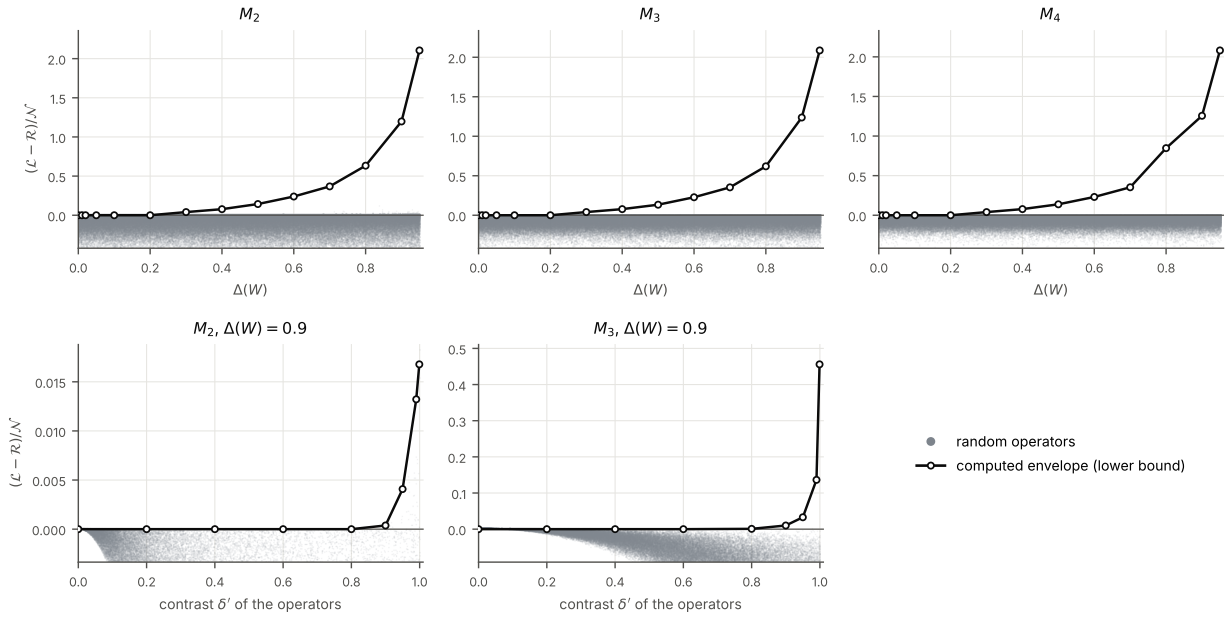


Figure 14: Item 6, $\varphi(A \natural B) \leq \sqrt{\varphi(A)\varphi(B)}$: random samples and the computed envelope (a certified lower bound); top: against $\Delta(W)$, bottom: against the contrast δ' of the operators at $\Delta(W) = 0.9$.

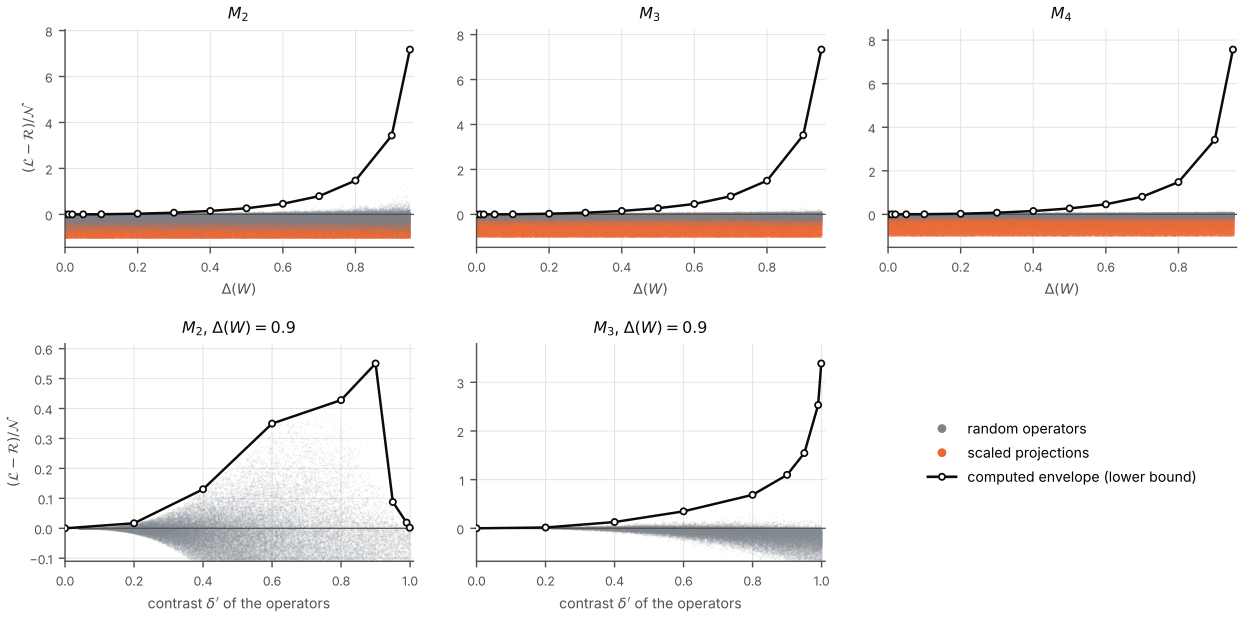


Figure 15: Item 7, $\varphi((B^{1/2}AB^{1/2})^2) \leq \varphi(BA^2B)$: random samples and the computed envelope (a certified lower bound); top: against $\Delta(W)$, bottom: against the contrast δ' of the operators at $\Delta(W) = 0.9$.

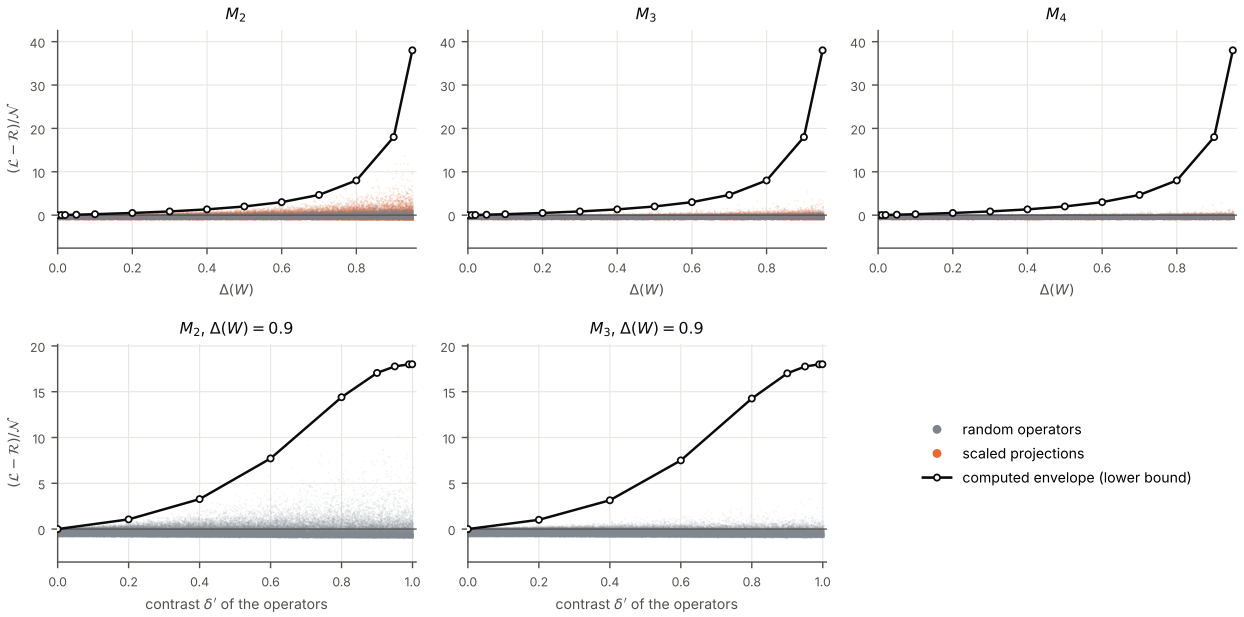


Figure 16: Item 8, $\varphi(|AB^*|) \leq \frac{1}{2}\varphi(|A|^2) + \frac{1}{2}\varphi(|B|^2)$: random samples and the computed envelope (a certified lower bound); top: against $\Delta(W)$, bottom: against the contrast δ' of the operators at $\Delta(W) = 0.9$.

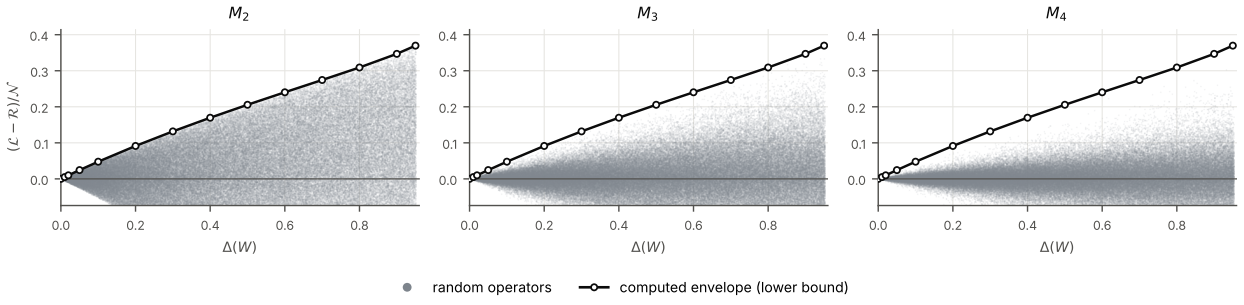


Figure 17: Item 9, $\varphi(PQ+QP) \leq 2\varphi(QPQ)$: random samples and the computed envelope (a certified lower bound) against $\Delta(W)$.

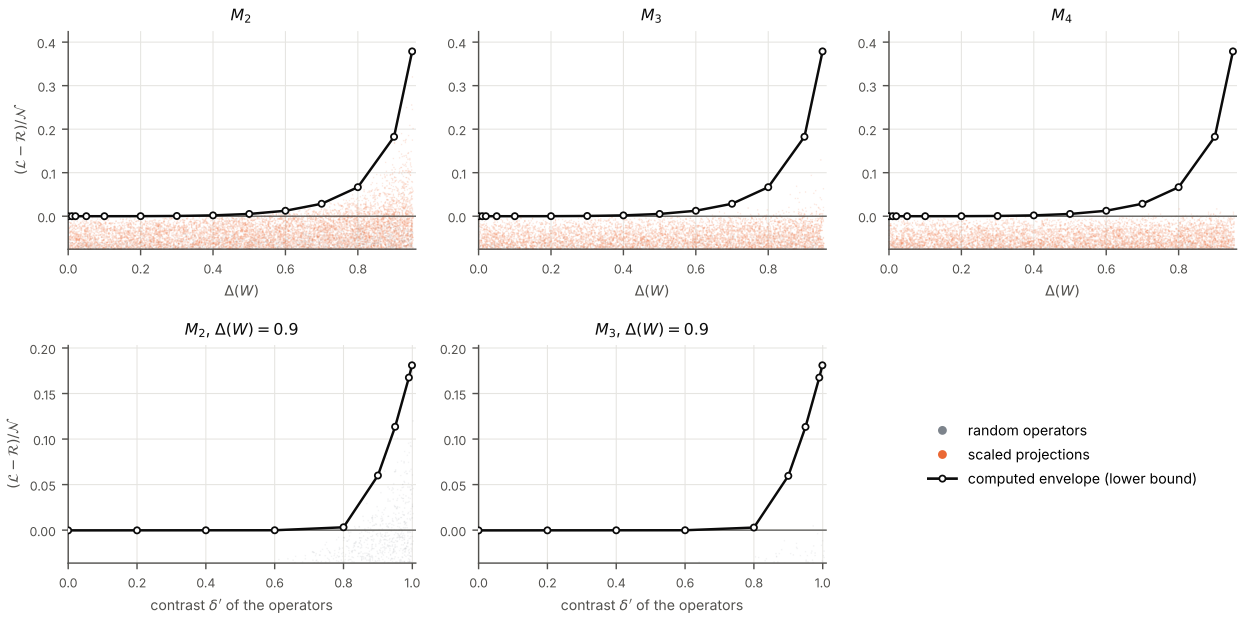


Figure 18: Item 10, $0 \le A \le B \Rightarrow \varphi(A^2) \le \varphi(B^2)$: random samples and the computed envelope (a certified lower bound); top: against $\Delta(W)$, bottom: against the contrast δ' of the operators at $\Delta(W) = 0.9$.

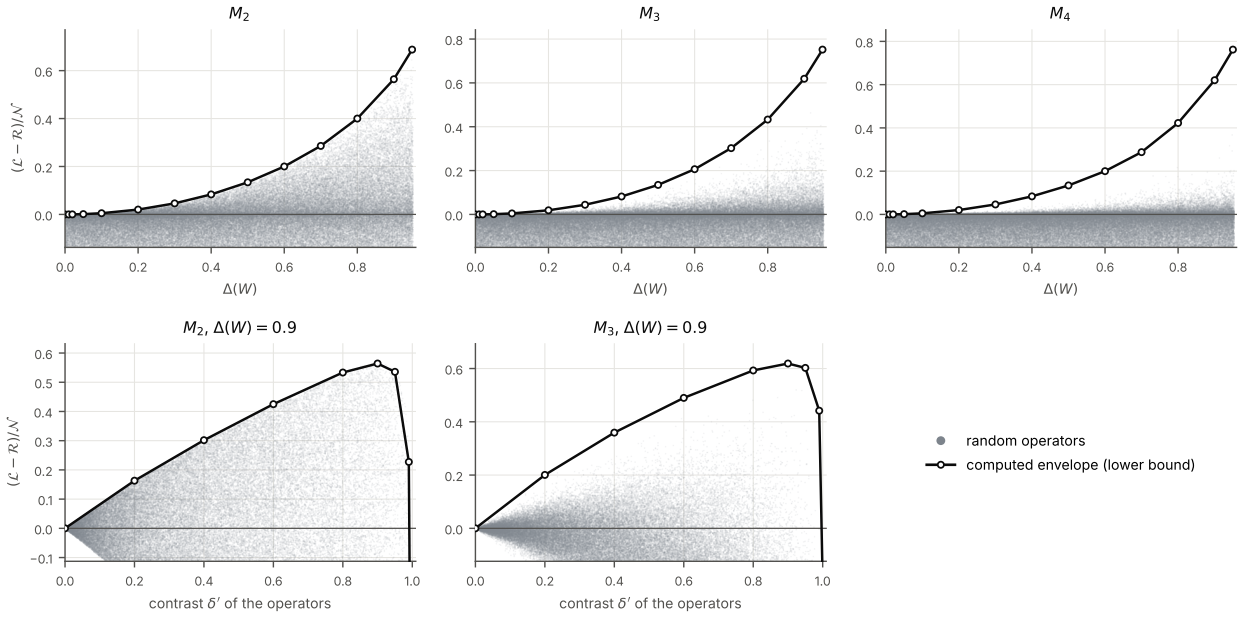


Figure 19: Item 11, $(\det A)^{1/n} \le \frac{1}{n}\varphi(A)$, $\varphi(1) = n$: random samples and the computed envelope (a certified lower bound); top: against $\Delta(W)$, bottom: against the contrast δ' of the operators at $\Delta(W) = 0.9$.

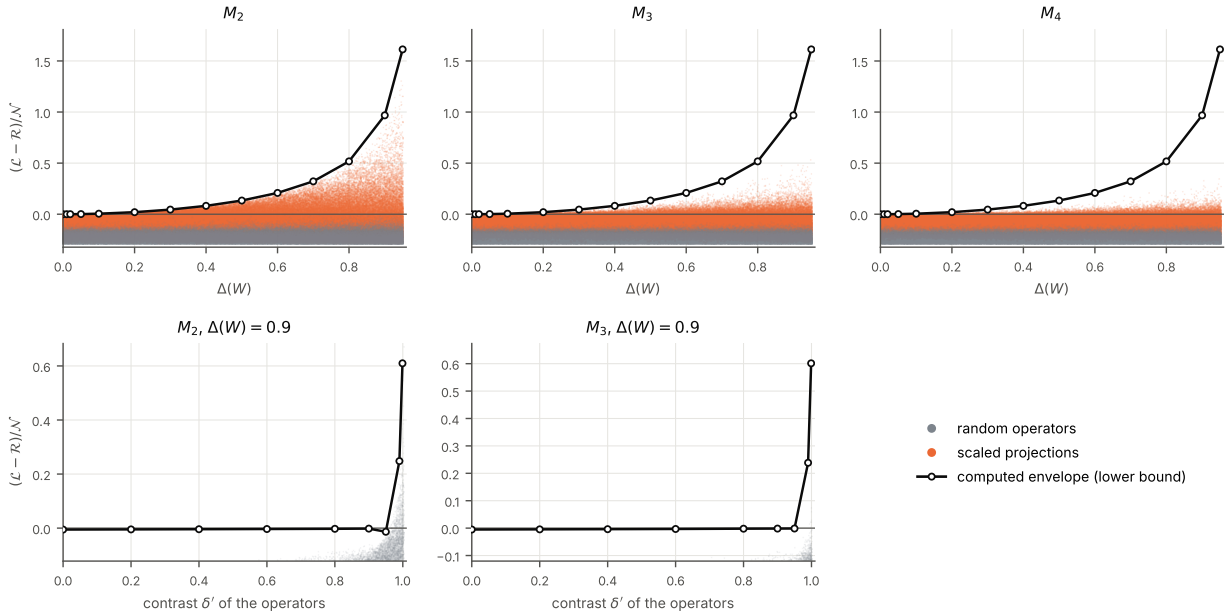


Figure 20: Item 12, $\varphi((A+B)^{1/2}) \leq \varphi(A^{1/2}) + \varphi(B^{1/2})$: random samples and the computed envelope (a certified lower bound); top: against $\Delta(W)$, bottom: against the contrast δ' of the operators at $\Delta(W) = 0.9$.

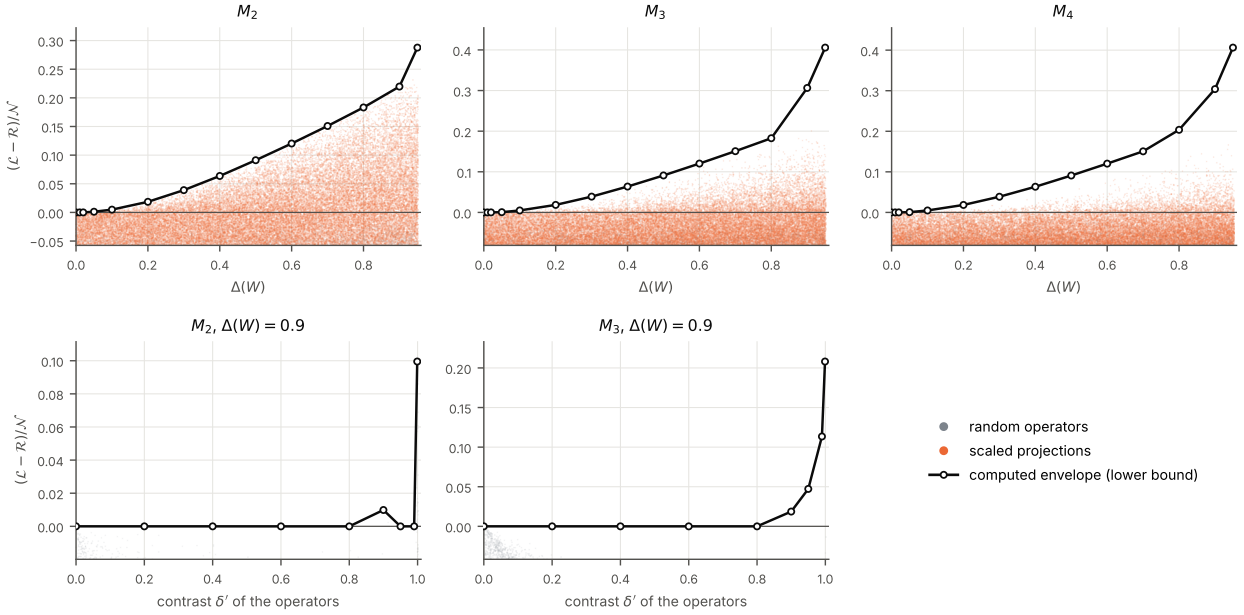


Figure 21: Item 13, $\varphi(A + B - |A - B|) \leq 2\varphi(A^{1/4}B^{1/2}A^{1/4})$: random samples and the computed envelope (a certified lower bound); top: against $\Delta(W)$, bottom: against the contrast δ' of the operators at $\Delta(W) = 0.9$.

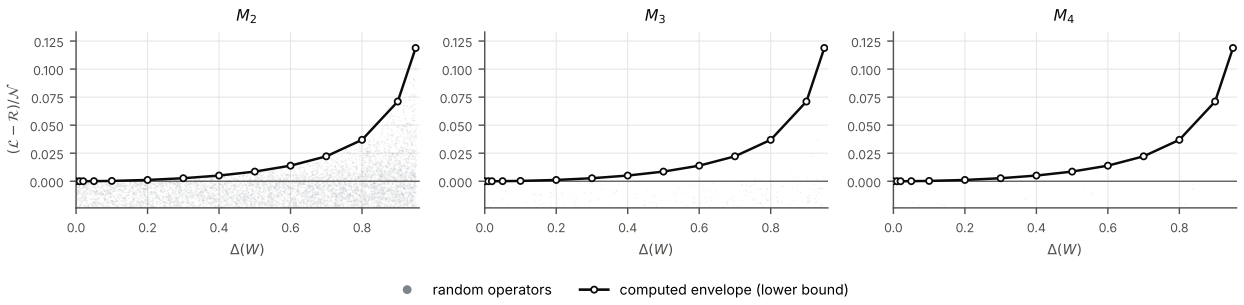


Figure 22: Item 14, $|\varphi(PQ)| \leq \varphi(P)^{1/3}\varphi(Q)^{2/3}$: random samples and the computed envelope (a certified lower bound) against $\Delta(W)$.

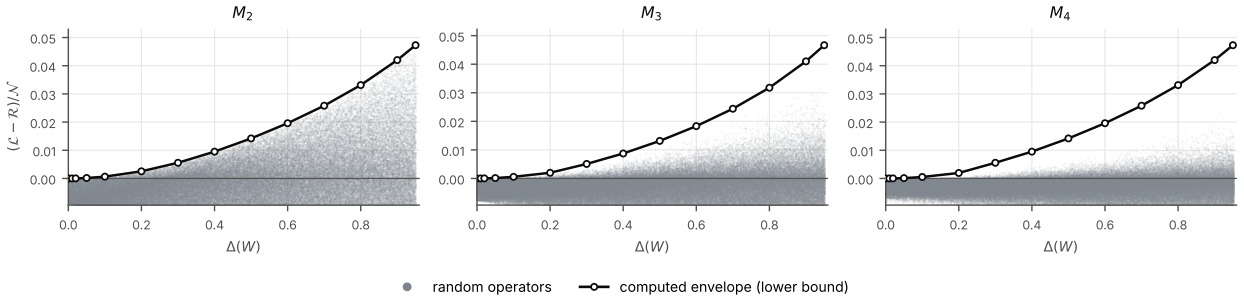


Figure 23: Item 15, $\varphi(e^{P+Q}) \leq \varphi(e^{P/2}e^{Q}e^{P/2})$: random samples and the computed envelope (a certified lower bound) against $\Delta(W)$.

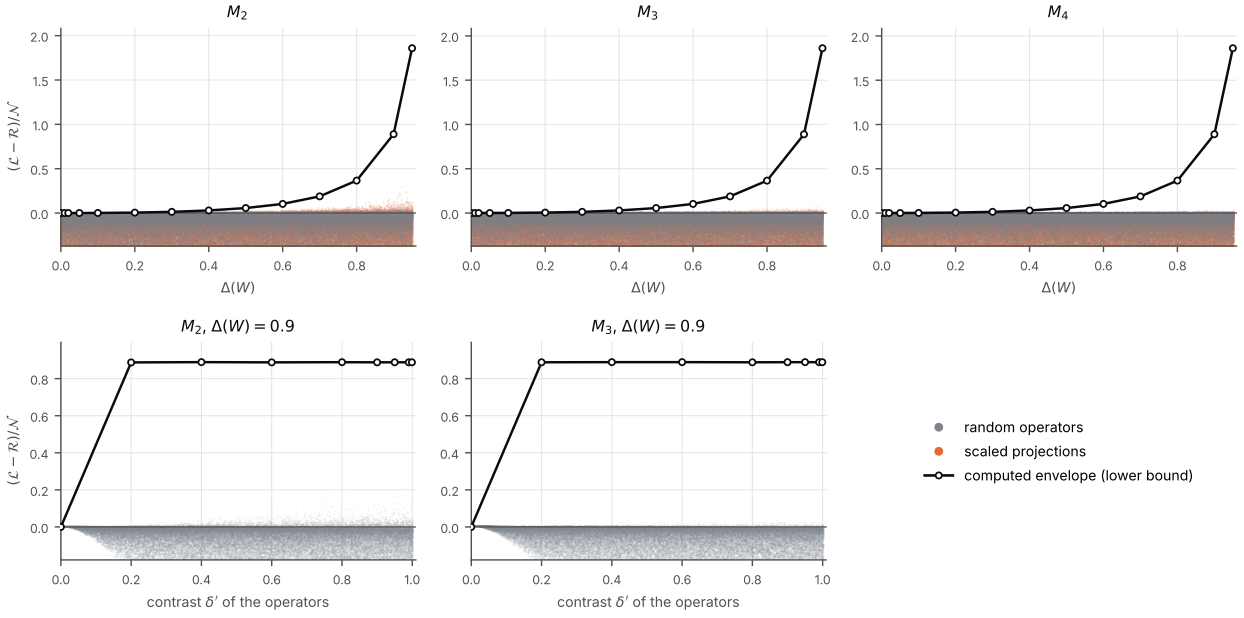


Figure 24: Item 16, $\varphi(e^{A})e^{\varphi(e^{A/2}Be^{A/2})/\varphi(e^{A})} \leq \varphi(e^{A+B})$: random samples and the computed envelope (a certified lower bound); top: against $\Delta(W)$, bottom: against the contrast δ' of the operators at $\Delta(W) = 0.9$.

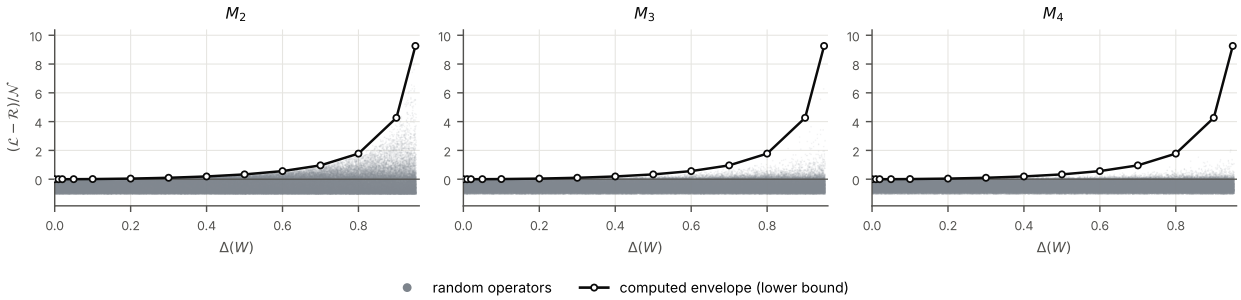


Figure 25: Item 17, $\varphi(PQP) \leq \varphi(Q)$: random samples and the computed envelope (a certified lower bound) against $\Delta(W)$.

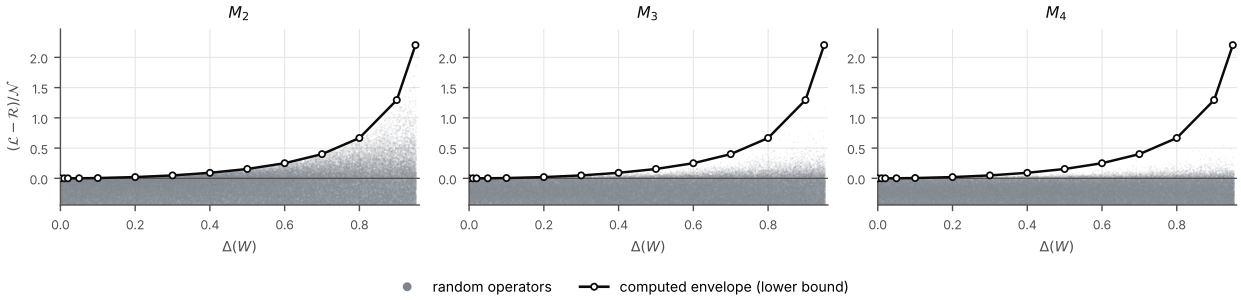


Figure 26: Item 18, $\varphi(|P-Q|) \leq \varphi(P+Q)$: random samples and the computed envelope (a certified lower bound) against $\Delta(W)$.

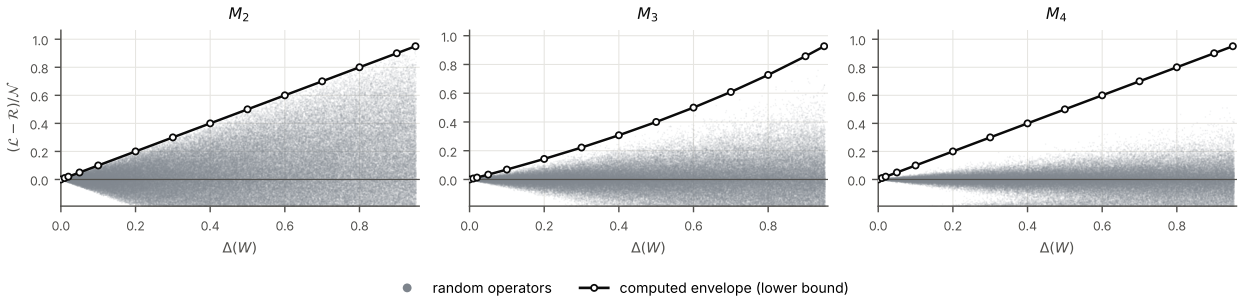


Figure 27: Item 19, $\varphi(P+Q) \leq \varphi(1+P\wedge Q)$: random samples and the computed envelope (a certified lower bound) against $\Delta(W)$.

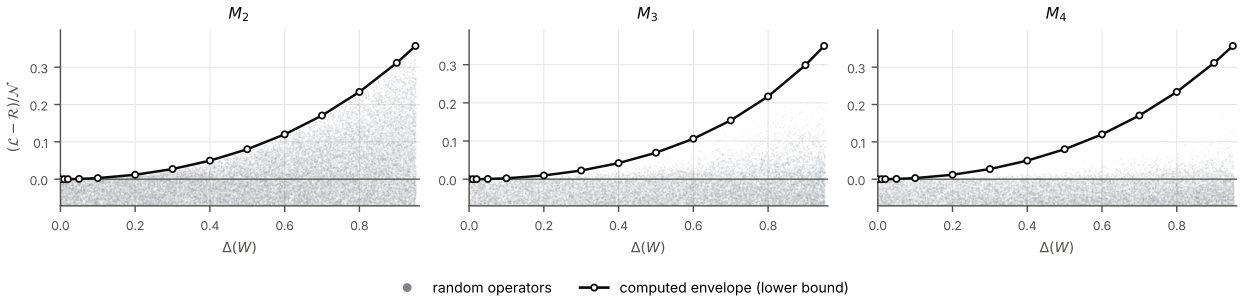


Figure 28: Item 20, $\varphi(e^{QPQ}) \leq \varphi(e^{P})$: random samples and the computed envelope (a certified lower bound) against $\Delta(W)$.

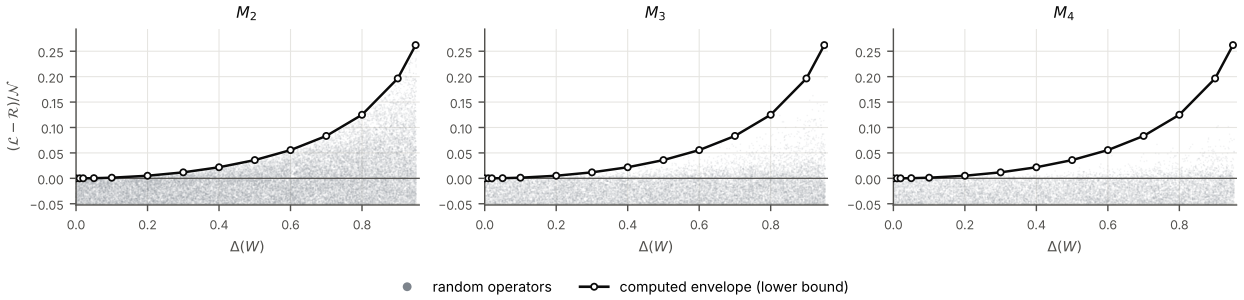


Figure 29: Item 21, $\varphi(\mathrm{Re}\, PQ) \geq 0$: random samples and the computed envelope (a certified lower bound) against $\Delta(W)$.

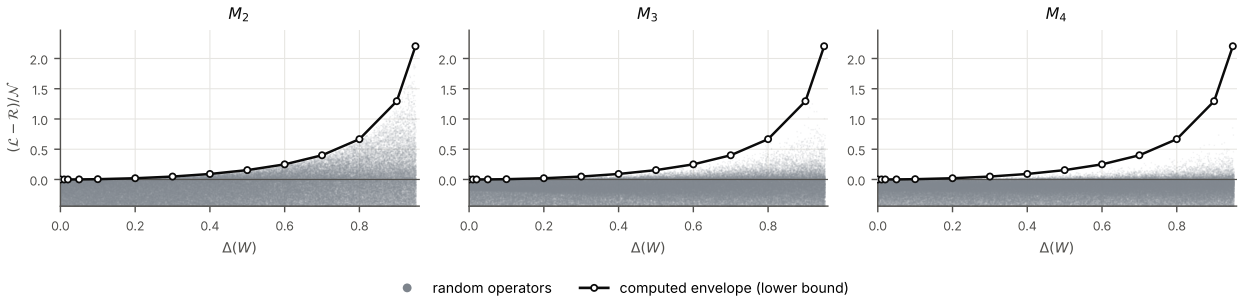


Figure 30: Item 22, $\varphi(\mathrm{Re}\, PQ) \leq \varphi(|PQ|)$: random samples and the computed envelope (a certified lower bound) against $\Delta(W)$.

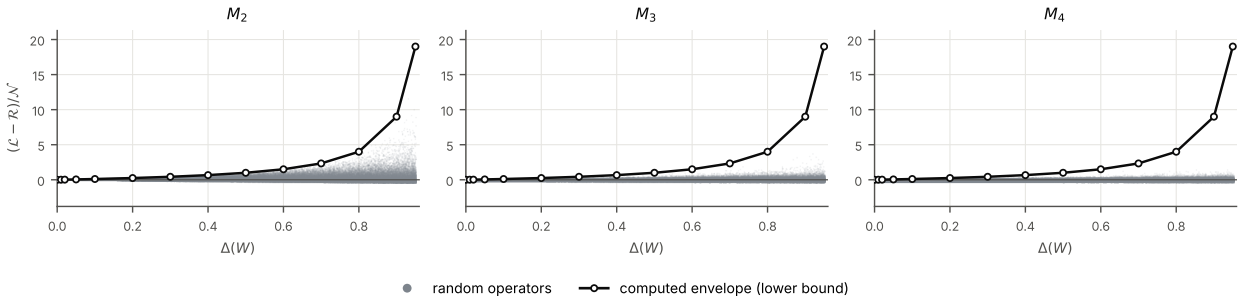


Figure 31: Item 23, $\varphi(P \vee Q) \leq \varphi(P) + \varphi(Q)$: random samples and the computed envelope (a certified lower bound) against $\Delta(W)$.

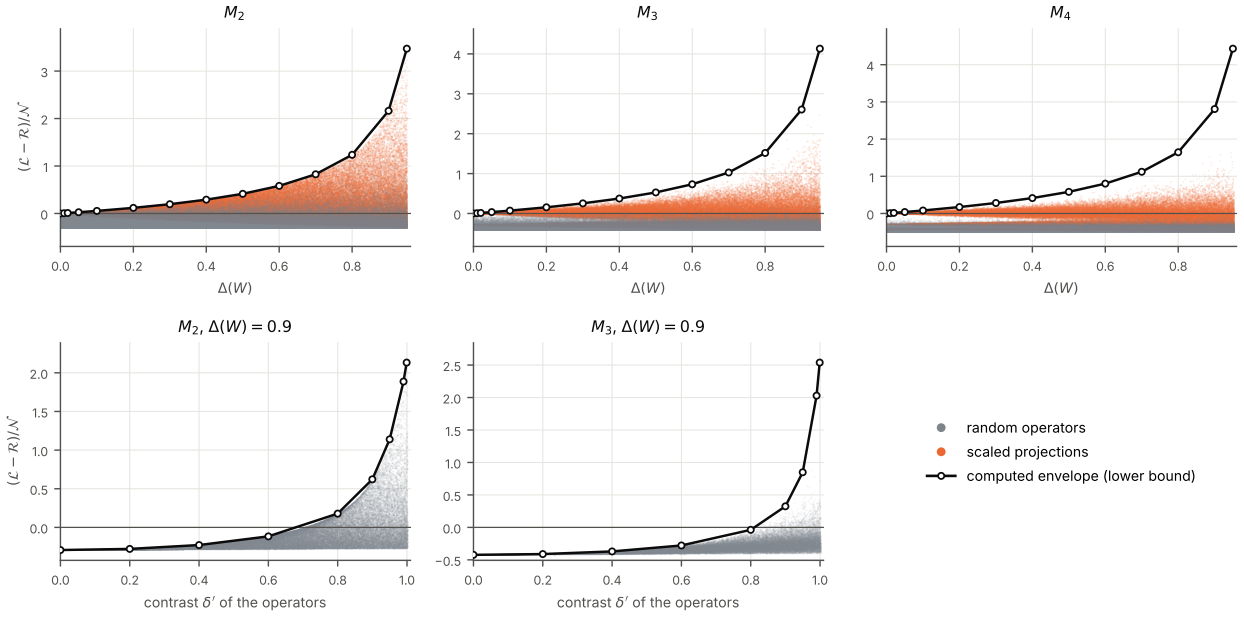


Figure 32: Item 24, $\varphi(A^2)^{1/2} \leq \varphi(A)$, $\varphi(1) = n$: random samples and the computed envelope (a certified lower bound); top: against $\Delta(W)$, bottom: against the contrast δ' of the operators at $\Delta(W) = 0.9$.

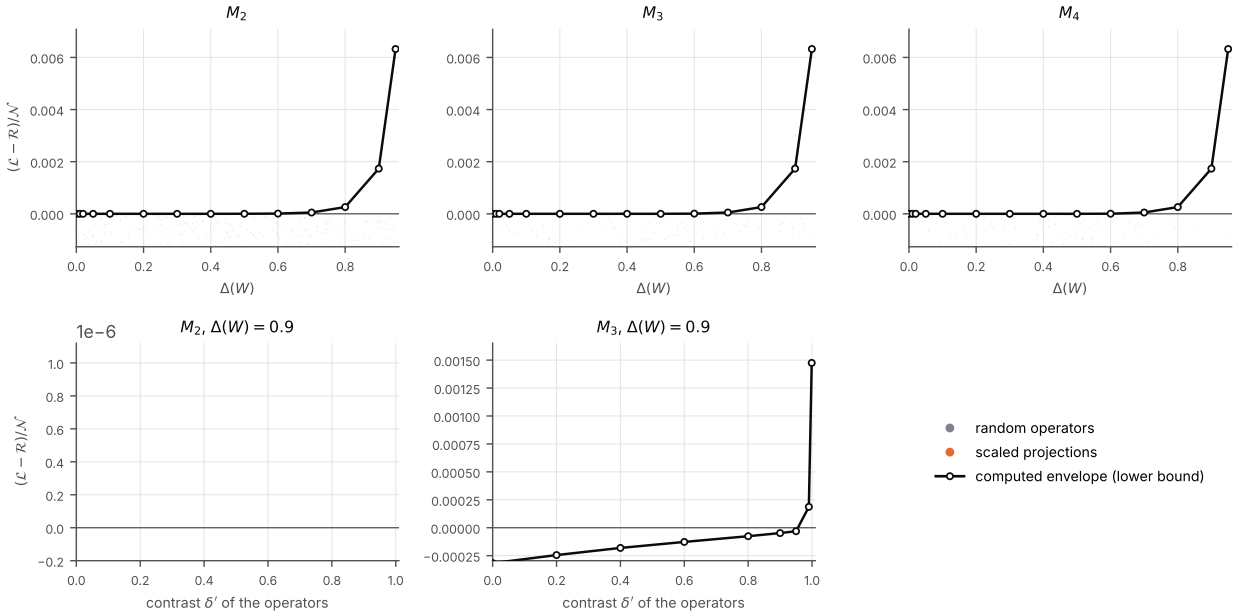


Figure 33: Item 25, $0 \leq A \leq B \Rightarrow \varphi(A^4) \leq \varphi(AB^2A)$: random samples and the computed envelope (a certified lower bound); top: against $\Delta(W)$, bottom: against the contrast δ' of the operators at $\Delta(W) = 0.9$.

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